



IT Green HPE Aruba Networking Combined 1-Day • 3 ก.ย. 2569

Aruba CX Switch + WLAN Basic

Agenda (1/3)

Part 1 : Aruba CX Switch (Wired)

09:15 Overview CX Switch & AOS-CX · CLI / Web UI

09:45 Access LAB (PNET) & Initial Setup

10:45 System Config · Checkpoint / Diff / Rollback

11:30 L2: VLAN, Access / Trunk, LAG

13:00 L3: SVI & Inter-VLAN Routing

Agenda (2/3)

Part 2 : Aruba WLAN (Wireless)

13:45 Overview WLAN Solution & Mobility Controller

14:15 Initial Setup MC & Network Setup

14:40 AP Provisioning

15:15 Authentication: PSK, Portal, 802.1X, MAC

16:15 Application Control (Policy) & Dashboard

Agenda (3/3)

Hands-on LAB — PNET Simulation

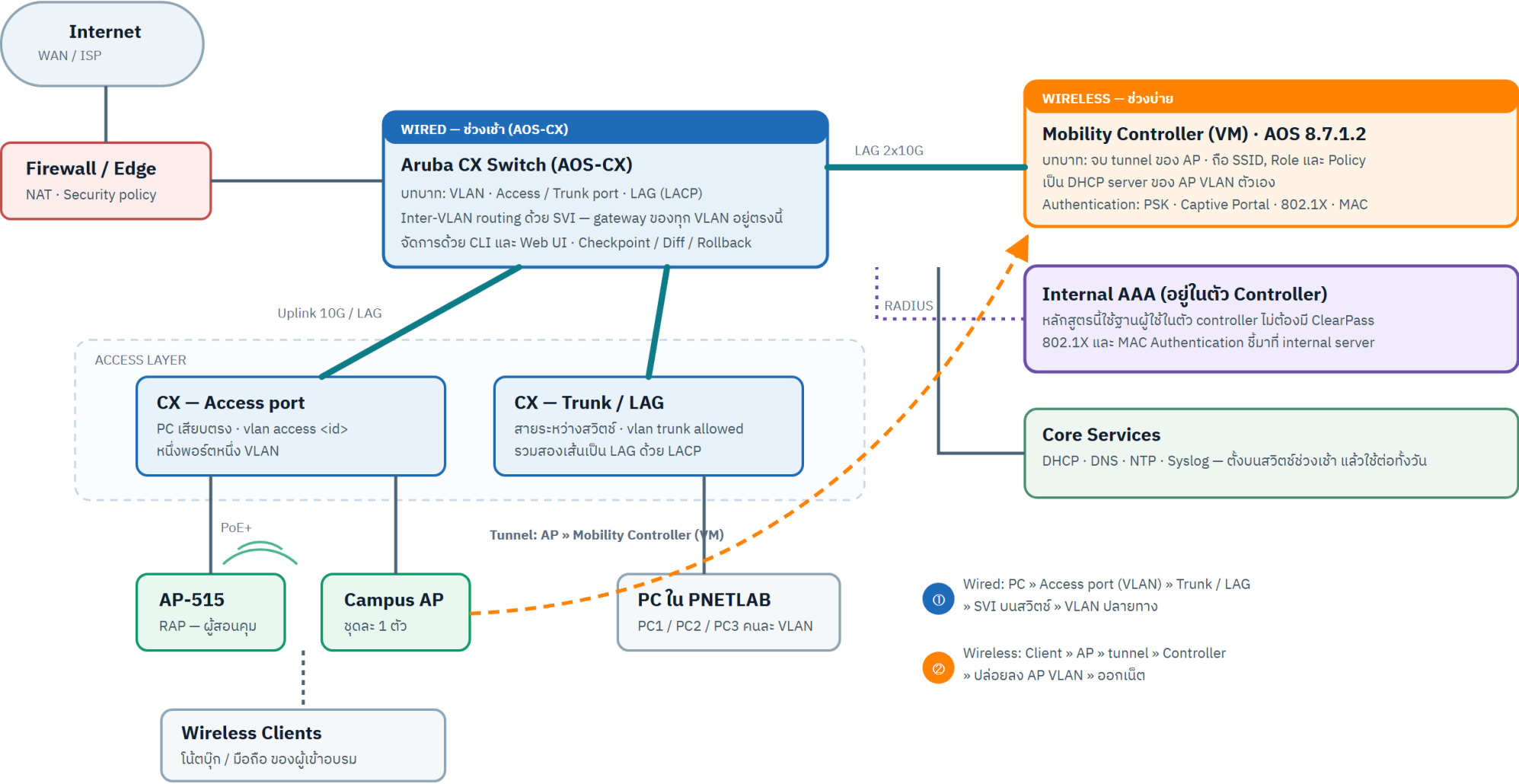
LAB 1–4 Initial setup, remote access, checkpoint

LAB 5–7 VLAN, Trunk, LAG, SVI & inter-VLAN routing

LAB 8–14 Controller, AP, SSID, Role & Policy

Aruba CX Switch + WLAN — ภาพรวมที่ประกอบขึ้นในวันเดียว

Wired วางโครงก่อน · Wireless ต่อกับบนโครงนั้น — Public Class / IT Green Public Company Limited · 3 ก.ย. 2569



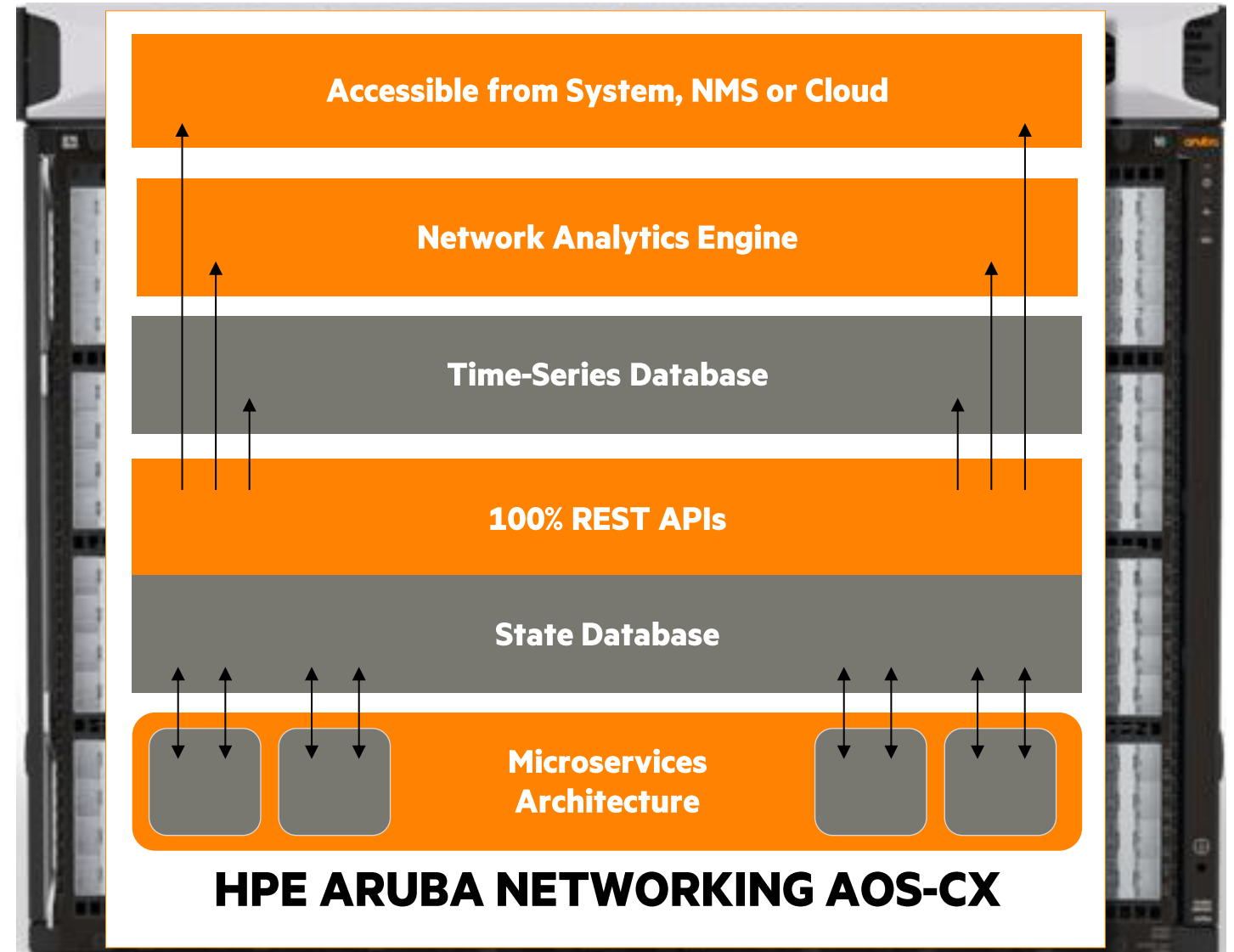
Cloud-native design brings innovation across the CX portfolio

Modularity - Faster innovation with independent processes

Programmability - Simplified operations with automation

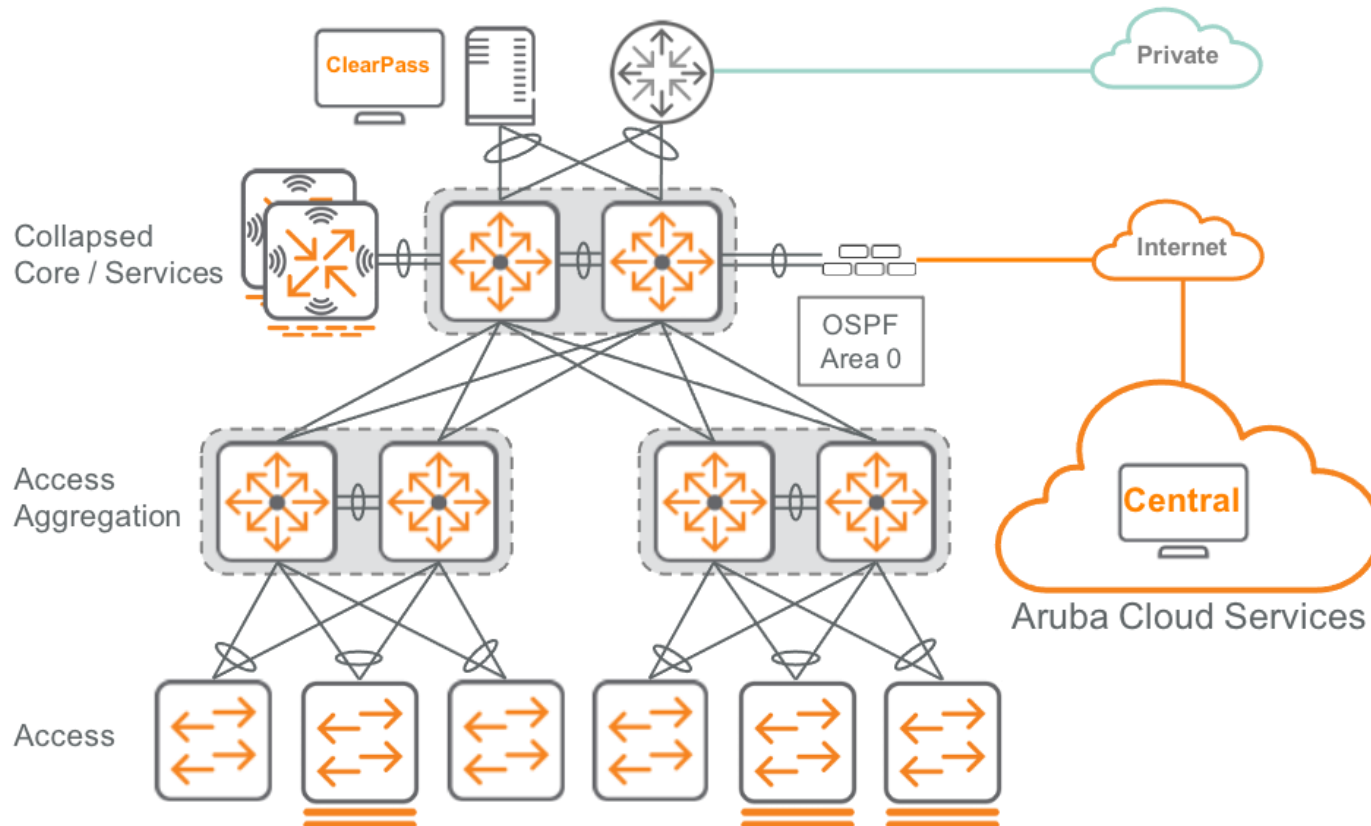
Resiliency - Stable and reliable microservices design

Elasticity - One operating model from edge access to data center



CX switching solutions for all networks

For branch, campus, mid and large enterprises, SMB, and data center networks



Data Center

Enterprise Core

Aggregation and Distribution

Campus and Branch Access

Ruggedized

AOS-CX Operating System: quality and non-stop reliability by design

HPE Aruba Networking commitment to quality

“Shift left” test philosophy

Exhaustive scale testing

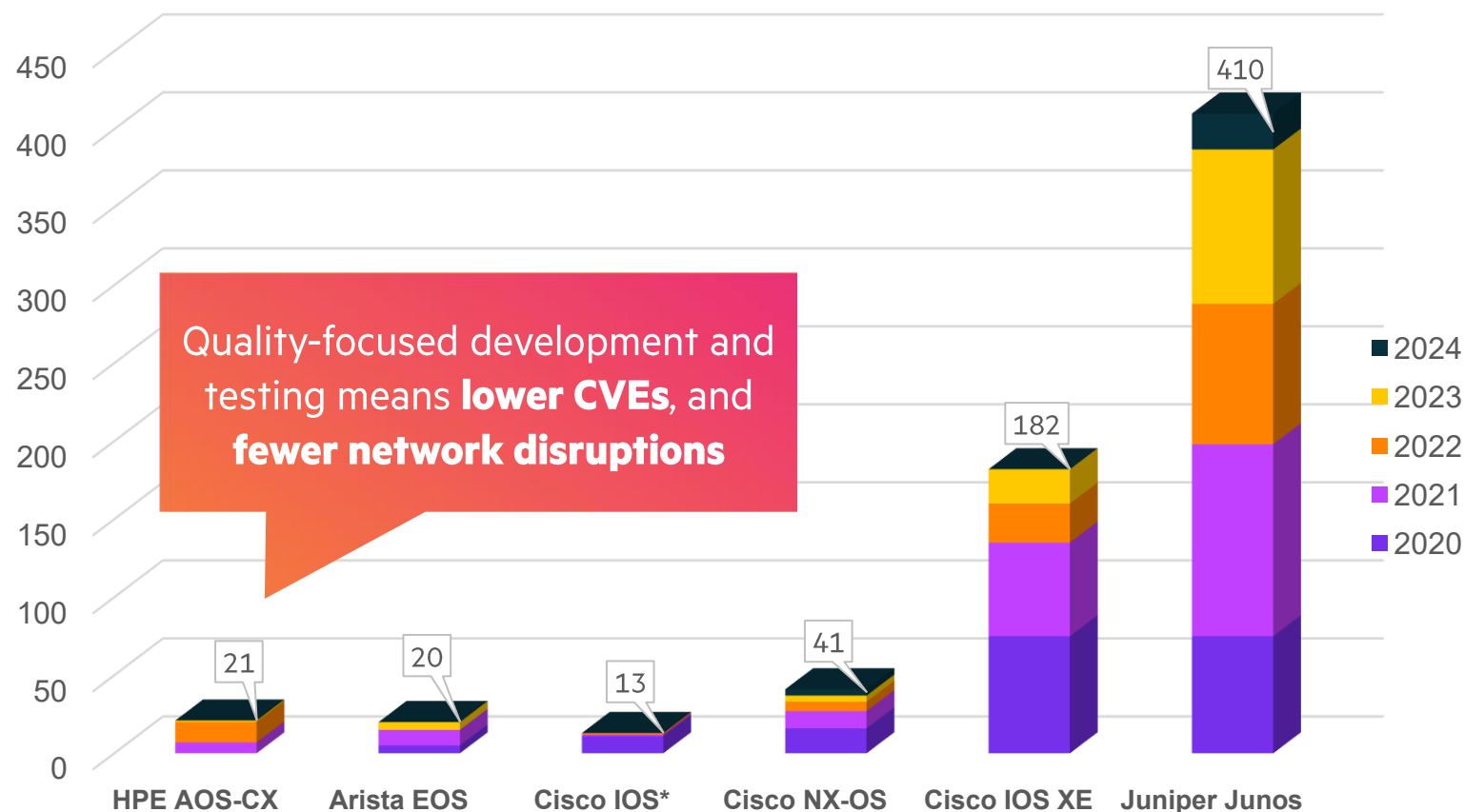
~50,000 test cases per day

~85% test case automation

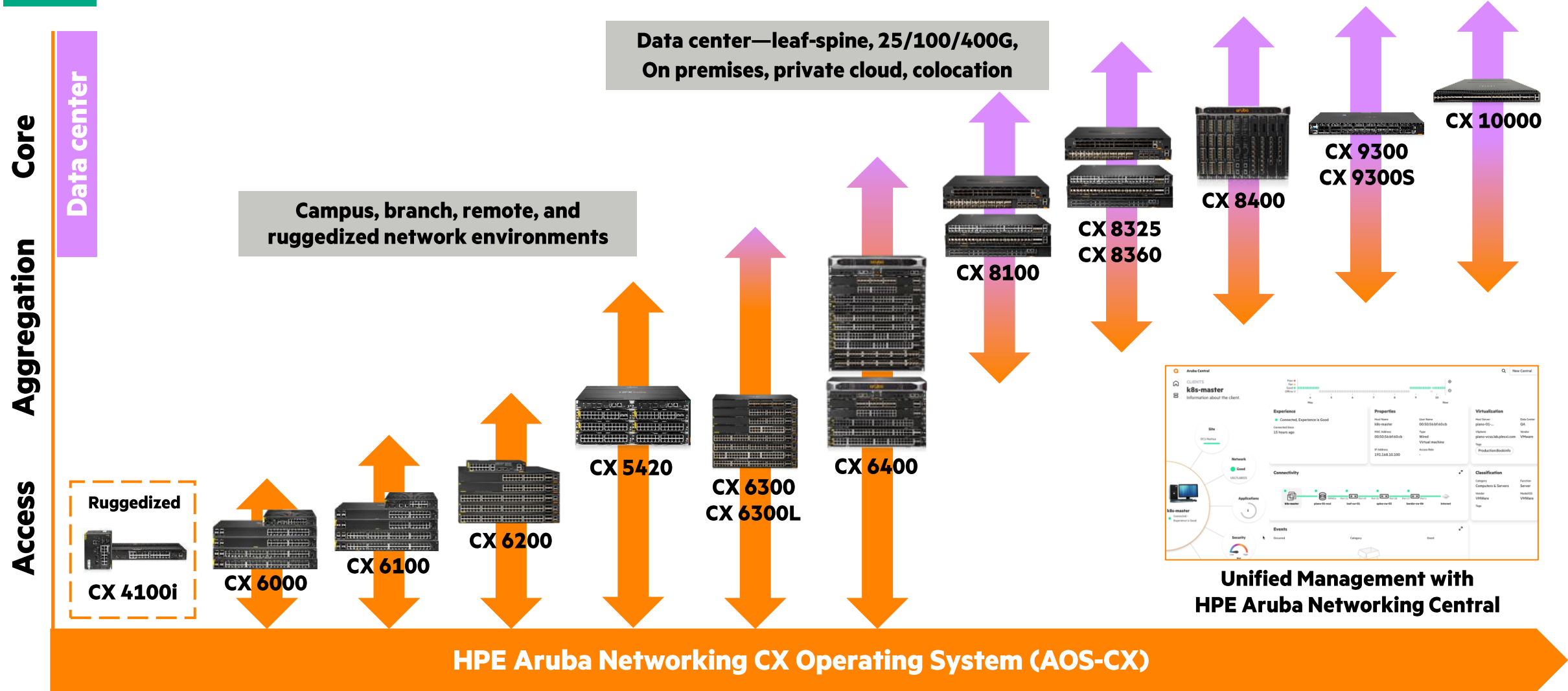
10,000+ automated test cases for
over 40 protocols

Common Vulnerability Enumeration (CVE)

Sponsored by DHS and CISA



HPE Aruba Networking CX switch portfolio



One common operating model from campus edge to DC

CX 6000 and CX 6100 Switch Series

Smart, secure, Layer 2 access switches for branch offices, mid-market enterprises, and SMB networks

CX 6000 for **L2 Gigabit** connectivity

CX 6100 adds or L2 Gigabit connectivity and **10G uplinks**

ACLs, robust QoS, static routing, and IPv6

Power IoT & Wi-Fi devices with support for up to **740W Class 4 PoE**

Simple to deploy 12, 24 and 48 port models

Fanless 12 port model with built-in power supply

CX portfolio benefits

Advanced configs and programmability with AOS-CX

Single pane of glass Central across wired, wireless, WAN

Flexible management via Web GUI, CLI, NetEdit, Central

Limited lifetime warranty and options for adv.TAC support

6
Fixed switches

2 to 4
Built-in 1GbE or 1/10GbE uplinks

740W
30W PoE



CX 6100 Switch Series

Smart, secure, entry level layer 2 gigabit access switches with **10 gig uplinks**

Delivers L2 Gigabit connectivity with
fast 10GbE uplinks

ACLs, robust QoS, static routing, and IPv6

Power IoT & Wi-Fi devices with support for up to
740W Class 4 PoE

Simple to deploy 12, 24 and 48 port models

Fanless 12 port model with built-in power supply

CX portfolio benefits

Advanced
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Single pane of
glass Central
across wired,
wireless, WAN

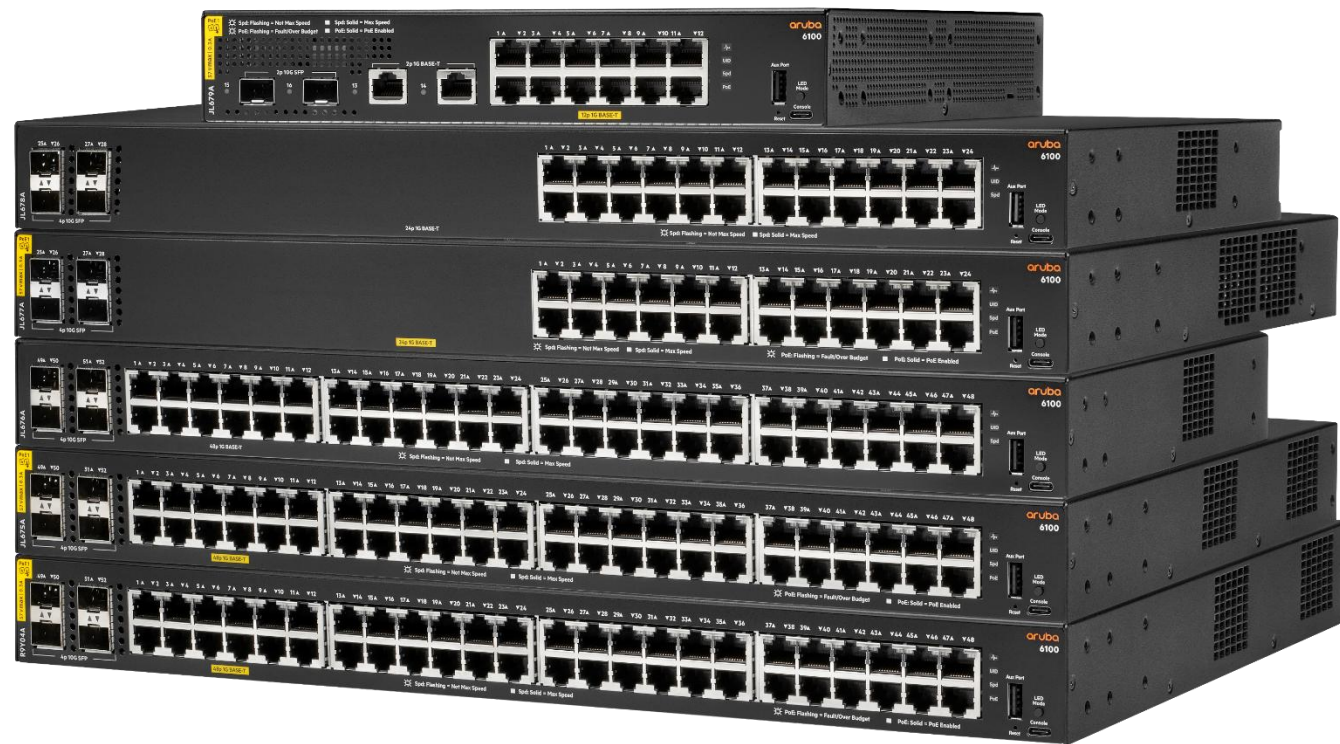
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2 to 4
Built-in 1/10GbE
uplinks

740W
30W PoE



HPE Aruba Networking CX Switches – End-to-end portfolio

Always-on Reliability

- HA with ISSU, HW, SW resiliency
- Proactive analytics

Switching Solution for Any Network

Strong, scalable switching portfolio for branch, campus, mid and large enterprises, SMB, and data center networks

Secure Access

- Dynamic Segmentation
- Campus- branch fabric EVPN/VXLAN
- NetConductor

Simplified Operations

- Single OS from access to core to DC
- Smart automation for flawless configs with NetEdit
- Built-in analytics with deep visibility and clear alerts
- Management flexibility (Cloud, on-prem, Web GUI, CLI, NetEdit, AFC)
- Unified cloud-based single pane of glass management for wired, WLAN, SD-WAN

HPE Aruba Networking CX Switches



Technology Innovation

- AOS-CX: Programmable, modern, designed to scale access to DC
- NAE: built-in analytics monitor, baseline, troubleshoot
- NetEdit: error-free automated rollouts
- VSX live upgrades, VSF rolling stack, & Hot Patching: HA during upgrades
- In-house ASICs: high-performance, programmable agility
- Central: Unified, AIOps, NetConductor

Faster Troubleshooting

- NAE built-in for monitoring, alerts, troubleshooting and root cause analysis
- Central integration with AI insights

Scalable, High-Performance Enterprise Switching

High-density fixed and modular switches with non-blocking architecture, speeds from 1G to 400G, extended table sizes, high power PoE, multi-gig, 80 Plus power supplies, fiber and copper transceivers and DACs

Customer First, Customer Last

- Industry leading warranty
- Global support and services
- Consumption choice: GL4A, NaaS
- Financing: HPEFS

ArubaOS-CX switch CLI contexts

Operator (>)

- Enables execution of commands to view—but not change—the configuration
- Switch prompt:
 - switch>

Manager (#)

- From manager context (#), execute commands that do not require saving changes to configuration
- Switch prompt:
 - switch#

Global configuration (config)#

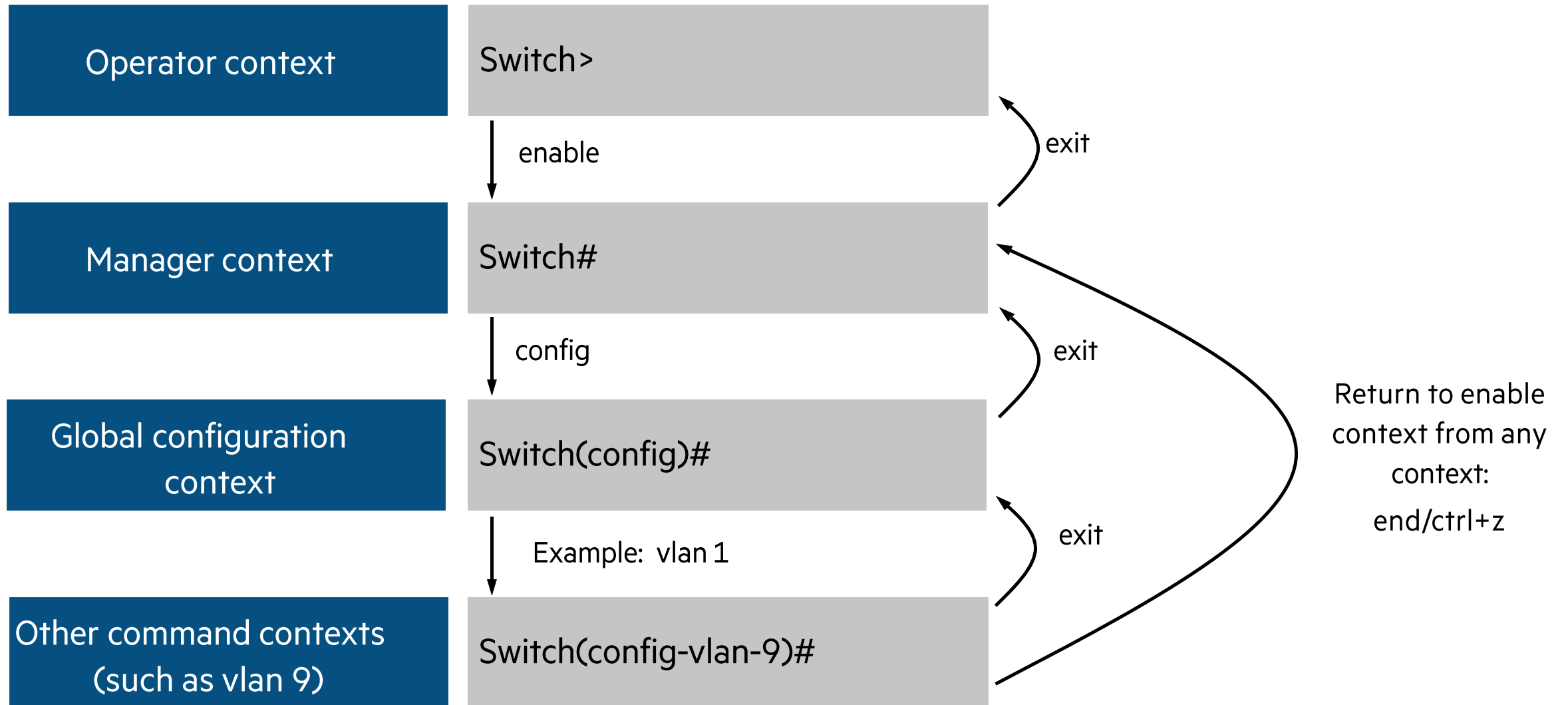
- From global configuration context (config), execute commands that change configuration of switch
- Switch prompt:
 - switch(config)#

Other configuration

- switch(config-if)#
- switch(config-router)#
- switch(config-vlan-100)#

Note: If you log into a device with the admin role you are immediately put into enable context. Type 'exit' at enable to disconnect (not be put into operator mode).

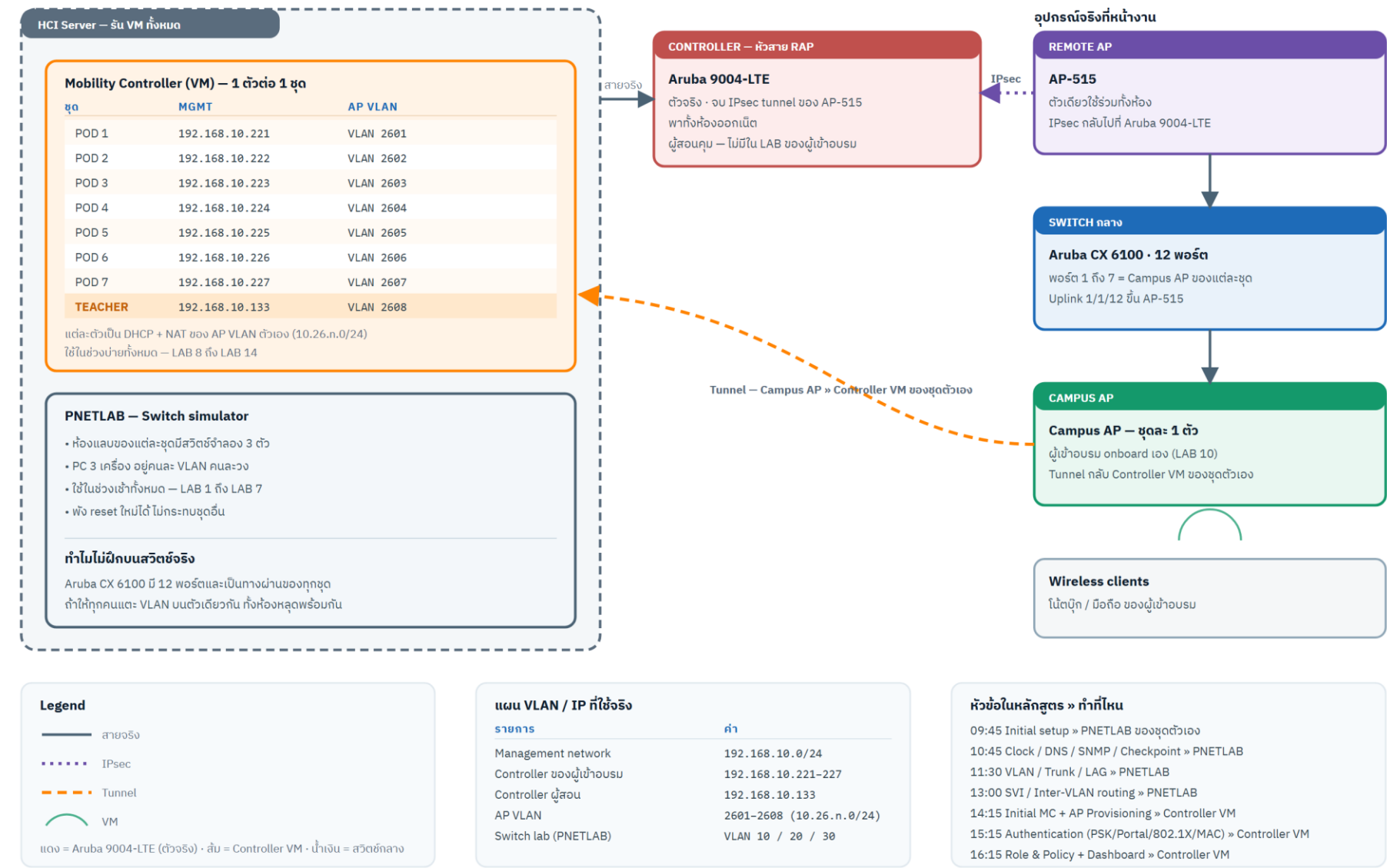
ArubaOS-CX switch CLI contexts



Switch Labs

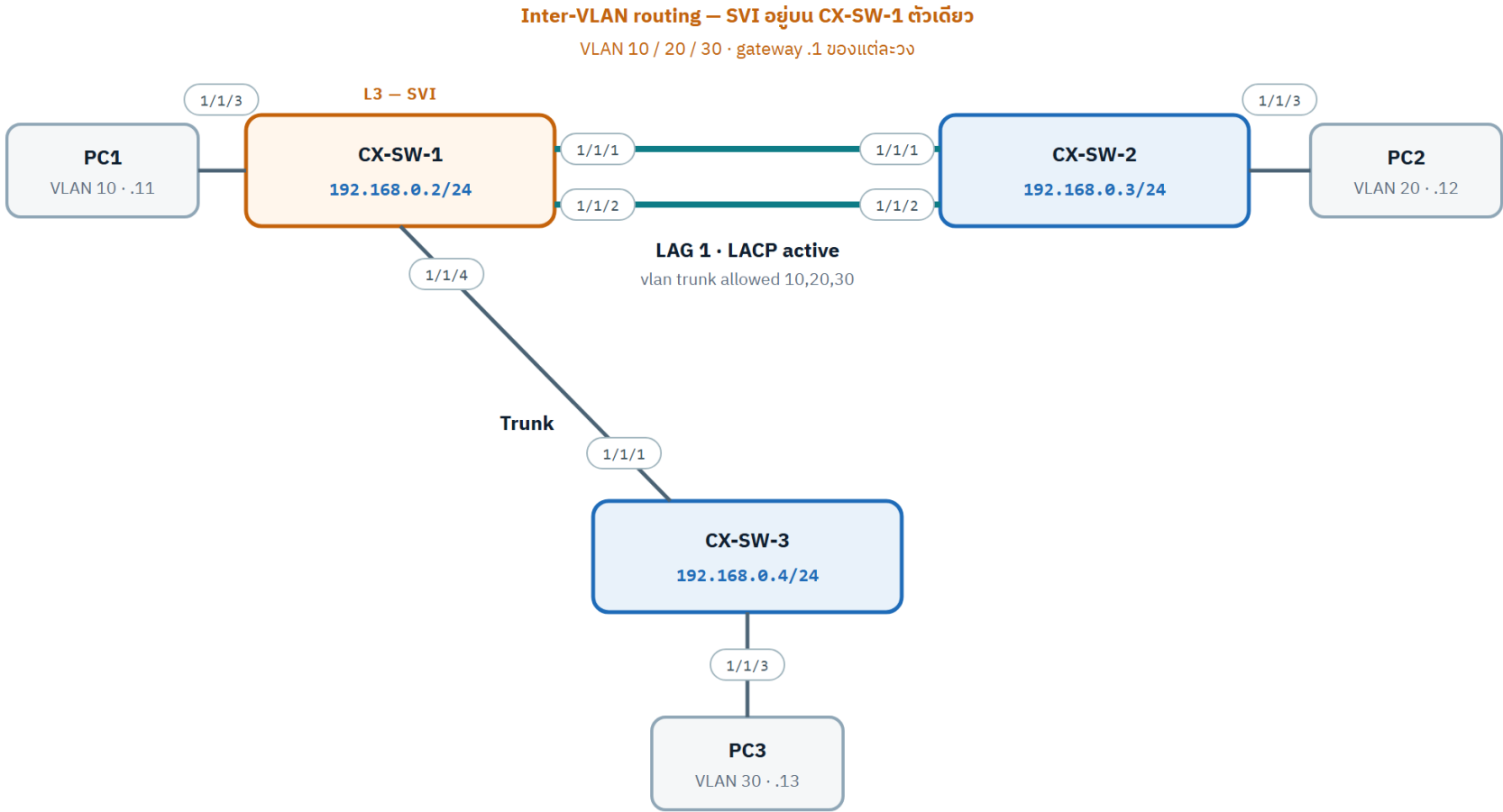
Aruba CX Switch + WLAN Basic – Lab Topology

IT Green — Authorized Aruba (HPE) Networking Training · 3 ก.ย. 2569 · ครึ่งเช้าอยู่ใน PNETLAB · ครึ่งบ่ายอยู่บน Controller VM ของชุด



Switch Lab uu PNETLAB

IT Green — Authorized Aruba (HPE) Networking Training · 3 ก.ย. 2569 · VLAN + Trunk + LAG · PC ควบ VLAN เพื่อทดสอบ inter-VLAN routing



เข้าใช้งาน PNETLAB

URL <http://10.69.11.10/>
ผู้ใช้ pod1 - pod7
รหัสผ่าน P@ssw0rd
ทุกคนในชุดใช้สิทธิ์กันเดียวกัน — ตกลงกันก่อนว่าใครกดอะไร

ค่าที่ใช้ในผังนี้

รายการ	ค่า
CX-SW-1 (L3)	192.168.0.2 · ถือ SVI ทั้งสามวง
CX-SW-2 · CX-SW-3	192.168.0.3 · 192.168.0.4 (L2)
PC1 · VLAN 10	10.10.10.11 » gw 10.10.10.1
PC2 · VLAN 20	10.10.20.12 » gw 10.10.20.1
PC3 · VLAN 30	10.10.30.13 » gw 10.10.30.1
LAG 1 (SW1-SW2)	1/1/1 + 1/1/2 · LACP active
Trunk (SW1-SW3)	SW1 1/1/4 » SW3 1/1/1

อ่านผังนี้อย่างไร

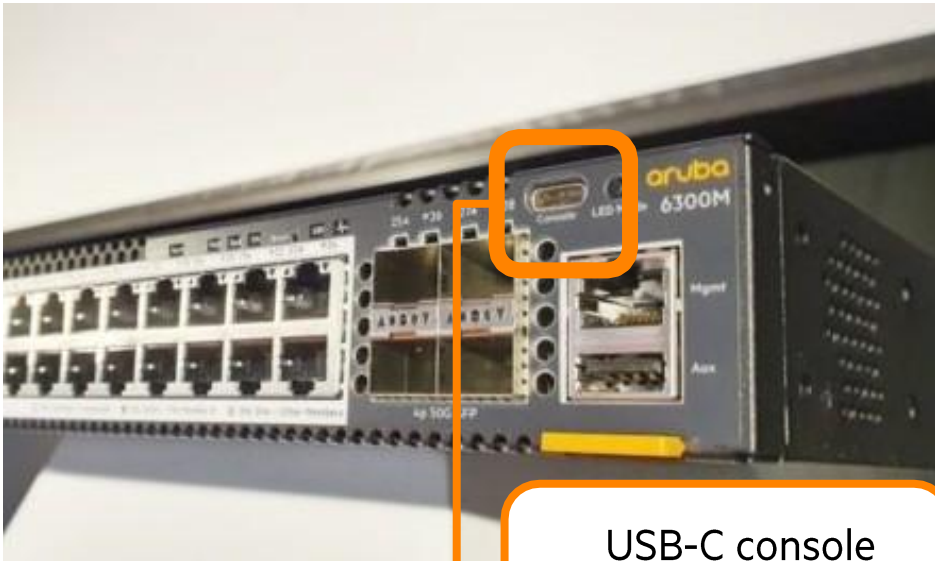
- สวิตช์สามตัวเป็น VM คนละชุดกับเพื่อน — พังได้ reset ใหม่ได้
- SVI อยู่บน CX-SW-1 ตัวเดียว อีกสองตัวเป็น L2 ส่วน ๆ
- LAG กับ Trunk ต่างกันแค่จำนวนสาย — allowed VLAN ต้องตรงกันสองฝั่ง
- หลักสูตรนี้ยังไม่แตะ VSX / Active Gateway — อยู่ในหลักสูตร Advance

LAG 1 — สองสายรวมเป็นเส้นเดียว

Trunk / Access — สายเดี่ยว

กล่องส้ม = สวิตช์ที่ทำ routing · กล่องฟ้า = L2 ส่วน

Connecting to Console Port



USB-C console
No RJ45 console
(CX6300)



CX Switch

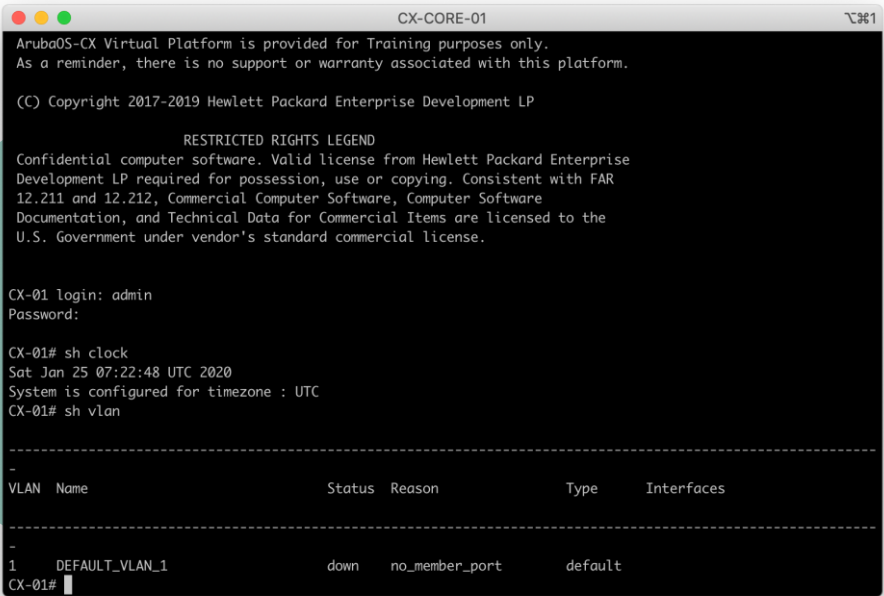
Direct serial connection to console port



CX Switch	Serial Console Type
8400, 8325-48Y8C	RJ45 and Micro USB-B
8320, 8325-32C	RJ45
6400	RJ45 and USB-C
6300	USB-C
6200 (12 port)	USB-C, RJ45

Terminal software settings

- Speed: 115200 bps
- Data bits: 8
- Stop bits: 1
- Parity: None
- Flow control: None



User Accounts

- admin is the default user with **no password**
- Administrators
 - Users with administrator rights can execute any command.
 - Default setting will have admin user account without password, and you are required to enter a **new password after login**
- Operators
 - Users with operator rights can execute commands in the operator context (>) only
- Auditors
 - Users with auditor rights can execute commands in the auditor context (**auditor>**) only
 - Limited view of show command

Lab 1.1 – Initialize Admin Account and Change Hostname

```
switch login: admin  
Password:
```

```
Please configure the 'admin' user account  
password. Enter new Password: ****  
Confirm new password: ****
```

1. Default setting will have admin user account without password, and you are required to enter a new password after login
2. Enter any password string you want, we will change to the new one in the next lab

```
Switch# config terminal  
Switch(config)# hostname CX-CORE-01  
CX-CORE-01(config)#
```

1. Change switch name to CX-CORE-01

Lab 1.2 - Creating additional user accounts

```
switch# configure terminal
switch(config)# user admin
password Enter password: *****
Confirm password: *****
```

1. Reset the password for admin account to P@ssw0rd

```
switch(config)# user operator group operators password
Enter password: ****
Confirm password: ****
```

```
switch(config)# user auditor group auditors password
Enter password: *****
Confirm password: *****
```

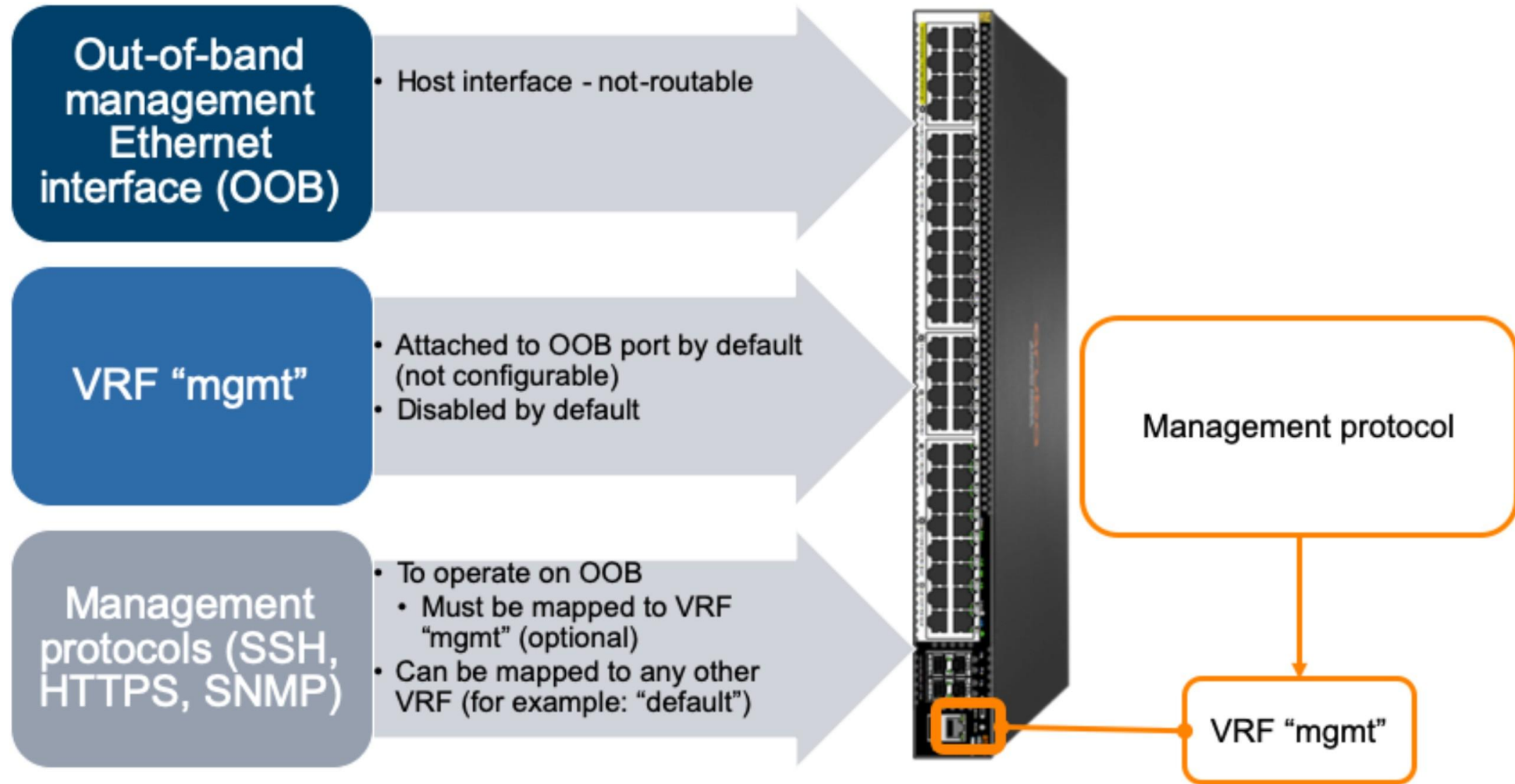
1. Create a user operator and assign to group of “operators” with password oper@tor123
2. Also create another user auditor and assign to group “auditors” with password @uditor123

```
Switch# show user information
```

```
Username          :admin
Authentication    type   : Local
-----
User group        : administrator
-----
User privilege    level : 15
```

1. After you create the user, use the exit command to logout then login as new user (operator and auditor), use the ? Command to see what are the command available to these different user

Out of band Management



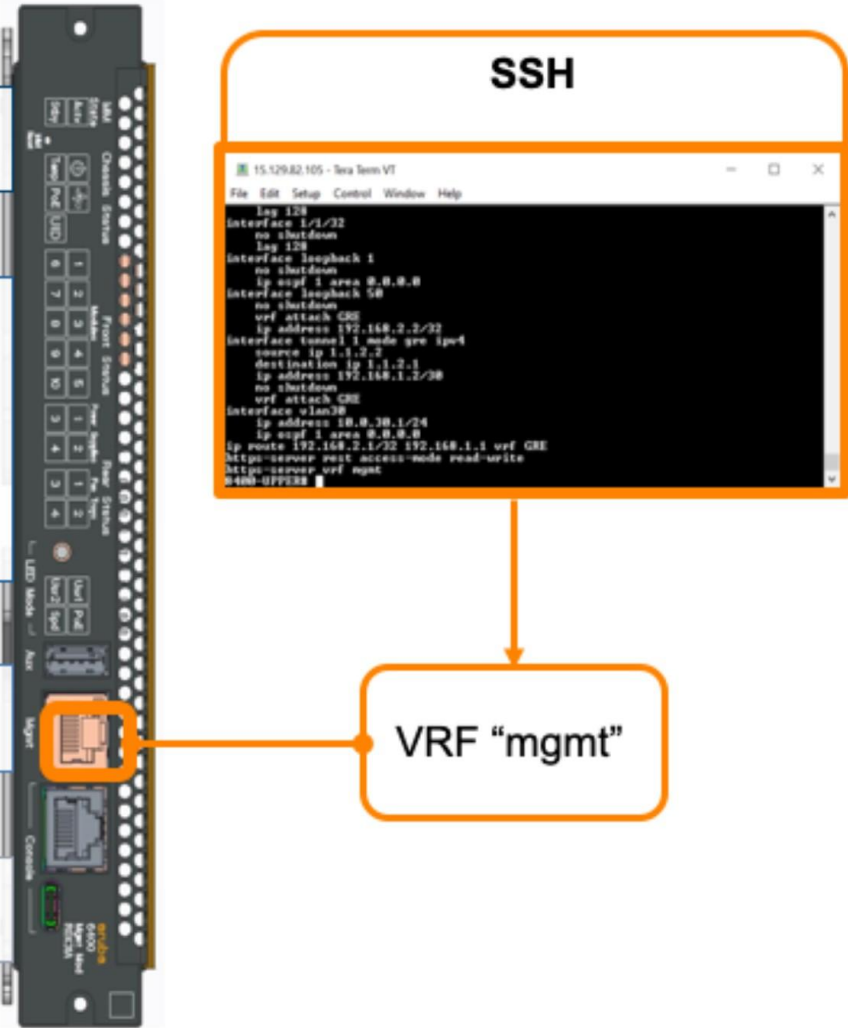
have a password

Must be attached to a VNI

- “mgmt” for OOB management
- “default”
- Any other

for multiple VRFs

supported



HTTPS Server - WebUI

Requires the admin user to have a password

Must be attached to a VRF

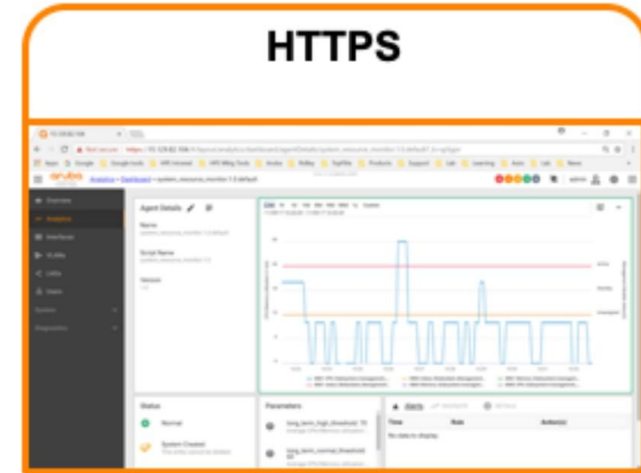
- “mgmt” for OOB management
- “default”
- Any other

Can be assigned simultaneously to multiple VRFs

API is accessible in read-only or read-write

Web-UI is disabled by default

Note: HTTP server is not supported



VRF “mgmt”

REST Interface

Requires admin user to have a password

REST reached via HTTPS

Access can be read-write or read-only (default read-only)



REST

ArubaOS-CX REST API

RESTful interface for ArubaOS-CX switch software

AAA_Server_Group	Show/Hide	Lock Operations	Expand Operations
AAA_Server_Group_Profile	Show/Hide	Lock Operations	Expand Operations
ACL	Show/Hide	Lock Operations	Expand Operations
ACL_Entry	Show/Hide	Lock Operations	Expand Operations
Aggregate_address	Show/Hide	Lock Operations	Expand Operations
SGP_Path_Filter	Show/Hide	Lock Operations	Expand Operations
SGP_Path_Filter_Entry	Show/Hide	Lock Operations	Expand Operations
SGP_Community_Filter	Show/Hide	Lock Operations	Expand Operations
SGP_Community_Filter_Entry	Show/Hide	Lock Operations	Expand Operations
SGP_Neighbor	Show/Hide	Lock Operations	Expand Operations
SGP_Route	Show/Hide	Lock Operations	Expand Operations
SGP_Router	Show/Hide	Lock Operations	Expand Operations

VRF "mgmt"

Lab 1.3 - Enable Remote Management Access

```
CX-CORE-01# configure terminal
CX-CORE-01(config)# https-server vrf mgmt
CX-CORE-01(config)# ssh server vrf mgmt
CX-CORE-01(config)# https-server rest access-mode read-only
CX-CORE-01(config)# interface mgmt
CX-CORE-01(config-if-mgmt)# ip static 10.1.10.101-103/24
CX-CORE-01(config-if-mgmt)# default-gateway 10.1.10.1
CX-CORE-01(config-if-mgmt)# nameserver 8.8.8.8
CX-CORE-01(config-if-mgmt)# no shutdown
```

- 1. Bind the https and ssh server with VRF “mgmt.”
- 2. Assign IP address, Gateway and DNS to the management interface

- 1. Open the browser and see if you can access CX switches
- 2. SSH CX switches and make sure you have the access
- 3. Ping to the switch IP address on mgmt VRF

MGMT IP Address

- Group1 - CX-CORE = 10.1.10.101-103/24
- Group2 - CX-CORE = 10.1.10.104-106/24
- Group3 - CX-CORE = 10.1.10.107-109/24
- Group4 - CX-CORE = 10.1.10.110-112/24

ping 10.1.10.101
PING 10.1.10.101 (10.1.10.101) 100(128) bytes of data.

108	bytes	from	10.1.10.101:	icmp_seq=1	ttl=128	time=1.04 ms
108	bytes	from	10.1.10.101:	icmp_seq=2	ttl=128	time=0.454 ms
108	bytes	from	10.1.10.101:	icmp_seq=3	ttl=128	time=0.763 ms
108	bytes	from	10.1.10.101:	icmp_seq=4	ttl=128	time=0.681 ms
108	bytes	from	10.1.10.101:	icmp_seq=5	ttl=128	time=0.604 ms

Lab 1.3 - Enable Remote Management Access

Verify connectivity to all three CX switches

```
$ ssh -l admin 10.1.10.101
```

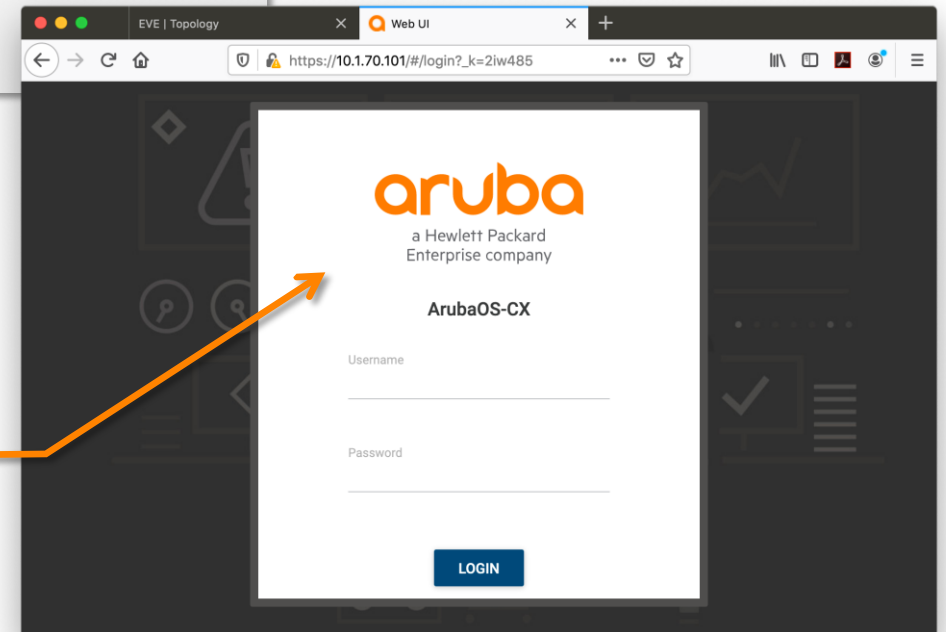
```
The authenticity of host '10.1.X0.101 (10.1.X0.101)' can't be established.  
ECDSA key fingerprint is SHA256:4AeXnXNh0mHGnNHWRo1kyVDm9nyhdrTOrfcpJxlDUbQ.  
Are you sure you want to continue connecting (yes/no)? yes  
Warning: Permanently added '10.1.X0.101' (ECDSA) to the list of known hosts.
```

```
admin@10.1.X0.101's password:
```

```
Last login: 2020-01-28 06:12:19 from the console  
User "admin" has logged in 2 times in the past 30 days  
CX-CORE-01#
```

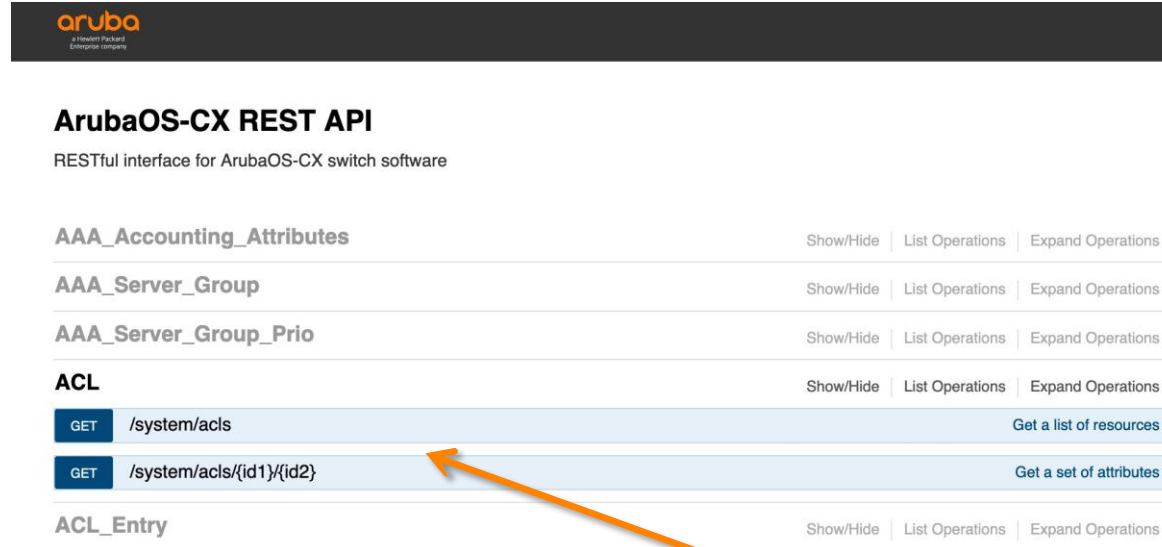
1. Test SSH to switch with your ssh client (eg. **Putty**) from your PC

2. Test Web UI to switch with your browser (<https://10.1.10.101>) from your PC



Lab 1.3 - Enable Remote Management Access (cont.)

Verify connectivity to all three CX switches



ArubaOS-CX REST API		
RESTful interface for ArubaOS-CX switch software		
AAA_Accounting_Attributes	Show/Hide	List Operations Expand Operations
AAA_Server_Group	Show/Hide	List Operations Expand Operations
AAA_Server_Group_Prio	Show/Hide	List Operations Expand Operations
ACL	Show/Hide	List Operations Expand Operations
GET /system/acls		Get a list of resources
GET /system/acls/{id1}/{id2}		Get a set of attributes
ACL_Entry	Show/Hide	List Operations Expand Operations

1. Next, we will verify REST API connectivity to all three CX switches
2. Point your web browser to <https://10.1.X0.10Y/api>
3. Select APIs shown in the screen to see what REST method we can use to contact to switch
4. You will notice that there is only GET method while actually CX support others as well such as POST and DELETE.
5. What's the problem?

LAB 1 เข้า PNETLAB และตั้งค่าเริ่มต้นให้สวิตช์

PNETLAB (สวิตช์จำลอง) เข้าห้องแล็บของชุดตัวเองได้ เห็นสวิตช์ครบสามตัว ตั้งรหัส admin และ hostname ให้ครบทุกตัว

1. เปิด <http://10.69.11.10/> แล้วล็อกอินด้วยผู้ใช้ของชุดตัวเอง — POD 1 ใช้ pod1 รหัส P@ssw0rd
2. เปิด lab ของชุดตัวเอง กด START แล้วรอให้สวิตช์ทั้งสามตัวขึ้นครบ
3. คลิกที่สวิตช์ตัวแรกเพื่อเปิด console — ครั้งแรกล็อกอินด้วย admin แล้วเคาะ Enter ผ่านช่องรหัส จากนั้นระบบจะบังคับให้ตั้งรหัสใหม่ ตั้งเป็น Aruba123! ให้เหมือนกันทั้งชุดจะได้ไม่ลืม
4. ตั้ง hostname ให้ตรงกับผัง FIG.03

```
configure terminal
hostname CX-SW-1
exit
write memory
```
5. ทำซ้ำข้อ 3–4 กับอีกสองตัว ตั้งชื่อเป็น CX-SW-2 และ CX-SW-3
6. จดไว้ใช้ทั้งวัน: สวิตช์ 192.168.0.2 / .3 / .4 · VLAN 10 / 20 / 30 · Controller ของชุดคือ 192.168.10.22n

ตรวจสอบ

```
show version
show running-config | include hostname
show interface brief
```

ระวัง

ทุกคนในชุดใช้ล็อกอิน PNETLAB เดียวกัน — ถ้าสองคนกดพร้อมกันจะแย่ง console ตกลงกันก่อนว่าใครทำสวิตช์ตัวไหน

LAB 2 สร้าง User เพิ่ม และเปิด Remote Management (1/2)

PNETLAB (สวิตช์จำลอง) เข้าสวิตช์ได้โดยไม่ต้องผ่าน console อีก — SSH และ Web UI ใช้ได้จริงจาก PC ในแล็บ

1. ที่ CX-SW-1 สร้างผู้ใช้เพิ่มอีกหนึ่งคน แยกจาก admin
configure terminal
user netadmin group administrators password plaintext Aruba123!
2. เปิด SSH server — เปิดทั้ง VRF default (ฝั่งข้อมูล) และ mgmt (ฝั่งพอร์ตจัดการ)
ssh server vrf default
ssh server vrf mgmt
3. เปิด Web UI ด้วย HTTPS — หัวข้อ CLI / Web UI ตอนเข้าใช้หน้านี้
https-server vrf default
https-server vrf mgmt

ตรวจสอบผล

```
show ssh server vrf default
show https-server vrf default
show ip interface brief
```

ระวัง

SSH เข้าไม่ได้ให้ดูก่อนว่าเปิดถูก VRF หรือเปล่า — เปิดแต่ vrf mgmt แล้วนั่ง ssh มาทาง VLAN 1 ยังไงก็ไม่ติด

LAB 2 สร้าง User เพิ่ม และเปิด Remote Management (2/2)

PNETLAB (สวิตช์จำลอง) เข้าสวิตช์ได้โดยไม่ต้องผ่าน console อีก — SSH และ Web UI ใช้ได้จริงจาก PC ในแล็บ

4. ให้ SVI ของ VLAN 1 มี IP ไว้ก่อน จะได้ ping และ SSH เข้ามาได้

```
vlan 1
interface vlan 1
    ip address 192.168.0.2/24
exit
write memory
```

5. จาก PC ในแล็บลอง ssh netadmin@192.168.0.2 และเปิด https://192.168.0.2 ต้องเข้าได้ทั้งคู่

6. ทำซ้ำกับ CX-SW-2 (192.168.0.3) และ CX-SW-3 (192.168.0.4)

ตรวจสอบผล

```
show ssh server vrf default
show https-server vrf default
show ip interface brief
```

ระวัง

SSH เข้าไม่ได้ให้ดูก่อนว่าเปิดถูก VRF หรือเปล่า — เปิดแต่ vrf mgmt แล้วนั่ง ssh มาทาง VLAN 1 ยังไงก็ไม่ติด

Lab 1.4 – Configure Clock, Timezone, DNS and SNMP

```
CX-CORE-01 # configure terminal
CX-CORE-01(config)# clock timezone Asia/Bangkok
CX-CORE-01(config)# ip dns server-address 8.8.8.8 vrf mgmt
CX-CORE-01(config)# ntp server 0.th.pool.ntp.org
CX-CORE-01(config)# ntp vrf mgmt
CX-CORE-01(config)# ntp enable
CX-CORE-01(config)# snmp-server system-description CX-CORE-01
CX-CORE-01(config)# snmp-server system-contact Administrator
CX-CORE-01(config)# snmp-server system-location CX_LAB
CX-CORE-01(config)# snmp-server community cx_snmp
CX-CORE-01(config)# snmp-server vrf mgmt
CX-CORE-01(config)# snmp-server host x.x.x.x trap version v2c community cx_snmp
CX-CORE-01(config)# logging x.x.x.x vrf mgmt
```

- Some of the basic configuration like DNS and Logging supports VRF binding

- SNMP trap and log server are dummy IP address. It is intended for student to practice how to config these two settings.

```
CX-CORE-01# sh ntp associations
```

ID	NAME	REMOTE	REF-ID	ST	LAST	POLL	REACH
* 1	0.th.pool.ntp.org	202.12.97.45	.PPS.	1	7	64	377

```
CX-CORE-01# show clock
Tue Jan 28 13:50:39 ICT 2020
System is configured for timezone : Asia/Bangkok
```

- Verify date, time and timezone

Configuration Management

- “**write memory**” to save running configuration into the Startup configuration
- **Types of checkpoints**
 - **System generated checkpoints:** Checkpoints that the system automatically generates whenever a configuration change is detected.
 - **User generated checkpoints:** Checkpoints that the user manually generates
- Total limit of checkpoint is **64** including the startup configuration; **32** each type
- After the limit the system will overwrite the oldest checkpoint
- For each additional configuration change, the **300** second timeout counter is reset. Once the timeout limit is reached with no additional configuration changes detected, the system will automatically generate a new checkpoint
- This system generated checkpoints are listed with a time stamp format CPC<**YYYYMMDDHHMMSS**>
- The checkpoints can be applied using the “**checkpoint rollback**” or “**copy checkpoint**” command
- **Difference** between two configuration can be viewed between two different checkpoints or between any checkpoint and the startup/running configuration

Lab 1.5 – Creating Configuration Checkpoint

```
CX-CORE-01 # copy running-config checkpoint initial_config  
Large configuration changes will take time to process, please be patient.
```

```
CX-CORE-01# show checkpoint  
CPC20200128072408  
CPC20200128075706  
initial_config
```

- Use this command to show all saved configuration checkpoint

```
CX-CORE-01# show checkpoint initial_config  
.  
.  
.
```

- Use this command to show initial_config checkpoint

Lab 1.6 – Checkpoint Difference

```
CX-CORE-01# config t
CX-CORE-01(config)# session-timeout 0
CX-CORE-01(config)# banner motd ^
CX-CORE-01(config-banner-motd)# #####
CX-CORE-01(config-banner-motd)# TEST BANNER
CX-CORE-01(config-banner-motd)# #####
CX-CORE-01(config-banner-motd)# ^
CX-CORE-01(config)# no snmp-server community cx_snmp
```

- 
- Perform some configuration changes for testing

Lab 1.6 – Checkpoint Difference (cont.)

```
CX--CORE-01# checkpoint diff initial_config running-config
--- /tmp/initial_config1580201859074 2020-01-28 15:57:39.073618177 +0700
+++ /tmp/running_config1580201866934 2020-01-28 15:57:46.934719256 +0700
@@ -2,6 +2,11 @@
 !Version ArubaOS-CX Virtual.10.04.0001
 !export-password: default
 hostname CX-CORE-01
+banner motd !
+#####
+TEST BANNER
+#####
+!
 led locator on
 clock timezone asia/bangkok
@@ -9,6 +14,8 @@
 ntp server pool.ntp.org prefer
 ntp enable
 ntp vrf mgmt
+cli-session
+  timeout 0
 !
@@ -18,7 +25,6 @@
 snmp-server system-description CX-CORE-01
 snmp-server system-location CX_LAB
 snmp-server system-contact Administrator
-snmp-server community cx_snmp
 ssh server vrf mgmt
 !
```

- 
- Configuration changes between checkpoint “initial_config” and the recent running-config

Lab 1.7 – Checkpoint Rollback

Use the **same CX switch in lab 1.6** to do this lab

```
CX-CORE-01# sh session-timeout
session-timeout: 0 minute
```

```
CX-CORE-01# sh banner motd
#####
TEST BANNER
#####
```

```
CX-CORE-01# sh snmp community
-----
SNMP communities
-----
public
```

1. Record the show output result and verify the difference after perform a configuration rollback

2. After that, issue a rollback command. You have two way to rollback your configuration to any checkpoint
3. Verify the show out result after rollback your configuration. What is the difference?

```
CX-CORE-01# copy checkpoint initial_config running-config
Configuration changes will take time to process, please be patient.
```

Option 1

```
CX-CORE-01# checkpoint rollback initial_config
Configuration changes will take time to process, please be patient.
```

Option 2

Checkpoint Auto Mode

Checkpoint auto mode lets you configure the switch with failover support. Checkpoint auto mode helps to prevent configuration changes that might cause undesired results, such as connectivity loss or system failure.

Procedure

1. Enable the checkpoint auto mode.
2. After entering `checkpoint auto <time>` mode, you will have a set amount of time to add, remove, or modify the switch configuration
3. To save the configuration, enter the `checkpoint auto confirm` command before the specified time set in step 1.

Lab 1.8 - Checkpoint Auto Rollback

1. Set the checkpoint auto timer to 1min (it execute only once)
2. In this lab, we will test by modifying IP management out of mgmt subnet that in the real world would make us lost connectivity to this switch. (You will not kicked out because you are working on the console not remote session)
3. But this feature would revert IP management **back after 1 min.**
4. **Wait for 1 min** and you will notice that configuration will **revert back** and any remote connectivity will be **restored**

```
CX-CORE-01# checkpoint auto 1
CX-CORE-01# config t
CX-CORE-01(config)# int mgmt
CX-CORE-01(config-if-mgmt)# ip static 10.1.222.222/24
. . .

CX-CORE-01# WARNING: Restoring configuration. Do NOT
add any new configuration.
Configuration changes will take time to process,
please be patient.
Restoration successful
```

Ping lost

Ping restored

Perform ping test from our PC to MGMT interface

```
$ ping 10.1.10.101
PING 10.1.10.101 (10.1.10.101): 56 data bytes
64 bytes from 10.1.10.101: icmp_seq=0 ttl=64 time=25.416 ms
64 bytes from 10.1.10.101: icmp_seq=1 ttl=64 time=28.695 ms
64 bytes from 10.1.10.101: icmp_seq=2 ttl=64 time=30.762 ms
64 bytes from 10.1.10.101: icmp_seq=3 ttl=64 time=29.446 ms
64 bytes from 10.1.10.101: icmp_seq=4 ttl=64 time=30.288 ms
64 bytes from 10.1.10.101: icmp_seq=5 ttl=64 time=22.854 ms
Request timeout
Request timeout
Request timeout
Request timeout
Request timeout
Request timeout
64 bytes from 10.1.10.101: icmp_seq=55 ttl=64 time=22.836 ms
64 bytes from 10.1.10.101: icmp_seq=56 ttl=64 time=29.004 ms
64 bytes from 10.1.10.101: icmp_seq=57 ttl=64 time=31.020 ms
64 bytes from 10.1.10.101: icmp_seq=58 ttl=64 time=29.094 ms
64 bytes from 10.1.10.101: icmp_seq=59 ttl=64 time=28.441 ms
```

Lab 1.8 - Checkpoint Auto Confirm (Cont.)

1. Set the checkpoint auto timer for 1min (it execute only once)
2. Disable session timeout on CX switches
3. Run the command “checkpoint auto confirm” to save the configurations before 1 min
4. This will tell the switch to retain configuration and not revert back after 1 min

```
CX-CORE-01# checkpoint auto 1
CX-CORE-01(config)# session-timeout 0
CX-CORE-01(config)# exit
CX-CORE-01# checkpoint auto confirm
Large configuration changes will take time to process, please be patient.
checkpoint AUTO20200214052353 created
```

Don't forgot to run this command

```
CX-CORE-01 # copy running-config checkpoint initial_config
Large configuration changes will take time to process, please be patient.
```



Knowledge Check #1

Which console interfaces and default baud rates are supported on the HPE Aruba CX 6200F 12-port model? (Select two.)

- a. USB-A, baud rate = 115200
- b. USB-C, baud rate = 9600
- c. RJ-45, baud rate = 115200
- d. USB-C, baud rate = 115200
- e. RJ-45, baud rate = 112500

Knowledge Check #2

Does the HPE Aruba 6200F model support only two VRF?

- a. True.
- b. False.

LAB 3 Clock, Timezone, DNS และ SNMP (1/2)

PNETLAB (สวิตช์จำลอง) ให้เวลาบนสวิตช์ตรง และตั้งค่าที่ระบบ monitoring ต้องใช้ — log ที่เวลาผิดคือ log ที่ไล่ปัญหาไม่ได้

1. ตั้ง timezone แล้วค่อยตั้งเวลา — สลับลำดับกันเมื่อไหร่เวลาจะเพี้ยนไปตามโซนเดิม

```
configure terminal
clock timezone asia/bangkok
clock date 2026-09-03
clock time 09:00:00
```

2. ซี่ NTP ไว้ด้วย จะได้ไม่ต้องมาตั้งมือทุกครั้งที reset

```
ntp server pool.ntp.org
ntp enable
```

3. ตั้ง DNS

```
ip dns server-address 8.8.8.8
ip dns server-address 1.1.1.1
ip dns domain-name lab.local
```

ตรวจสอบผล

```
show clock
show ntp status
show ip dns
show snmp-server
```

ระวัง

clock timezone ต้องพิมพ์เป็นตัวเล็กทั้งหมดแบบ
asia/bangkok — พิมพ์ Asia/Bangkok สวิตช์จะไม่รู้จัก

LAB 3 Clock, Timezone, DNS และ SNMP (2/2)

PNETLAB (สวิตช์จำลอง) ให้เวลาบนสวิตช์ตรง และตั้งค่าที่ระบบ monitoring ต้องใช้ — log ที่เวลาผิดคือ log ที่ไล่ปัญหาไม่ได้

4. ตั้ง SNMP สำหรับอ่านค่า

```
snmp-server community public
snmp-server system-location "Training Room"
snmp-server system-contact "IT Green"
exit
write memory
```

5. เช็คว่าเวลาขยับจริง — เคาะ show clock สองครั้งห่างกันสักครู่ ตัวเลขต้องเดิน

ตรวจสอบผล

```
show clock
show ntp status
show ip dns
show snmp-server
```

ระวัง

clock timezone ต้องพิมพ์เป็นตัวเล็กทั้งหมดแบบ
asia/bangkok — พิมพ์ Asia/Bangkok สวิตช์จะไม่รู้จัก

LAB 4 Checkpoint, Diff และ Rollback (1/2)

PNETLAB (สวิตช์จำลอง) เซฟจุดที่ config ยังดีอยู่ แล้วย้อนกลับมาได้เมื่อพังในแลบถัดไป — นี่คือตาข่ายรองของทั้งวัน

1. เซฟ checkpoint ก่อนเริ่มแตะ VLAN ตั้งชื่อให้รู้เรื่อง

```
configure terminal
exit
copy running-config checkpoint BEFORE-VLAN
```

2. ลองทำอะไรสักอย่างให้ต่างจากเดิม

```
configure terminal
vlan 99
    name TEST-DIFF
exit
```

3. เปรียบดูว่าต่างจากจุดที่เซฟไว้ตรงไหน — ต้องเห็นแค่ VLAN 99 ที่เพิ่งเพิ่ม

```
checkpoint diff BEFORE-VLAN running-config
```

4. ย้อนกลับไปจุดเดิม แล้ว diff ซ้ำ คราวนี้ต้องไม่มีอะไรต่าง

```
checkpoint rollback BEFORE-VLAN
checkpoint diff BEFORE-VLAN running-config
```

ตรวจสอบผล

```
show checkpoint
show checkpoint BEFORE-VLAN
show checkpoint auto
```

ระวัง

checkpoint auto ที่ลืม confirm จะย้อน config กลับเจียบ ๆ ตอนหมดเวลา — ถ้าอยู่ ๆ ของที่เพิ่งตั้งหายไป ให้สงสัยข้อนี้ก่อน

LAB 4 Checkpoint, Diff และ Rollback (2/2)

PNETLAB (สวิตช์จำลอง) เซฟจุดที่ config ยังดีอยู่ แล้วย้อนกลับมาได้เมื่อพังในแลบถัดไป — นี่คือตาข่ายรองของทั้งวัน

5. ลองโหมด auto rollback — สั่งแล้วมีเวลานับถอยหลัง ถ้าไม่ confirm มันจะย้อนกลับเอง เหมือนกับตอนแก้ config ผ่าน SSH ที่อาจตัดตัวเองหลุด

```
configure terminal
checkpoint auto 5
exit
```

6. ยืนยันว่ายังคุยกันได้อยู่ ให้ config ที่แก้ยู่ต่อ
checkpoint auto confirm

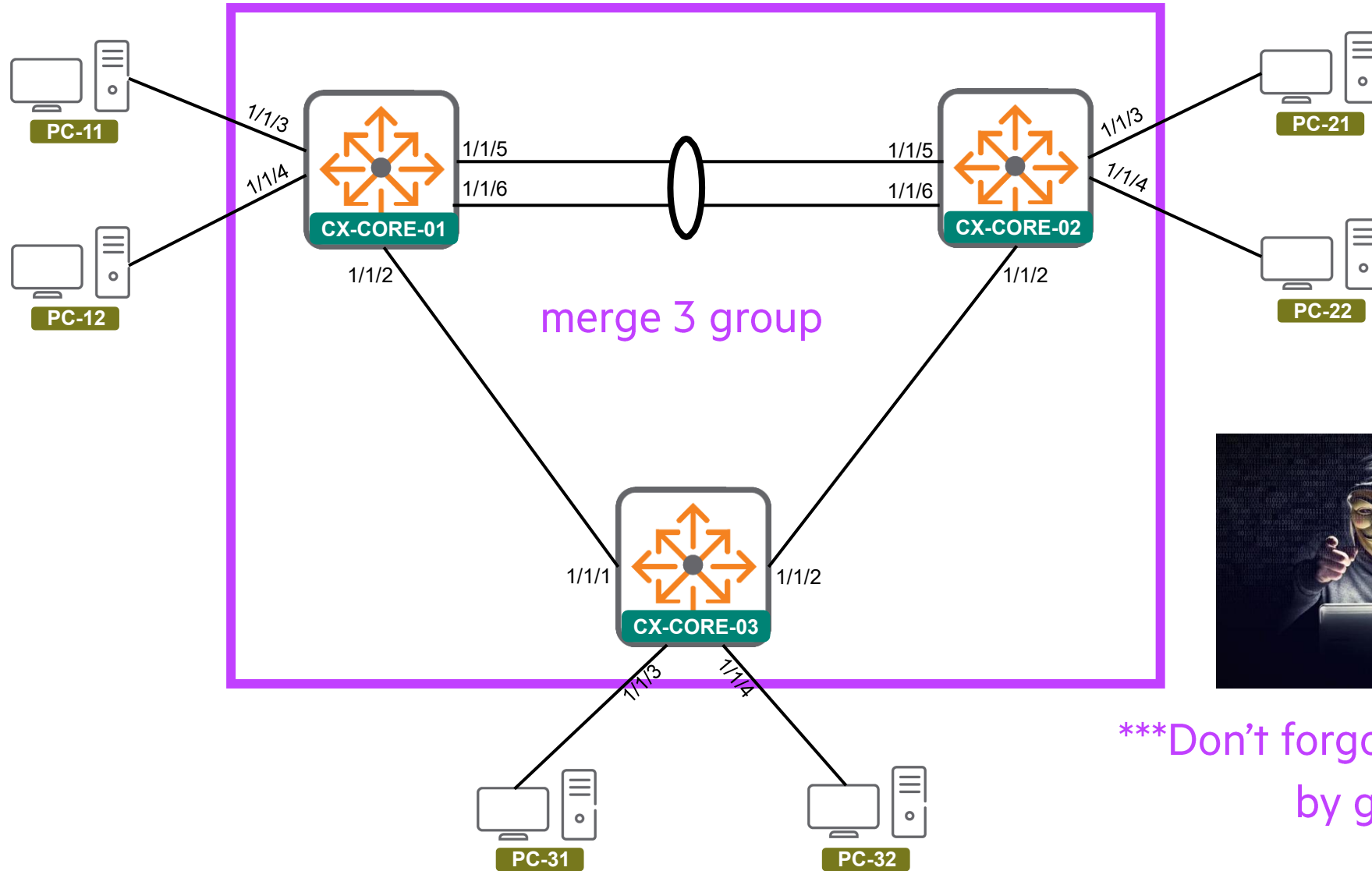
ตรวจสอบผล

```
show checkpoint
show checkpoint BEFORE-VLAN
show checkpoint auto
```

ระวัง

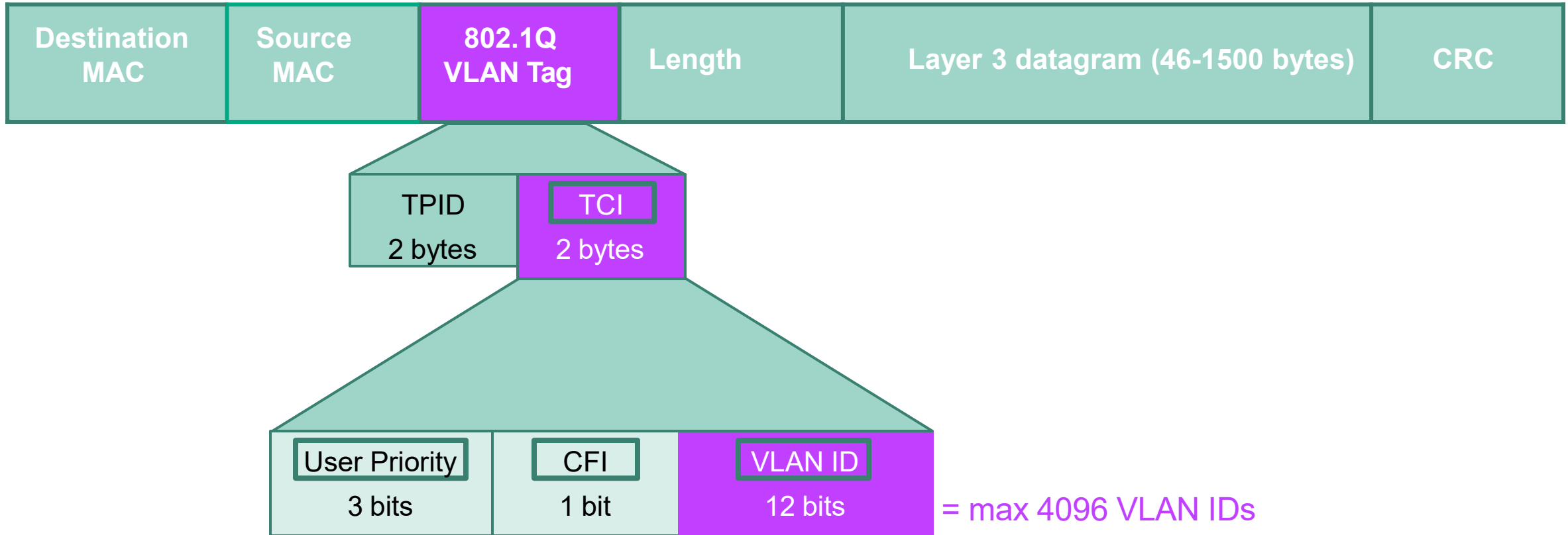
checkpoint auto ที่ลืม confirm จะย้อน config กลับเจียบ ๆ ตอนหมดเวลา — ถ้าอยู่ ๆ ของที่เพิ่งตั้งหายไป ให้สงสัยข้อนี้ก่อน

Lab 2 Topology

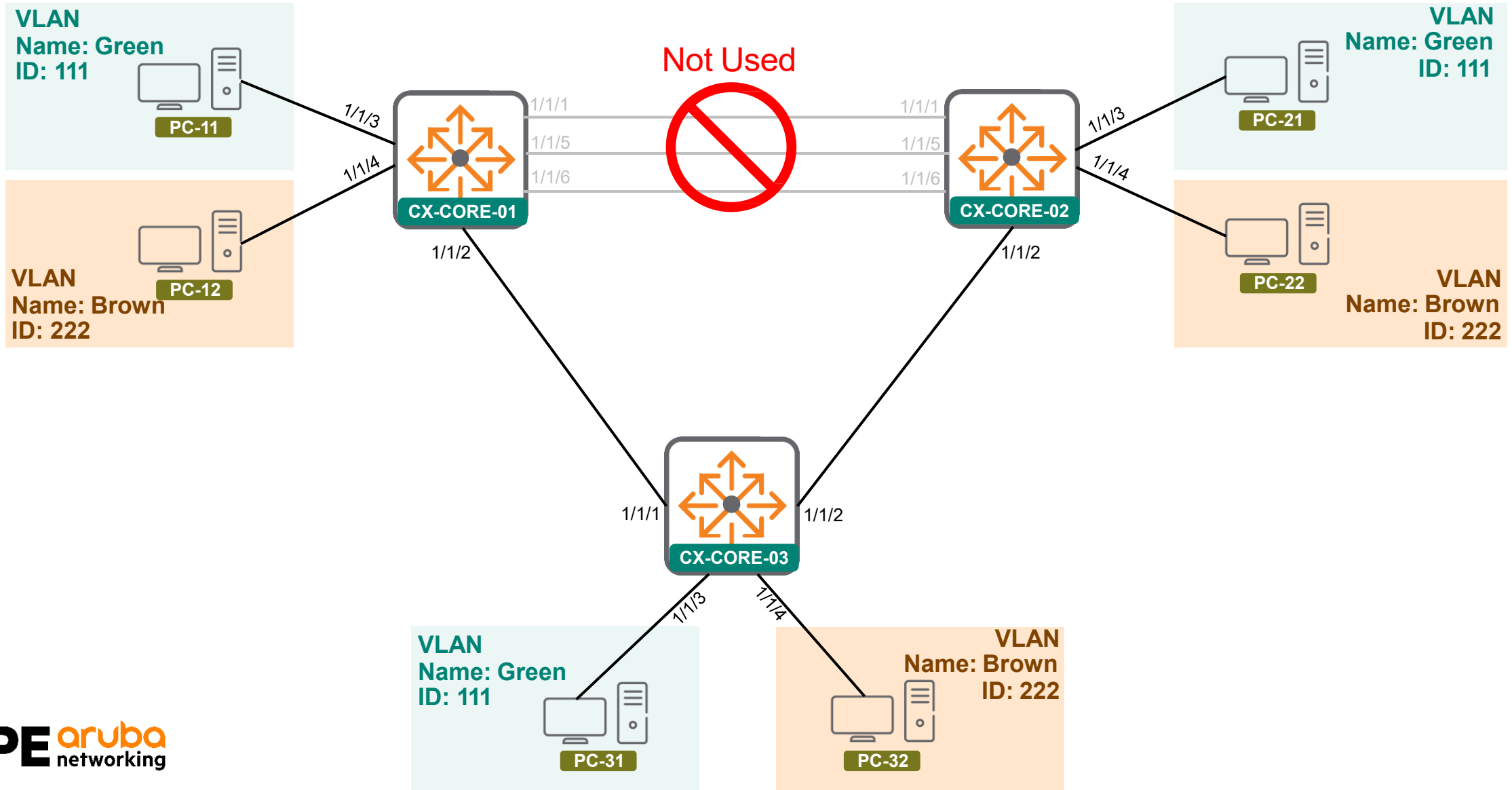


***Don't forgot rename switch
by group ID

Ethernet Frame Structure



Lab 2.4 – VLAN Manipulation



Lab 2.4 – VLAN Manipulation (Cont.)

- In this lab, interface lag 20, 1/1/1, 1/1/5 and 1/1/6 between CX-CORE-01 and CX-CORE-02 will be shutdown and not used

```
CX-CORE-01# config t
CX-CORE-01(config)# int lag 20
CX-CORE-01(config-lag-if)# shutdown
CX-CORE-01(config)# int 1/1/1,1/1/5-1/1/6
CX-CORE-01(config-if-<1/1/1,1/1/5-1/1/6>)# shutdown
CX-CORE-01(config-if-<1/1/1,1/1/5-1/1/6>)# end
CX-CORE-01#
```

- Repeat with CX-CORE-02 as well

Lab 2.4 – VLAN Manipulation (Cont.)

Do this lab on all 3 CX switches

```
CX-CORE-02# config t
CX-CORE-02(config)# vlan 111
CX-CORE-02(config-vlan-111)# name Green
CX-CORE-02(config-vlan-111)# vlan 222
CX-CORE-02(config-vlan-222)# name Brown
CX-CORE-02(config-vlan-222)# end
CX-CORE-02#
```

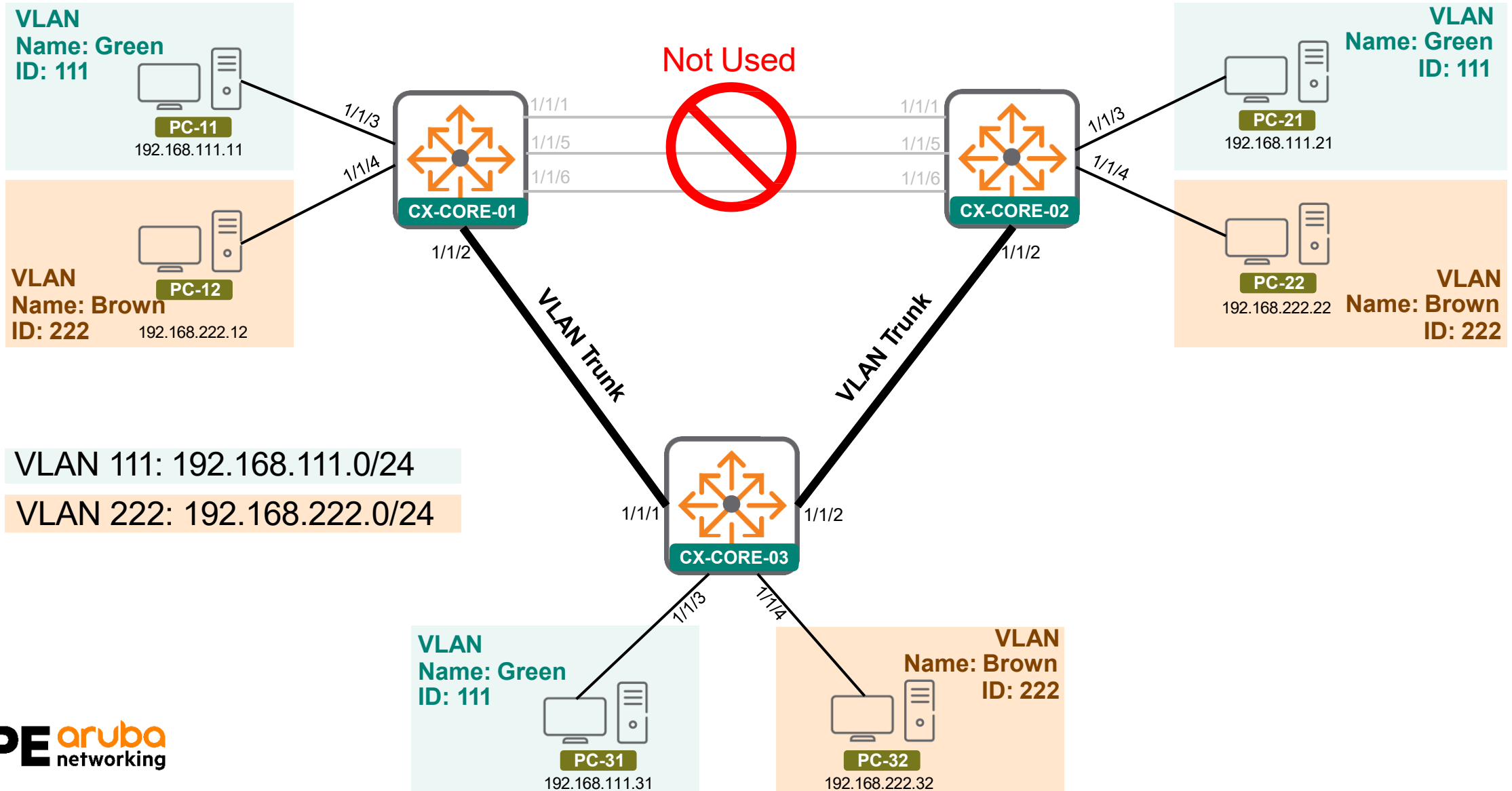
1. Create VLAN 111 and 222 on all three CX switches

```
CX-CORE-02# show vlan
```

VLAN	Name	Status	Reason	Type	Interfaces
1	DEFAULT_VLAN_1	down	no_member_port	default	
111	Green	down	no_member_port	static	
222	Brown	down	no_member_port	static	

2. Verify VLAN database on the switches

Lab 2.5 – Extending the VLAN across multiple switches



Lab 2.5 - Extending the VLAN across multiple switches (Cont.)

- Assign VLANs and port operation mode according to the lab topology on all three CX switches which is already summarized in the table below

Device	Interface	Port Mode
CX-CORE-01	1/1/2	Trunk VLAN 111, 222
	1/1/3	Access VLAN 111
	1/1/4	Access VLAN 222
CX-CORE-02	1/1/2	Trunk VLAN 111, 222
	1/1/3	Access VLAN 111
	1/1/4	Access VLAN 222
CX-CORE-03	1/1/1	Trunk VLAN 111, 222
	1/1/2	Trunk VLAN 111, 222
	1/1/3	Access VLAN 111
	1/1/4	Access VLAN 222

Lab 2.5 - Extending the VLAN across multiple switches (Cont.)

- Example of how to config VLAN trunk interface on CX-CORE-01

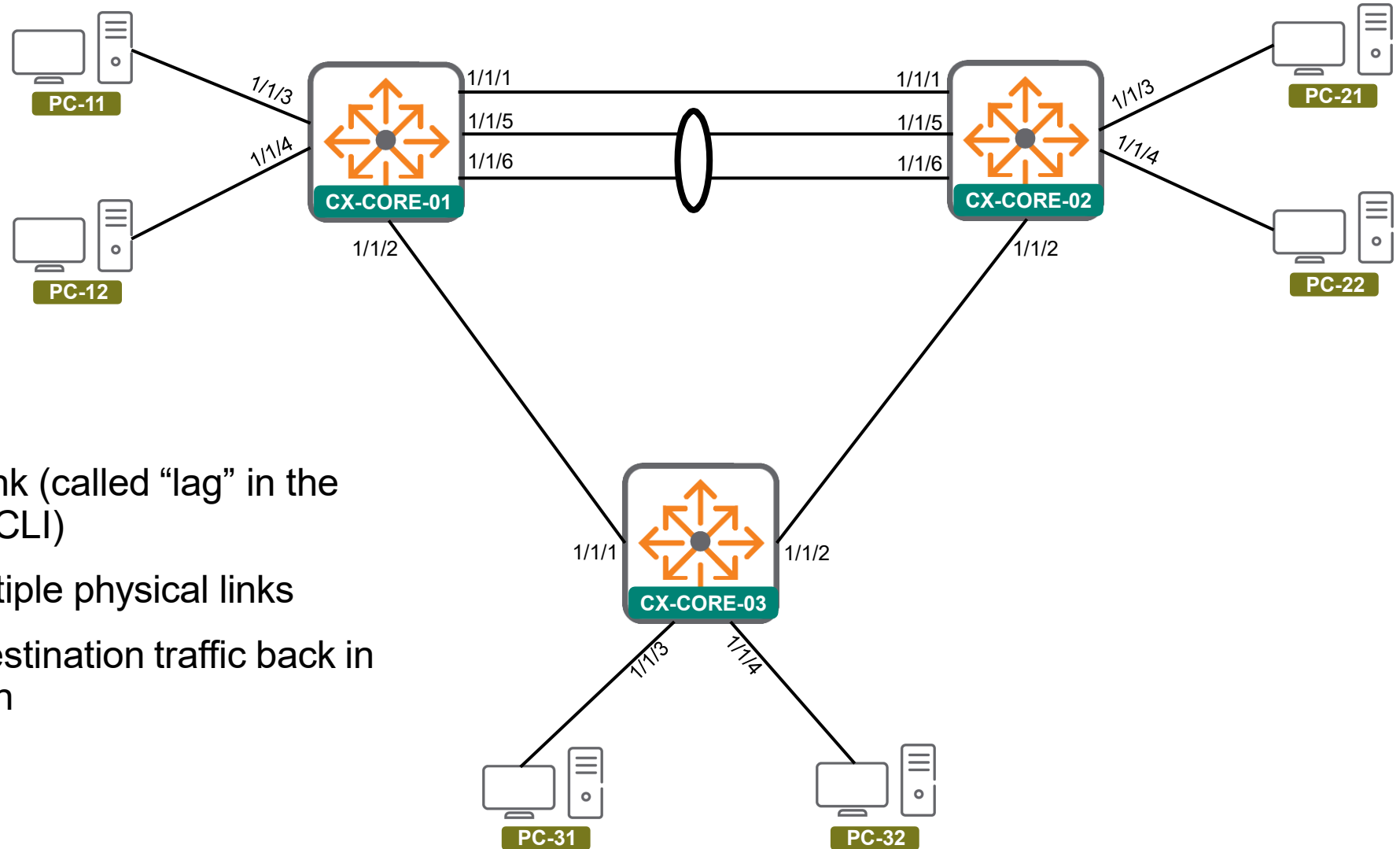
```
CX-CORE-01# config t
CX-CORE-01(config)# int 1/1/2
CX-CORE-01(config-if)# no routing
CX-CORE-01(config-if)# vlan trunk native 1
CX-CORE-01(config-if)# vlan trunk allowed 1,111,222
CX-CORE-01(config-if)# no shut
```

- Example of how to config VLAN access interface for connecting to host device on CX-CORE-01

```
CX-CORE-01# config t
CX-CORE-01(config)# int 1/1/3
CX-CORE-01(config-if)# no routing
CX-CORE-01(config-if)# vlan access 111
CX-CORE-01(config-if)# no shut
CX-CORE-01(config-if)# int 1/1/4
CX-CORE-01(config-if)# no routing
CX-CORE-01(config-if)# vlan access 222
CX-CORE-01(config-if)# no shut
```

- Assign VLANs and port operation mode according to the lab topology on all three CX switches

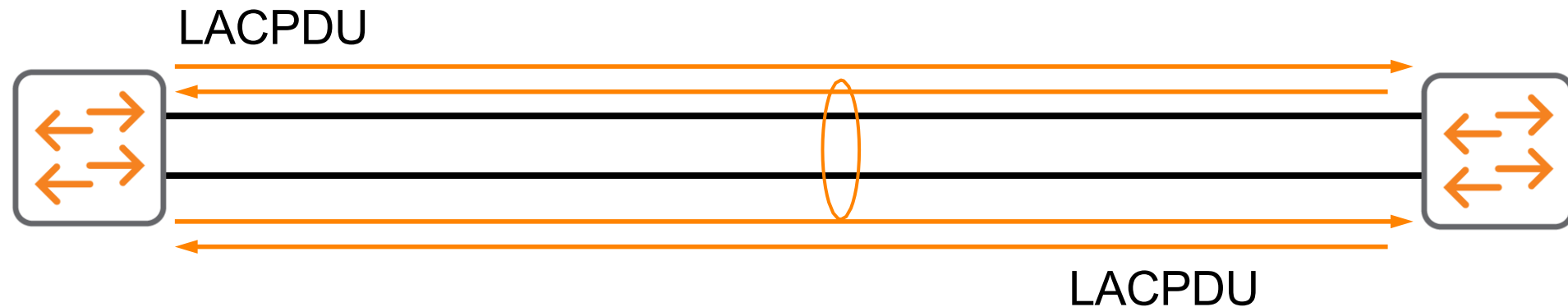
Link Aggregation



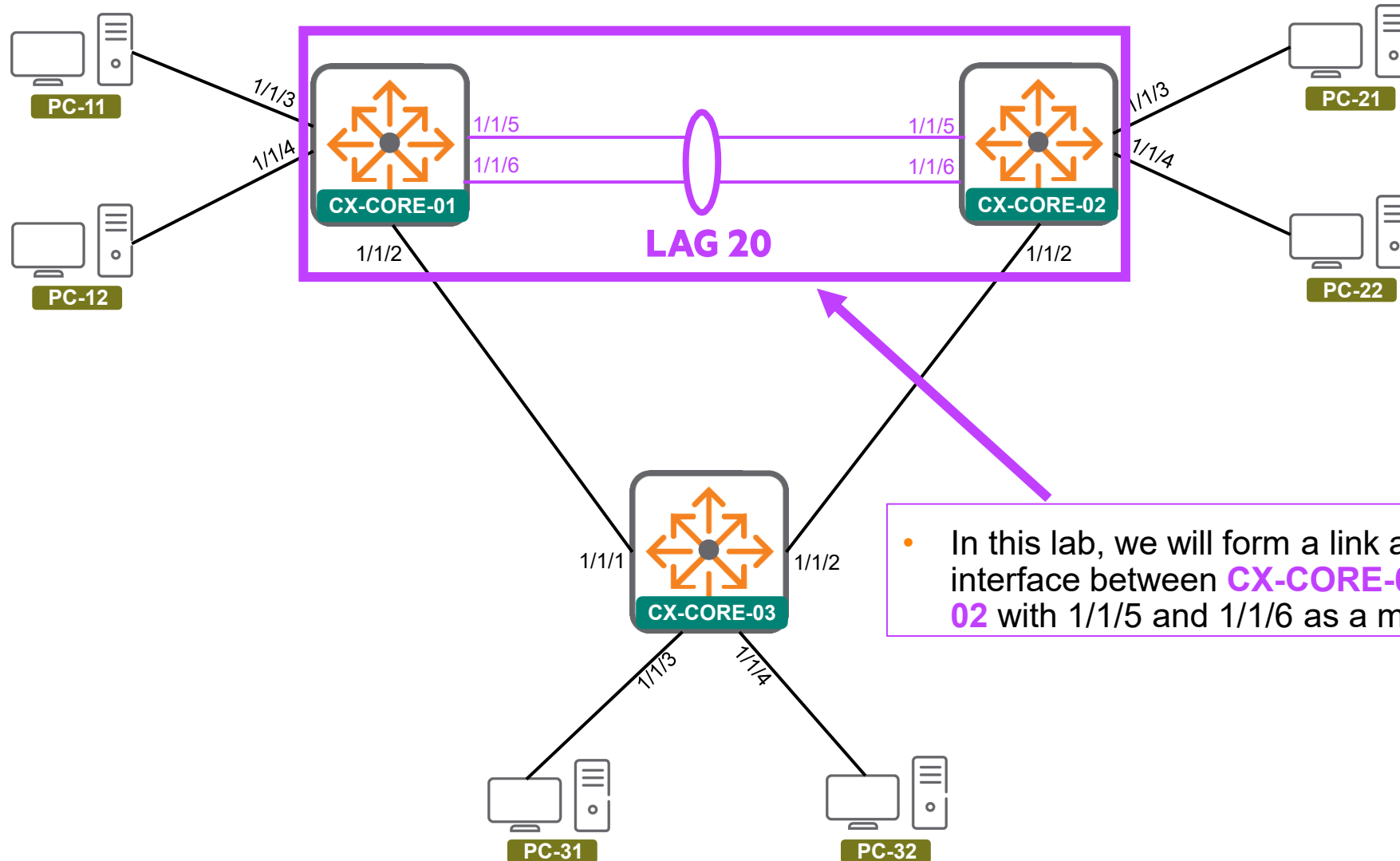
- Acts as one logical link (called “lag” in the ArubaOS-CX switch CLI)
- Carries traffic on multiple physical links
- Never floods multi-destination traffic back in the same aggregation

LACP

- Uses messages to establish the link aggregation
- Ensures that all links are connected to the same peer (system ID) and link aggregation (operational key)
- Ensures compatibility for other settings
- Manages adding and removing links



Lab 2.3 – Dynamic Link Aggregation



Lab 2.3 – Dynamic Link Aggregation (Cont.)

```
CX-CORE-01# config t
CX-CORE-01(config)# int lag 20
CX-CORE-01(config-lag-if)# lacp mode active
CX-CORE-01(config-lag-if)# no shut
CX-CORE-01(config-lag-if)# int 1/1/5-1/1/6
CX-CORE-01(config-if-<1/1/5-1/1/6>)# no lag 10
CX-CORE-01(config-if-<1/1/5-1/1/6>)# lag 20
CX-CORE-01(config-if-<1/1/5-1/1/6>)# no shut
```

1. Repeat above steps with CX-CORE-02 as well

```
CX-CORE-02# config t
CX-CORE-02(config)# int lag 20
CX-CORE-02(config-lag-if)# lacp mode active
CX-CORE-02(config-lag-if)# no shut
CX-CORE-02(config-lag-if)# int 1/1/5-1/1/6
CX-CORE-02(config-if-<1/1/5-1/1/6>)# no lag 10
CX-CORE-02(config-if-<1/1/5-1/1/6>)# lag 20
CX-CORE-02(config-if-<1/1/5-1/1/6>)# no shut
```

2. Next, verify the link aggregation 20 status on both CX-CORE-01 and CX-CORE-02

Lab 2.3 – Dynamic Link Aggregation (Cont.)

```
CX-CORE-01# show lacp interfaces
```

State abbreviations :

A - Active P - Passive F - Aggregable I - Individual
S - Short-timeout L - Long-timeout N - InSync O - OutofSync
C - Collecting D - Distributing
X - State m/c expired E - Default neighbor state

Actor details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key	Forwarding State
1/1/5	lag20	6	1	ALFNCD	08:00:09:e8:ae:71	65534	20	up
1/1/6	lag20	7	1	ALFNCD	08:00:09:e8:ae:71	65534	20	up

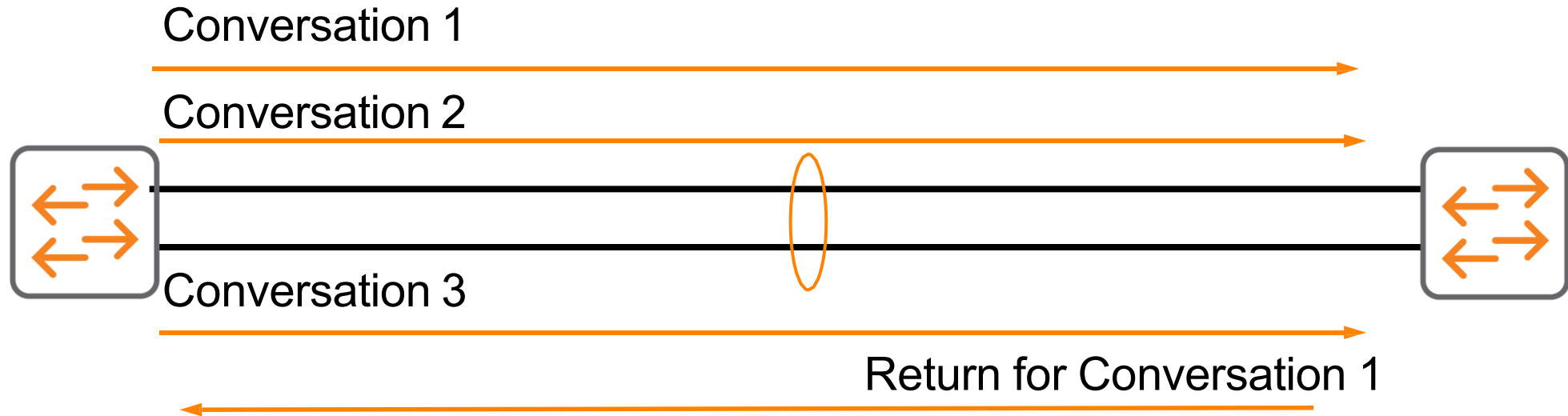
Partner details of all interfaces:

Intf	Aggr Name	Port Id	Port Pri	State	System-ID	System Pri	Aggr Key
1/1/5	lag20	6	1	ALFNCD	08:00:09:13:ed:f1	65534	20
1/1/6	lag20	7	1	ALFNCD	08:00:09:13:ed:f1	65534	20

- All member interface should be up

Load-sharing traffic over a link aggregation

- The link aggregation uses a hash to select a link for each *conversation*:
 - Conversation defined by the load-sharing mode
 - MAC source and destination address (L2-based)
 - IP source and destination address (L3-based) = Default option
 - Source and destination UDP/TCP ports (L4-based) = Only 8400 series switches support L4 hashing
 - Link for a conversation stays constant
 - Traffic tends to be balanced more evenly as conversations increase



LAB 5 VLAN, Access port และ Trunk port (1/5)

PNETLAB (สวิตช์จำลอง) ให้ PC ทั้งสามเครื่องอยู่คนละ VLAN คนละวง และให้ VLAN เดินข้ามสวิตช์ได้

1. สร้าง VLAN ทั้งสามวงให้เหมือนกันทั้งสามตัว — เริ่มที่ CX-SW-1

```
configure terminal
vlan 10
    name PC-A
vlan 20
    name PC-B
vlan 30
    name PC-C
```

2. ที่ CX-SW-1 ตั้งพอร์ตที่ PC1 เสียบให้เป็น access VLAN 10

```
interface 1/1/3
    no shutdown
    no routing
    vlan access 10
```

ตรวจสอบ

```
show vlan
show interface brief
show vlan port 1/1/3
show mac-address-table
```

ระวัง

ตอนนี้ PC ทั้งสามยัง ping ข้ามกันไม่ได้ และ ping gateway ตัวเองก็ยังไม่ได้ เพราะยังไม่มี SVI — เป็นเรื่องปกติ เดี่ยวได้เห็นความต่างชัดใน LAB 7

LAB 5 VLAN, Access port และ Trunk port (2/5)

PNETLAB (สวิตช์จำลอง) ให้ PC ทั้งสามเครื่องอยู่คนละ VLAN คนละวง และให้ VLAN เดินข้ามสวิตช์ได้

3. ที่ CX-SW-1 ตั้งพอร์ตที่ลงไป CX-SW-3 ให้เป็น trunk

```
interface 1/1/4
  no shutdown
  no routing
  vlan trunk native 1
  vlan trunk allowed 10,20,30
exit
write memory
```

ตรวจสอบผล

```
show vlan
show interface brief
show vlan port 1/1/3
show mac-address-table
```

ระวัง

ตอนนี้ PC ทั้งสามยัง ping ข้ามกันไม่ได้ และ ping gateway ตัวเองก็ยังไม่ได้ เพราะยังไม่มี SVI — เป็นเรื่องปกติ เดี่ยวได้เห็นความต่างชัดใน LAB 7

LAB 5 VLAN, Access port และ Trunk port (3/5)

PNETLAB (สวิตช์จำลอง) ให้ PC ทั้งสามเครื่องอยู่คนละ VLAN คนละวง และให้ VLAN เดินข้ามสวิตช์ได้

4. ที่ CX-SW-2 สร้าง VLAN ทั้งสามวงเหมือนกัน แล้วตั้งพอร์ตของ PC2 เป็น access VLAN 20

```
configure terminal
vlan 10
vlan 20
vlan 30
interface 1/1/3
    no shutdown
    no routing
    vlan access 20
exit
write memory
```

ตรวจสอบผล

```
show vlan
show interface brief
show vlan port 1/1/3
show mac-address-table
```

ระวัง

ตอนนี้ PC ทั้งสามยัง ping ข้ามกันไม่ได้ และ ping gateway ตัวเองก็ยังไม่ได้ เพราะยังไม่มี SVI — เป็นเรื่องปกติ เดี่ยวได้เห็นความต่างชัดใน LAB 7

LAB 5 VLAN, Access port และ Trunk port (4/5)

PNETLAB (สวิตช์จำลอง) ให้ PC ทั้งสามเครื่องอยู่คนละ VLAN คนละวง และให้ VLAN เดินข้ามสวิตช์ได้

5. ที่ CX-SW-3 สร้าง VLAN ทั้งสามวง ตั้งพอร์ตของ PC3 เป็น access VLAN 30 และตั้งพอร์ตขึ้น CX-SW-1 เป็น trunk

```
configure terminal
vlan 10
vlan 20
vlan 30
interface 1/1/3
    no shutdown
    no routing
    vlan access 30
interface 1/1/1
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed 10,20,30
exit
write memory
```

ตรวจสอบผล

```
show vlan
show interface brief
show vlan port 1/1/3
show mac-address-table
```

ระวัง

ตอนนี้ PC ทั้งสามยัง ping ข้ามกันไม่ได้ และ ping gateway ตัวเองก็ยังไม่ได้ เพราะยังไม่มี SVI — เป็นเรื่องปกติ เดี่ยวได้เห็นความต่างชัดใน LAB 7

LAB 5 VLAN, Access port และ Trunk port (5/5)

PNETLAB (สวิตช์จำลอง) ให้ PC ทั้งสามเครื่องอยู่คนละ VLAN คนละวง และให้ VLAN เดินข้ามสวิตช์ได้

6. ตั้ง IP ให้ PC — PC1 10.10.10.11/24 · PC2 10.10.20.12/24 · PC3 10.10.30.13/24 gateway ของแต่ละเครื่องคือ .1 ของวงตัวเอง

ตรวจสอบผล

```
show vlan
show interface brief
show vlan port 1/1/3
show mac-address-table
```

ระวัง

ตอนนี้ PC ทั้งสามยัง ping ข้ามกันไม่ได้ และ ping gateway ตัวเองก็ยังไม่ได้ เพราะยังไม่มี SVI — เป็นเรื่องปกติ เดี่ยวได้เห็นความต่างชัดใน LAB 7

LAB 6 LAG ระหว่างสวิตช์ด้วย LACP (1/3)

PNETLAB (สวิตช์จำลอง) รวมสองสายระหว่าง CX-SW-1 กับ CX-SW-2 ให้เป็นเส้นเดียวที่เร็วขึ้นและขาดได้หนึ่งเส้น

1. ที่ CX-SW-1 สร้าง LAG 1 เป็น trunk แล้วเปิด LACP

```
configure terminal
interface lag 1
  no shutdown
  no routing
  vlan trunk native 1
  vlan trunk allowed 10,20,30
  lacp mode active
```

2. ใส่พอร์ต 1/1/1 และ 1/1/2 เข้า LAG

```
interface 1/1/1-1/1/2
  no shutdown
  lag 1
exit
write memory
```

ตรวจสอบผล

```
show lacp interfaces
show lacp aggregates
show interface lag 1
show vlan
```

ระวัง

LAG ไม่ขึ้นเกือบทุกครั้งเพราะสองฝั่งตั้งไม่ตรงกัน — เช็คสามอย่าง: lacp mode ต้อง active ทั้งคู่, vlan trunk allowed ต้องเหมือนกัน, native VLAN ต้องเท่ากัน

LAB 6 LAG ระหว่างสวิตช์ด้วย LACP (2/3)

PNETLAB (สวิตช์จำลอง) รวมสองสายระหว่าง CX-SW-1 กับ CX-SW-2 ให้เป็นเส้นเดียวที่เร็วขึ้นและขาดได้หนึ่งเส้น

3. ที่ CX-SW-2 ทำเหมือนกันทุกบรรทัด — LACP จะจับคู่กันเองก็ต่อเมื่อสองฝั่งตรงกัน

```
configure terminal
interface lag 1
    no shutdown
    no routing
    vlan trunk native 1
    vlan trunk allowed 10,20,30
    lacp mode active
interface 1/1/1-1/1/2
    no shutdown
    lag 1
exit
write memory
```

4. ดูว่า LAG ขึ้นจริง — ต้องเห็นสมาชิกสองพอร์ตอยู่ในสถานะ up ทั้งคู่
- ```
show lacp interfaces
```

### ตรวจสอบผล

```
show lacp interfaces
show lacp aggregates
show interface lag 1
show vlan
```

### ระวัง

LAG ไม่ขึ้นเกือบทุกครั้งเพราะสองฝั่งตั้งไม่ตรงกัน — เช็คสามอย่าง: lacp mode ต้อง active ทั้งคู่, vlan trunk allowed ต้องเหมือนกัน, native VLAN ต้องเท่ากัน

## LAB 6 LAG ระหว่างสวิตช์ด้วย LACP (3/3)

PNETLAB (สวิตช์จำลอง) รวมสองสายระหว่าง CX-SW-1 กับ CX-SW-2 ให้เป็นเส้นเดียวที่เร็วขึ้นและขาดได้หนึ่งเส้น

5. ทดสอบว่าขาดหนึ่งเส้นแล้วยังใช้ได้ — shutdown 1/1/1 แล้วดูว่า LAG ยังอยู่ เหลือสมาชิกหนึ่งพอร์ต

```
interface 1/1/1
shutdown
no shutdown
```

### ตรวจสอบผล

```
show lacp interfaces
show lacp aggregates
show interface lag 1
show vlan
```

### ระวัง

LAG ไม่ขึ้นเกือบทุกครั้งเพราะสองฝั่งตั้งไม่ตรงกัน — เช็คสามอย่าง: lacp mode ต้อง active ทั้งคู่, vlan trunk allowed ต้องเหมือนกัน, native VLAN ต้องเท่ากัน

# Layer 3 Interface

1. Change to global configuration mode with the command `config`.
2. Change to the interface configuration context for the interface with the command `interface`.
3. Enable routing support with the command `routing`.
4. Assign an IPv4 address with the command `ip address`, or an IPv6 address with the command `ipv6 address`.
5. Review interface configuration settings with the command `show interface`.

## Example: Assign IP address to physical interface (ROP: Routed Only Port)

```
switch(config)# interface 1/1/1
switch(config-if)# routing
switch(config-if)# ip address 192.168.100.1/24
```

## Example: Assign IP address to loopback interface

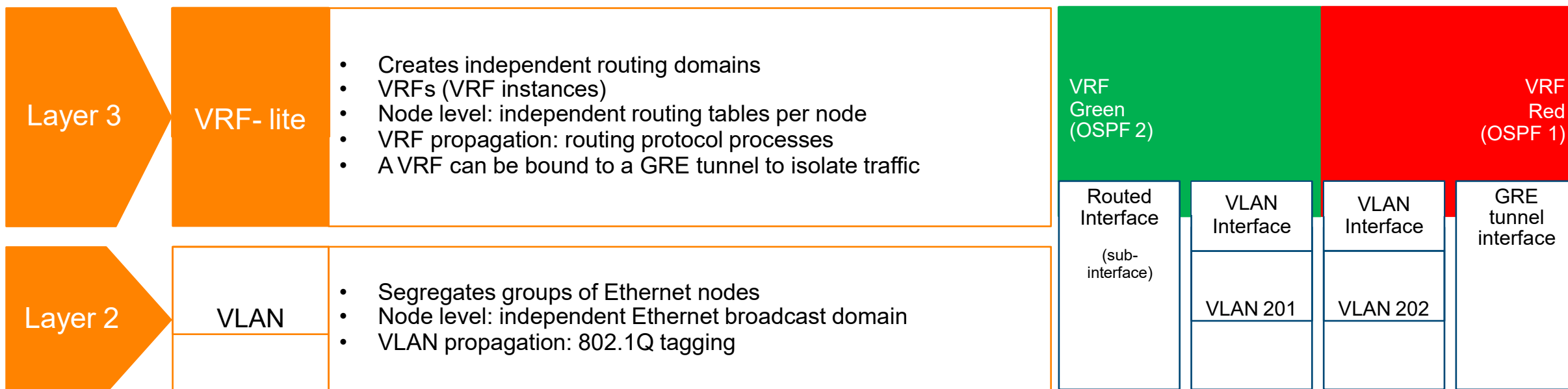
```
switch(config)# interface loopback 1
switch(config-loopback-if)# ip address 192.168.20.1/24
```

## Example: Assign IP address to VLAN interface (SVI: Switched Virtual Interface)

```
switch(config)# interface vlan 10
switch(config-loopback-if)# ip address 192.168.199.1/24
```

# Introduction

## Layer 2 and layer 3 network segmentation

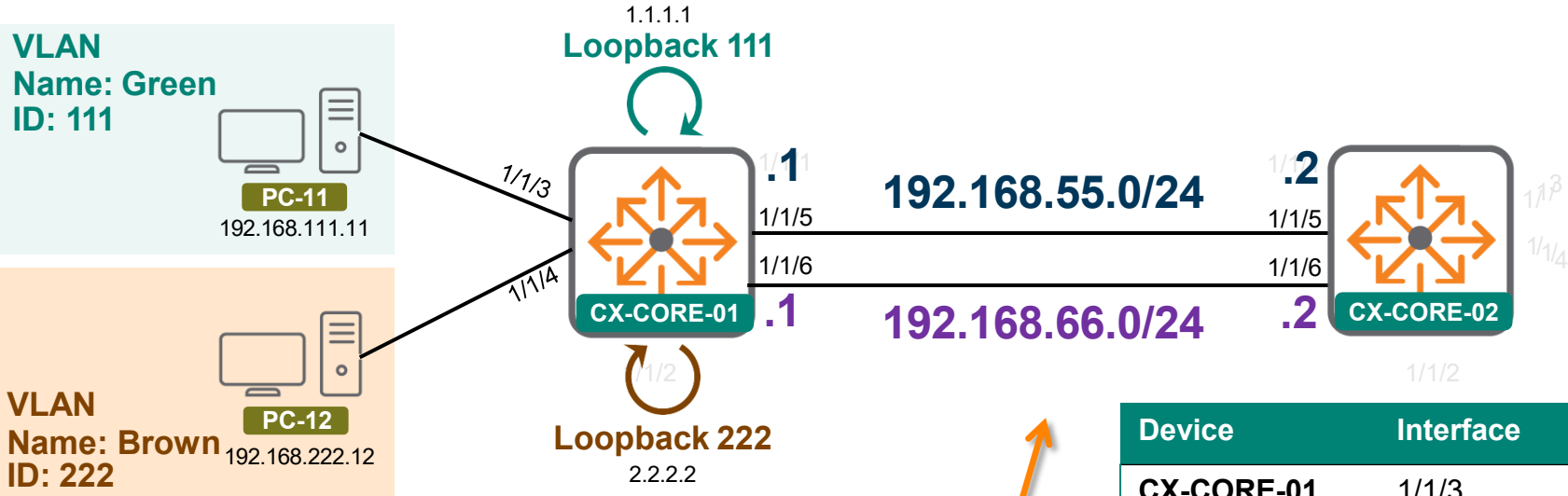


Note:

VRF (lite) = independent routing domain (MCE in Comware OS)  
VRF = MPLS L3VPN (not covered in this course)

- There is two default VRF called default and mgmt
- The management interface belongs to the mgmt
- All data L3 interfaces belong to the default
- **Maximum user defined VRFs: 62**

# Lab 3.1 – Assign L2 and L3 Interface Configuration



- In this lab, we will use **CX-CORE-01** and **CX-CORE-02** with 1/1/5 and 1/1/6 plus interface 1/1/3 and 1/1/4 connected to PC-11 and PC12 on CX-CORE-01. The others interfaces will not be used and remain shutdown

Note: VRF Default means you don't have to assign that port to any specific VRF

| Device     | Interface    | IP Address         | VLAN / VRF  |
|------------|--------------|--------------------|-------------|
| CX-CORE-01 | 1/1/3        | N/A – Layer 2 Port | VLAN Green  |
|            | 1/1/4        | N/A – Layer 2 Port | VLAN Brown  |
|            | 1/1/5        | 192.168.55.1/24    | VRF CE1     |
|            | 1/1/6        | 192.168.66.1/24    | VRF CE2     |
|            | Loopback 111 | 1.1.1.1/32         | VRF CE1     |
|            | Loopback 222 | 2.2.2.2/32         | VRF CE2     |
|            | Vlan 111     | 192.168.111.254/24 | VRF CE1     |
|            | Vlan 222     | 192.168.222.254/24 | VRF CE2     |
| CX-CORE-02 | 1/1/5        | 192.168.55.2/24    | VRF Default |
|            | 1/1/6        | 192.168.66.2/24    | VRF Default |

## Lab 3.1 – Assign L2 and L3 Interface Configuration (Cont.)



1. Revert our three CX switches configuration to checkpoint `initial_config` before continue to this lab  
`#copy checkpoint initial_config running-config`
2. Shutdown unused interfaces in this lab according to the topology diagram in the previous slide
3. In this lab you have to create VLANs and assign port 1/1/3 and 1/1/4 to VLAN according to the table in the topology diagram
4. Assign virtual PC IP address and gateway according to the table in the topology diagram
5. We will continue with layer 3 and VRF configuration as following . . .

```
CX-CORE-01# config t
CX-CORE-01(config)# vrf CE1
CX-CORE-01(config-vrf)# vrf CE2
```

1. Add VRF CE1 and CE2 to CX-CORE-01
2. Then, show VRF configuration. You will see both VRF appear but no interface member
3. Next, we will create L3 interfaced and assign VRF to them

```
CX-CORE-01# sh vrf
```

VRF Configuration:

| VRF Name   | : default | Status |
|------------|-----------|--------|
| Interfaces |           |        |

| VRF Name   | : CE1 | Status |
|------------|-------|--------|
| Interfaces |       |        |

| VRF Name   | : CE2 | Status |
|------------|-------|--------|
| Interfaces |       |        |

## Lab 3.1 – Assign L2 and L3 Interface Configuration (Cont.)

### Layer 3 configuration on CX-CORE-01

```
CX-CORE-01# config t
CX-CORE-01(config)# int vlan 111
CX-CORE-01(config-if-vlan)# vrf attach CE1
CX-CORE-01(config-if-vlan)# ip address 192.168.111.254/24
CX-CORE-01(config-if-vlan)# no shut
CX-CORE-01(config-if-vlan)# int vlan 222
CX-CORE-01(config-if-vlan)# vrf attach CE2
CX-CORE-01(config-if-vlan)# ip address 192.168.222.254/24
CX-CORE-01(config-if-vlan)# no shut
```

1. Create VLAN interface
2. Assign VRF and IP address

```
CX-CORE-01(config-if-vlan)# int loopback 111
CX-CORE-01(config-loopback-if)# vrf attach CE1
CX-CORE-01(config-loopback-if)# ip address 1.1.1.1/32
CX-CORE-01(config-loopback-if)# int loopback 222
CX-CORE-01(config-loopback-if)# vrf attach CE2
CX-CORE-01(config-loopback-if)# ip address 2.2.2.2/32
```

1. Create Loopback interface
2. Assign VRF and IP address

Note: You have to assign VRF to L3 interface before IP address, otherwise IP configuration will be wiped out

# Lab 3.1 – Assign L2 and L3 Interface Configuration (Cont.)

Layer 3 configuration on CX-CORE-01

```
CX-CORE-01# config t
CX-CORE-01(config)# int 1/1/5
CX-CORE-01(config-if)# vrf attach CE1
CX-CORE-01(config-if)# ip address 192.168.55.1/24
CX-CORE-01(config-if)# no shut
CX-CORE-01(config-if)# int 1/1/6
CX-CORE-01(config-if)# vrf attach CE2
CX-CORE-01(config-if)# ip address 192.168.66.1/24
CX-CORE-01(config-if)# no shut
```

- Assign VRF and IP address to interface 1/1/5 and 1/1/6

```
CX-CORE-01# sh vrf
VRF Configuration:

VRF Name : default
Interfaces Status

VRF Name : CE1
Interfaces Status

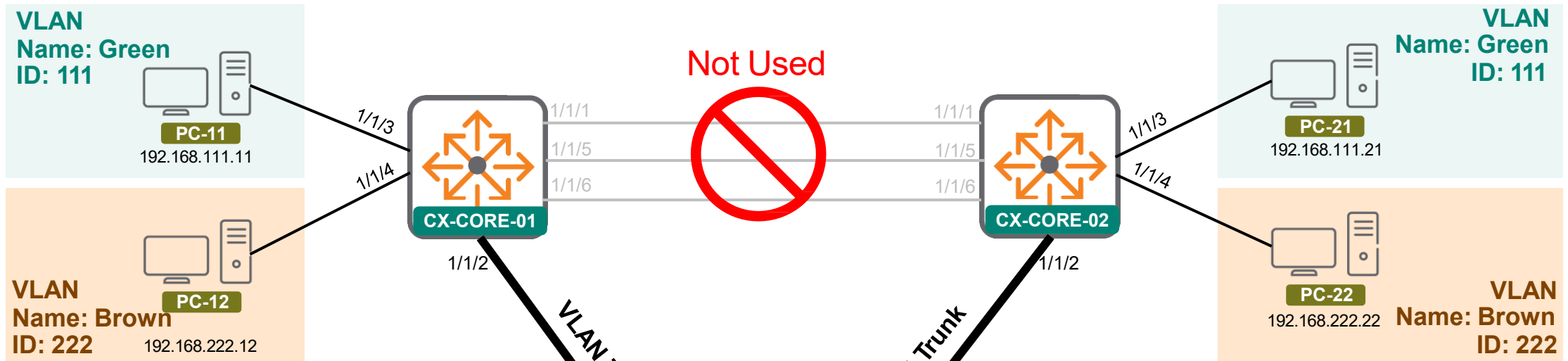
1/1/5 up
loopback111 up
vlan111 up
VRF Name : CE2
Interfaces Status

1/1/6 up
loopback222 up
vlan222 up
```

- Verify VRF member port assignment



# Lab 2.6 - Inter VLAN routing



## Assign Interface VLAN to CX-CORE-03

Interface VLAN 111: 192.168.111.254/24

Interface VLAN 222: 192.168.222.254/24

## Lab 2.6 - Inter VLAN routing (Cont.)

```
CX-CORE-03# config t
CX-CORE-03(config)# int vlan 111
CX-CORE-03(config-if-vlan)# ip address 192.168.111.254/24
CX-CORE-03(config-if-vlan)# no shut
CX-CORE-03(config-if-vlan)# int vlan 222
CX-CORE-03(config-if-vlan)# ip address 192.168.222.254/24
CX-CORE-03(config-if-vlan)# no shut
```

1. Create interface VLAN 111 and 222, then assigned IP address to each of them

```
CX-CORE-03# sh ip int br
```

| Interface | IP Address         | Interface Status<br>link/admin |
|-----------|--------------------|--------------------------------|
| vlan111   | 192.168.111.254/24 | up/up                          |
| vlan222   | 192.168.222.254/24 | up/up                          |

Do this lab on CX-CORE-03 only

2. Verify IP address setting of all interface in the switch

```
CX-CORE-03# sh ip route
```

Displaying ipv4 routes selected for forwarding

'[x/y]' denotes [distance/metric]

```
192.168.111.0/24, vrf default
 via vlan111, [0/0], connected
192.168.111.254/32, vrf default
 via vlan111, [0/0], local
192.168.222.0/24, vrf default
 via vlan222, [0/0], connected
192.168.222.254/32, vrf default
 via vlan222, [0/0], local
```

3. Verify routing table on CX-CORE-03 and check for 192.168.111.0/24 and 192.168.222.0/24 should be shown up, otherwise verify your configuration once again

## Lab 2.6 - Inter VLAN routing (Cont.)

1. Re-assign IP address and gateway to PCs in this lab based on their information in the topology diagram

3. Verify ping test to VLAN gateway

4. Verify ping test to other host on the different VLAN. They are all must be success, otherwise verify your configuration before proceed

```
PC> ip 192.168.111.31/24 192.168.111.254
Checking for duplicate address...
PC1 : 192.168.111.31 255.255.255.0 gateway 192.168.111.254
```

```
PC> save
Saving startup configuration to startup.vpc
. Done
```

```
PC> ping 192.168.111.254
84 bytes from 192.168.111.254 icmp_seq=1 ttl=64 time=12.371 ms
84 bytes from 192.168.111.254 icmp_seq=2 ttl=64 time=1.406 ms
84 bytes from 192.168.111.254 icmp_seq=3 ttl=64 time=1.265 ms
84 bytes from 192.168.111.254 icmp_seq=4 ttl=64 time=1.263 ms
84 bytes from 192.168.111.254 icmp_seq=5 ttl=64 time=2.788 ms
```

```
PC> ping 192.168.222.22
84 bytes from 192.168.222.22 icmp_seq=1 ttl=63 time=13.690 ms
84 bytes from 192.168.222.22 icmp_seq=2 ttl=63 time=3.108 ms
84 bytes from 192.168.222.22 icmp_seq=3 ttl=63 time=1.988 ms
84 bytes from 192.168.222.22 icmp_seq=4 ttl=63 time=1.960 ms
84 bytes from 192.168.222.22 icmp_seq=5 ttl=63 time=2.226 ms
```

```
PC> ping 192.168.222.12
84 bytes from 192.168.222.12 icmp_seq=1 ttl=63 time=6.830 ms
84 bytes from 192.168.222.12 icmp_seq=2 ttl=63 time=2.774 ms
84 bytes from 192.168.222.12 icmp_seq=3 ttl=63 time=2.030 ms
84 bytes from 192.168.222.12 icmp_seq=4 ttl=63 time=2.461 ms
84 bytes from 192.168.222.12 icmp_seq=5 ttl=63 time=2.263 ms
```

```
PC> ping 192.168.222.32
84 bytes from 192.168.222.32 icmp_seq=1 ttl=63 time=5.571 ms
84 bytes from 192.168.222.32 icmp_seq=2 ttl=63 time=1.511 ms
84 bytes from 192.168.222.32 icmp_seq=3 ttl=63 time=1.297 ms
84 bytes from 192.168.222.32 icmp_seq=4 ttl=63 time=0.990 ms
84 bytes from 192.168.222.32 icmp_seq=5 ttl=63 time=0.925 ms
```

# Knowledge Check #1

---

**Which protocol is used to advertise switch capabilities to directly connected devices?**

- a. STP
- b. LLDP (Link Layer Discovery Protocol)
- c. ARP
- d. DHCP

## Knowledge Check #2

---

**What feature allows combining multiple physical links into a single logical interface?**

- a. VLAN
- b. STP
- c. LACP (Link Aggregation Control Protocol)
- d. LLDP

## Knowledge Check #3

---

**What is the default spanning tree mode on the Aruba CX 6200F?**

- a. PVST+
- b. Rapid PVST+
- c. RSTP
- d. MSTP

## LAB 7 SVI และ Inter-VLAN routing (1/2)

PNETLAB (สวิตช์จำลอง) ให้ทั้งสาม VLAN มี gateway อยู่บน CX-SW-1 แล้วทำให้ PC ทั้งสามเครื่อง ping ข้าม VLAN หากันได้

1. ที่ CX-SW-1 สร้าง SVI ของทั้งสาม VLAN — เลข .1 คือ gateway ที่ PC ตั้งไว้แล้ว

```
configure terminal
interface vlan 10
 ip address 10.10.10.1/24
 no shutdown
interface vlan 20
 ip address 10.10.20.1/24
 no shutdown
interface vlan 30
 ip address 10.10.30.1/24
 no shutdown
exit
write memory
```

2. ดูตาราง routing — ต้องเห็นทั้งสามวงเป็น connected ถ้าขาดวงไหนแปลว่า SVI วงนั้นยังไม่ up  
show ip route

### ตรวจสอบผล

```
show ip interface brief
show ip route
show arp
show vlan
```

### ระวัง

ping ข้าม VLAN ไม่ผ่านทั้งที่ SVI ขึ้นครบ ให้ดูที่ PC ก่อน — gateway ที่ตั้งไว้ในเครื่องต้องเป็น .1 ของวงตัวเอง ไม่ใช่ .1 ของวงอื่น

## LAB 7 SVI และ Inter-VLAN routing (2/2)

PNETLAB (สวิตช์จำลอง) ให้ทั้งสาม VLAN มี gateway อยู่บน CX-SW-1 แล้วทำให้ PC ทั้งสามเครื่อง ping ข้าม VLAN หากันได้

3. ที่ PC1 ping gateway ตัวเองก่อน (10.10.10.1) ต้องผ่าน — ถ้าข้อนี้ไม่ผ่าน อย่าเพิ่งไปข้อถัดไป
4. ทดสอบ inter-VLAN routing จริง: PC1 ping PC2 (10.10.20.12) และ ping PC3 (10.10.30.13) ต้องผ่านทั้งคู่
5. ไล่ปัญหาตามลำดับนี้เมื่อ ping ไม่ผ่าน — PC ping gateway ตัวเองได้ไหม → show ip route เห็นวงปลายทางหรือยัง → show mac-address-table เห็น MAC ของ PC ปลายทางไหม → VLAN นั้นอยู่ใน vlan trunk allowed ของ LAG และ trunk หรือเปล่า

```
show ip route
show arp
show mac-address-table
show vlan
```
6. เซฟ checkpoint ปิดท้ายฝั่งสวิตช์ไว้ก่อนพักเที่ยง

```
copy running-config checkpoint END-OF-SWITCH
```

### ตรวจสอบผล

```
show ip interface brief
show ip route
show arp
show vlan
```

### ระวัง

ping ข้าม VLAN ไม่ผ่านทั้งที่ SVI ขึ้นครบ ให้ดูที่ PC ก่อน — gateway ที่ตั้งไว้ในเครื่องต้องเป็น .1 ของวงตัวเอง ไม่ใช่ .1 ของวงอื่น



# WLAN Fundamentals

---



# Generational Wi-Fi Naming Conventions

- To help folks identify device generations without understanding the IEEE alphabet soup, the Wi-Fi Alliance created a naming scheme using numbers to identify the various generations.
  - The IEEE organization works on the standard.
  - The Wi-Fi Alliance is responsible for the branding and certification of Wi-Fi products.

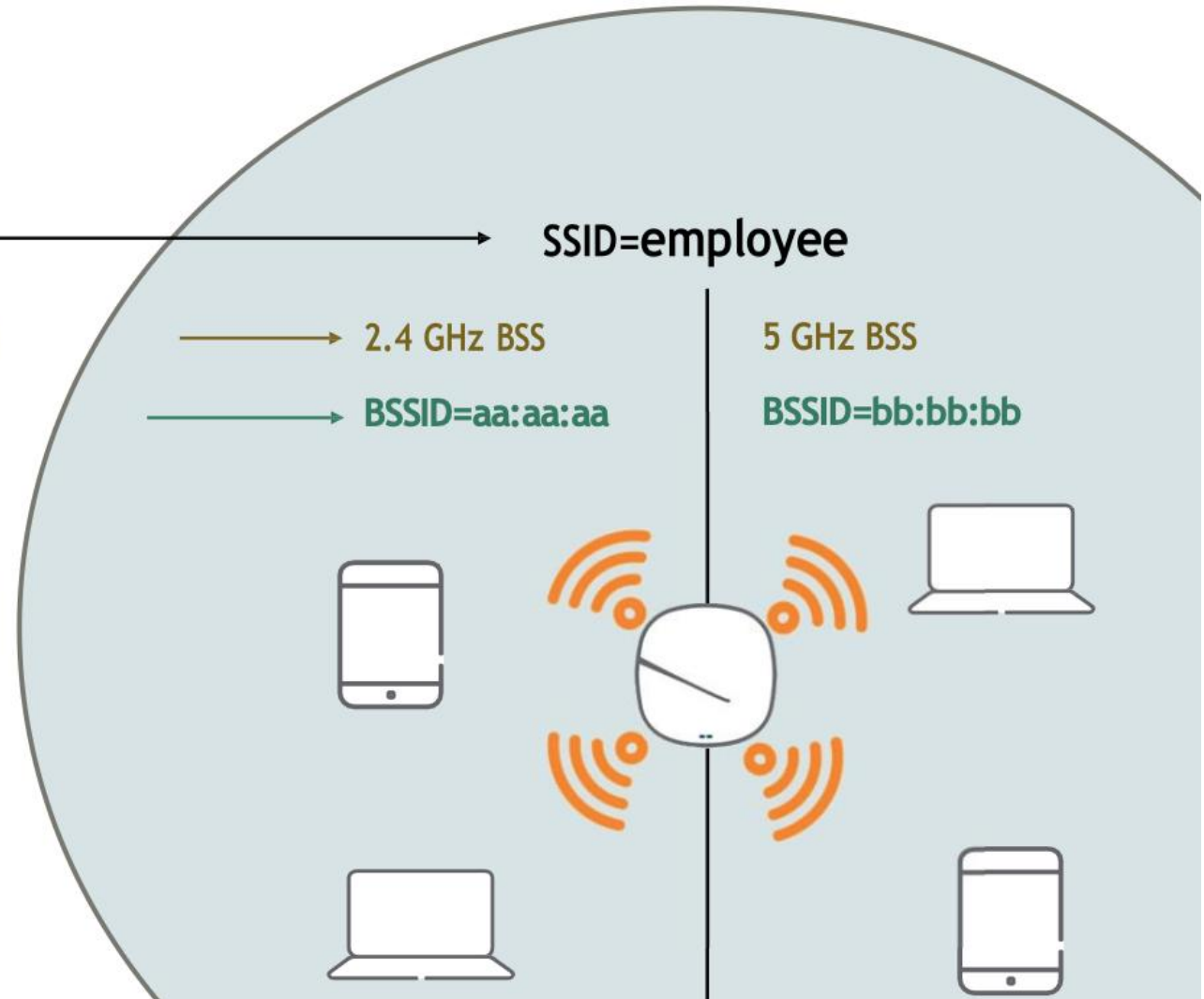
| IEEE Equivalent             | Wi-Fi Alliance Generation |
|-----------------------------|---------------------------|
| 802.11n                     | Wi-Fi 4                   |
| 802.11ac                    | Wi-Fi 5                   |
| 802.11ax                    | Wi-Fi 6                   |
| 802.11ax operating in 6 GHz | Wi-Fi 6E                  |
| 802.11be                    | Wi-Fi 7                   |

# **BSS: WLAN logical configuration**

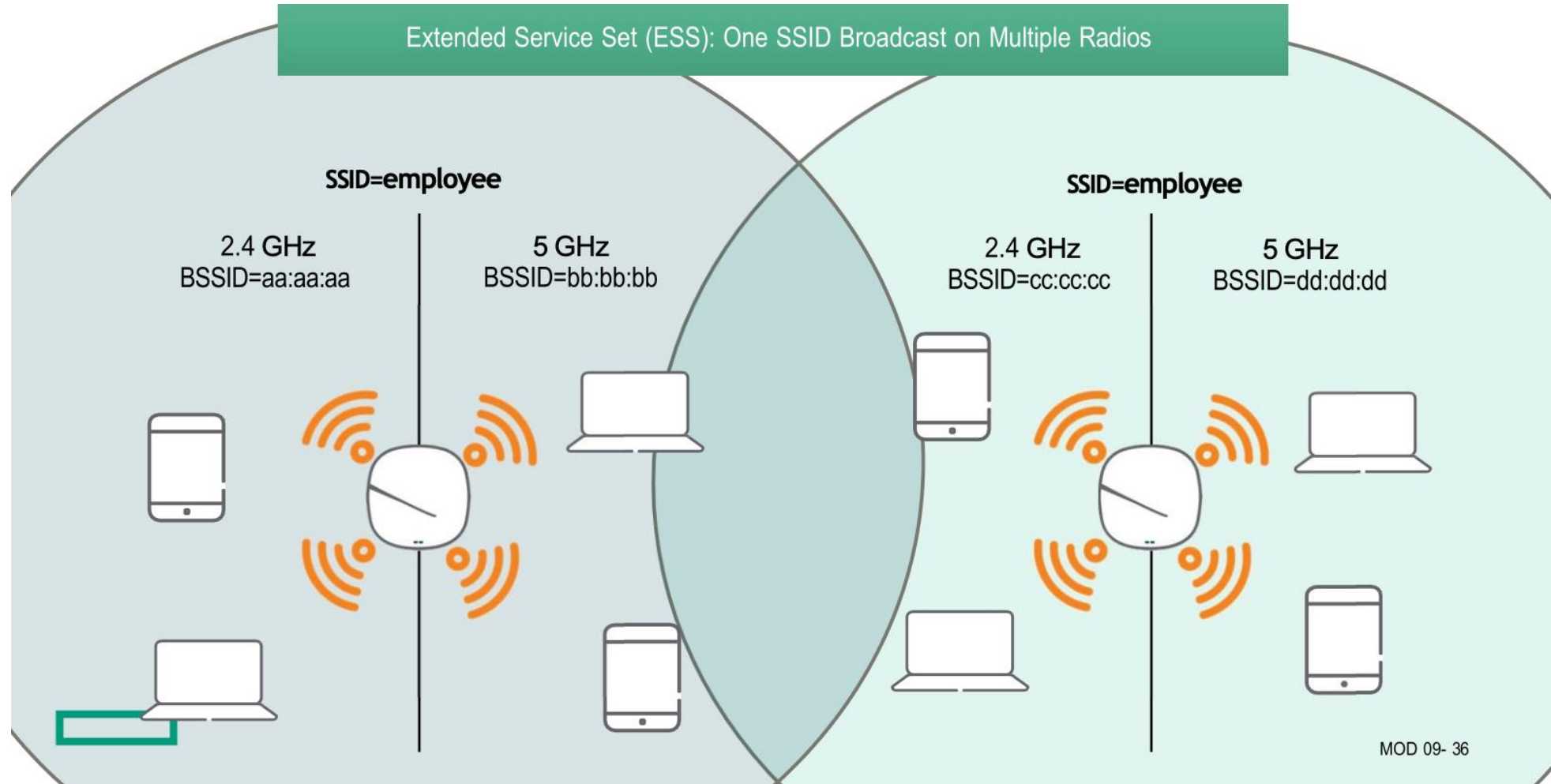
SSID/ESSID: Defines the logical WLAN

Basic Service Set (BSS): Radio, all associated clients

Basic Service Set ID (BSSID): Radio MAC Address



# ESS: WLAN logical configuration: ESS



# Airmatch function

HPE Aruba Networking Central hosted RF optimization service

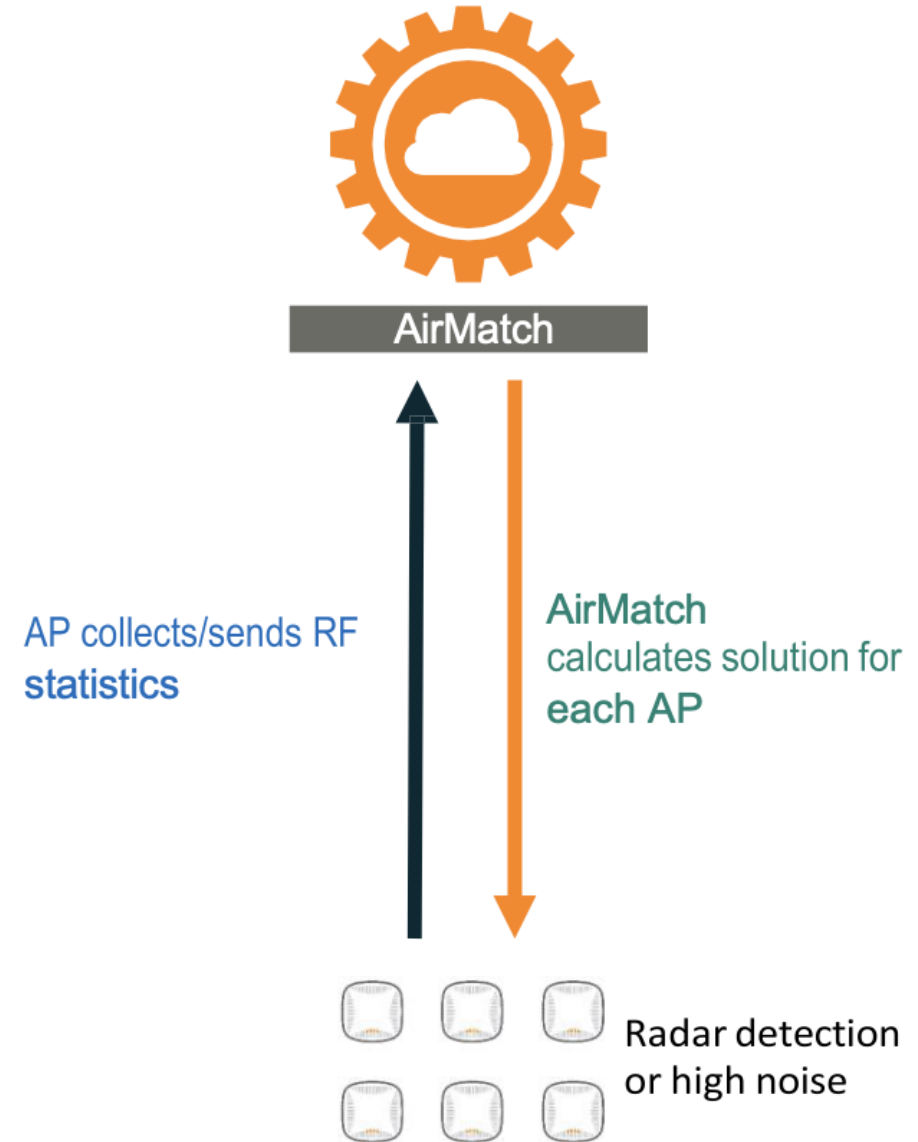
Generates an RF solution for APs

Dynamically adapts to the changing RF conditions

Past 24 hours RF info used for calculation

Channel & Power deployment only once a day

Minimizes channel coupling, channel width



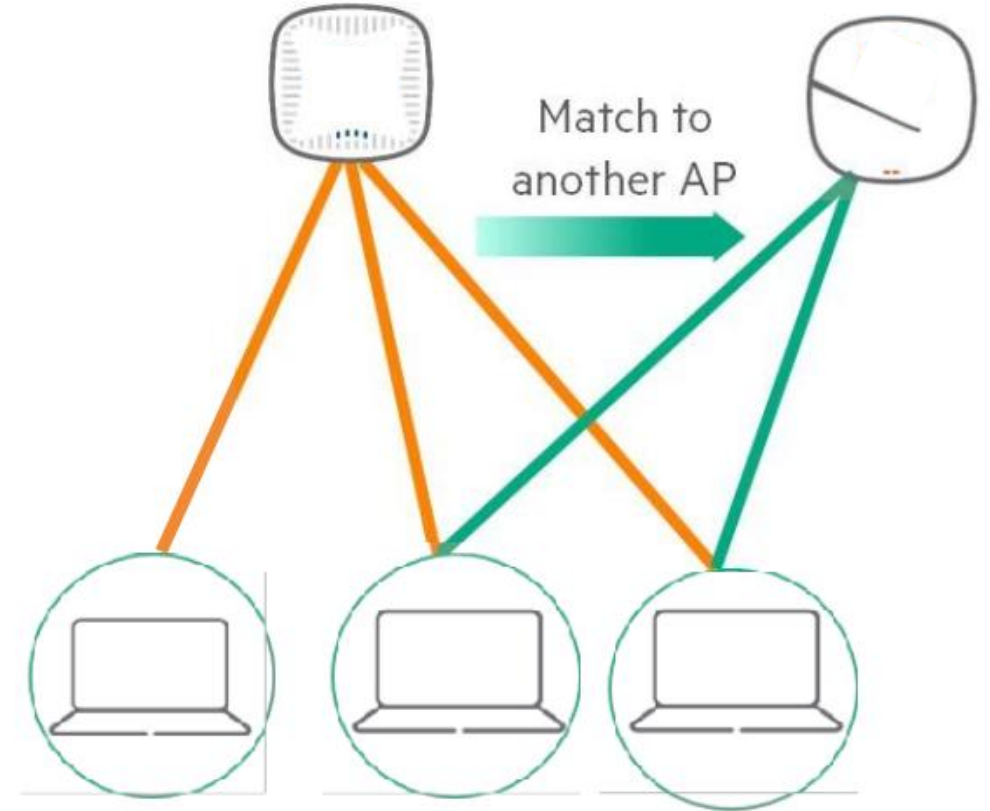
# What is client match?

HPE Aruba Networking Patented Technology

Load Balancing

Band Steering and Balancing

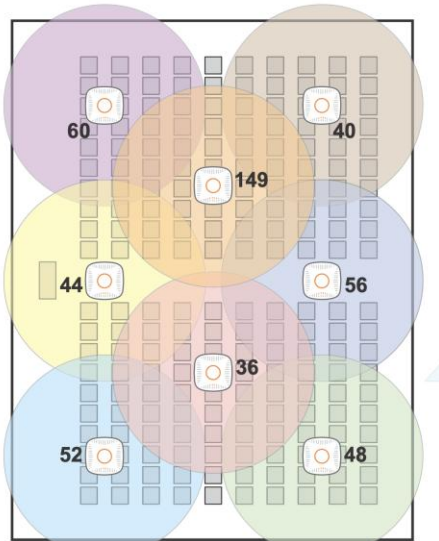
Sticky Clients



# Coverage vs Capacity design

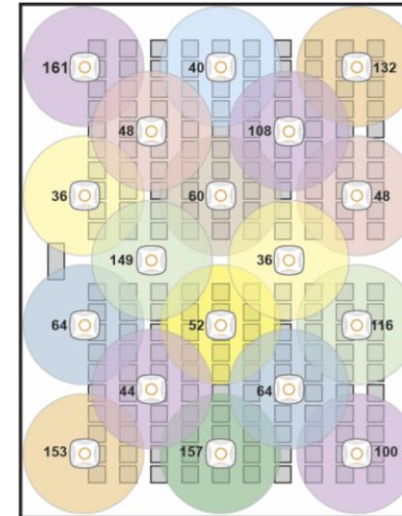
## Coverage designs:

- Fewer APs
- APs set to higher power
- Typical: Few clients
- Low Throughput needs



## Capacity design:

- More APs
- APs set to lower power
- Typical: Many clients
- High aggregate throughput needs



# AP platforms for comprehensive campus and branch connectivity

Wi-Fi 7

Wi-Fi 6E

Wi-Fi 6

AOS-10  
BASED



## Mid-range Wi-Fi 7 730 Series (AP-734/735)

Tri-band support  
2x2:2SS (8RU), 9.4Gbps,  
2x 5GE, 2xUSB, embedded GPS &  
barometer, 2 x IoT radios  
Ultra tri-band filtering

AOS-10  
BASED



## Flagship Wi-Fi 7 750 Series (AP-734/735)

Tri-band support  
4x4:4SS (8RU), 18.8Gbps,  
2x 10GE, 2xUSB, embedded GPS &  
barometer, 2 x IoT radios  
Ultra tri-band filtering



## Entry level Wi-Fi 6E 610 Series (AP-615)

Dual-radio/tri-band  
2x2:2SS (8RU), 3.6Gbps,  
1x 2.5GE, USB, embedded GPS



## Mid-range Wi-Fi 6E 630 Series (AP-634/635)

Tri-band support  
2x2:2SS (8RU), 3.9Gbps,  
2x 2.5GE, USB, embedded GPS, UTB



## Flagship Wi-Fi 6E 650 Series (AP-654/655)

Tri-band support  
4x4:4SS (8RU), 7.8Gbps,  
2x 5.0 GE, USB, embedded GPS, UTB



## Entry level Wi-Fi 6 500 Series (AP-503)

Dual-radio 2x2:2SS, 1.5Gbps, 1x  
1GE, USB



## Entry level Wi-Fi 6 500 Series (AP-505/504)

Dual-radio 2x2:2SS (8RU), 1.5Gbps, 1x  
1GE, USB, DC, BLE/Zigbee, IPM, FTM



## Mid-range Wi-Fi 6 510 Series (AP-515/514)

Dual-radio 2x2:2SS (2.4GHz), 4x4:4SS (5GHz),  
2.7Gbps, 1x2.5GE + 1xGE, USB, DC, BLE/Zigbee,  
IPM, FTM



## High-performance Wi-Fi 6 530 Series (AP-535/534)

Dual-radio 4x4:4SS, 3.98Gbps, 2x 5GE, USB,  
DC, BLE/Zigbee, IPM, FTM



## Flagship Wi-Fi 6 550 Series (AP-555/554)

Dual-radio 4x4:4SS (2.4GHz) 4x4:4SS x2 or  
8x8:8SS (5GHz), 5.4Gbps, 2x 5GE, USB, DC,  
BLE/Zigbee, IPM, FTM

< Low-density to high-density, demanding environments >



# AP platforms **purpose-built** for hospitality and remote work

Wi-Fi 6E

Wi-Fi 6

AOS-10  
BASED



## Entry level remote Wi-Fi 6 500R Series

Dual-radio 2x2:2SS (8RU),  
100Mbps crypto, 1.5Gbps  
1x 1GE + 2x 1GE, USB, DC,  
desk mount  
**(AOS 10 only)**



## Wi-Fi 6 RAP 500H Series

Dual radio 2x2:2SS 802.11ax  
(8RU), 100Mbps crypto, 1.5Gbps  
1x 1GE + 2x 1GE, BLE /  
802.15.4, 802.3af, DC, wall/desk



## Wi-Fi 6E 600H Series

Dual radio 2x2:2SS 802.11ax  
(8RU), 500Mbps crypto, 3.6Gbps  
1x 2.5GE + 1x 1GE, 3x PSE,  
USB, BLE / 802.15.4, IPM  
802.3af/at/bt, DC, wall/desk

AOS-10  
BASED



## High performance Wi-Fi 6E

### 600R Series Remote AP with modular cellular radio support

Dual-radio / tri-band 2x2:2SS (8RU),  
500Mbps crypto, 3.6Gbps  
1x 2.5GE + 4x 1GE, 1x PSE, USB,  
BLE/ 802.15.4, DC, desk mount  
Bundle option  
**(AOS 10 only)**

<< Entry-level to advanced features for remote power users >>



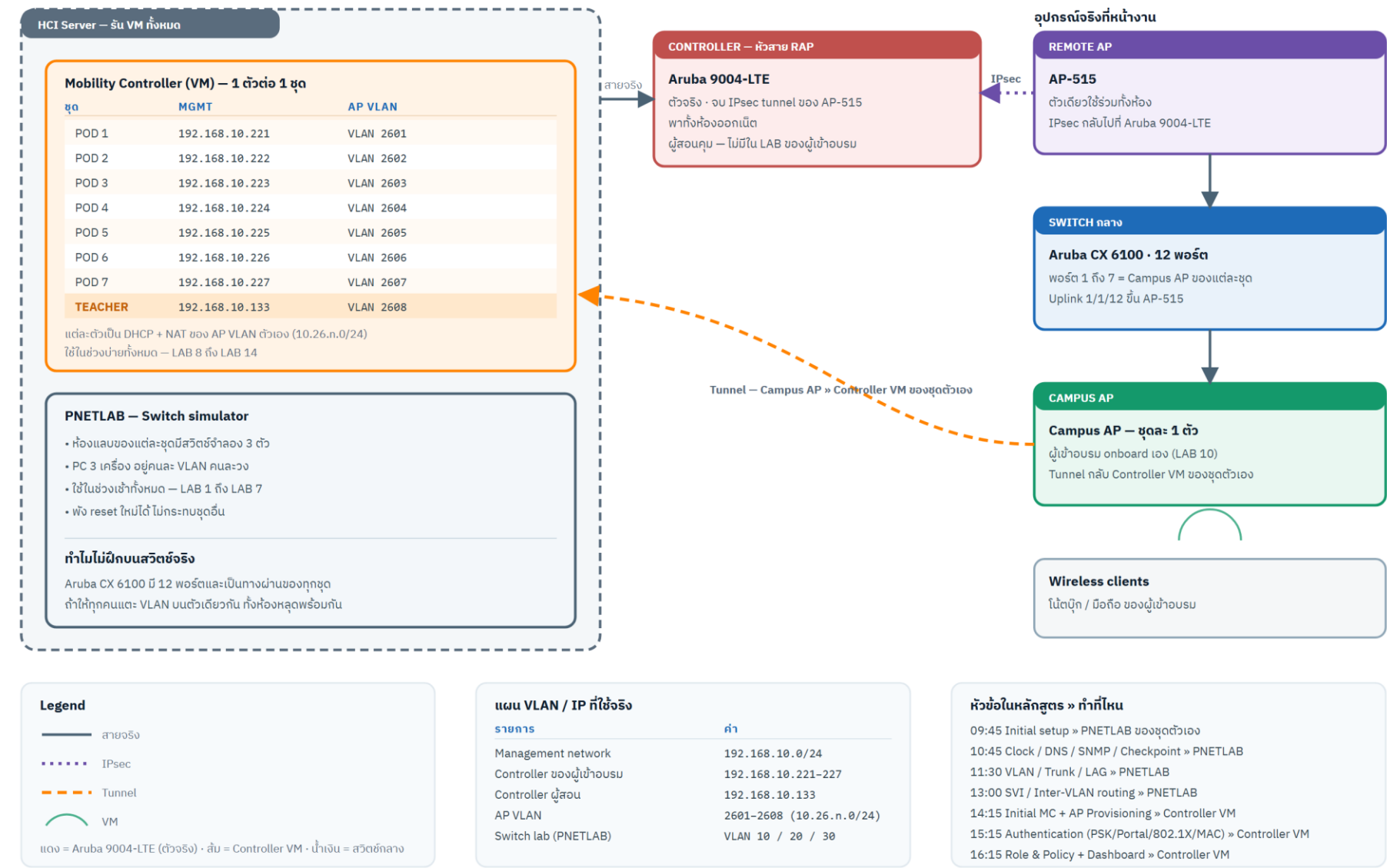
# WLAN Labs

---



# Aruba CX Switch + WLAN Basic – Lab Topology

IT Green — Authorized Aruba (HPE) Networking Training · 3 ก.ย. 2569 · ครึ่งเช้าอยู่ใน PNETLAB · ครึ่งบ่ายอยู่บน Controller VM ของชุด



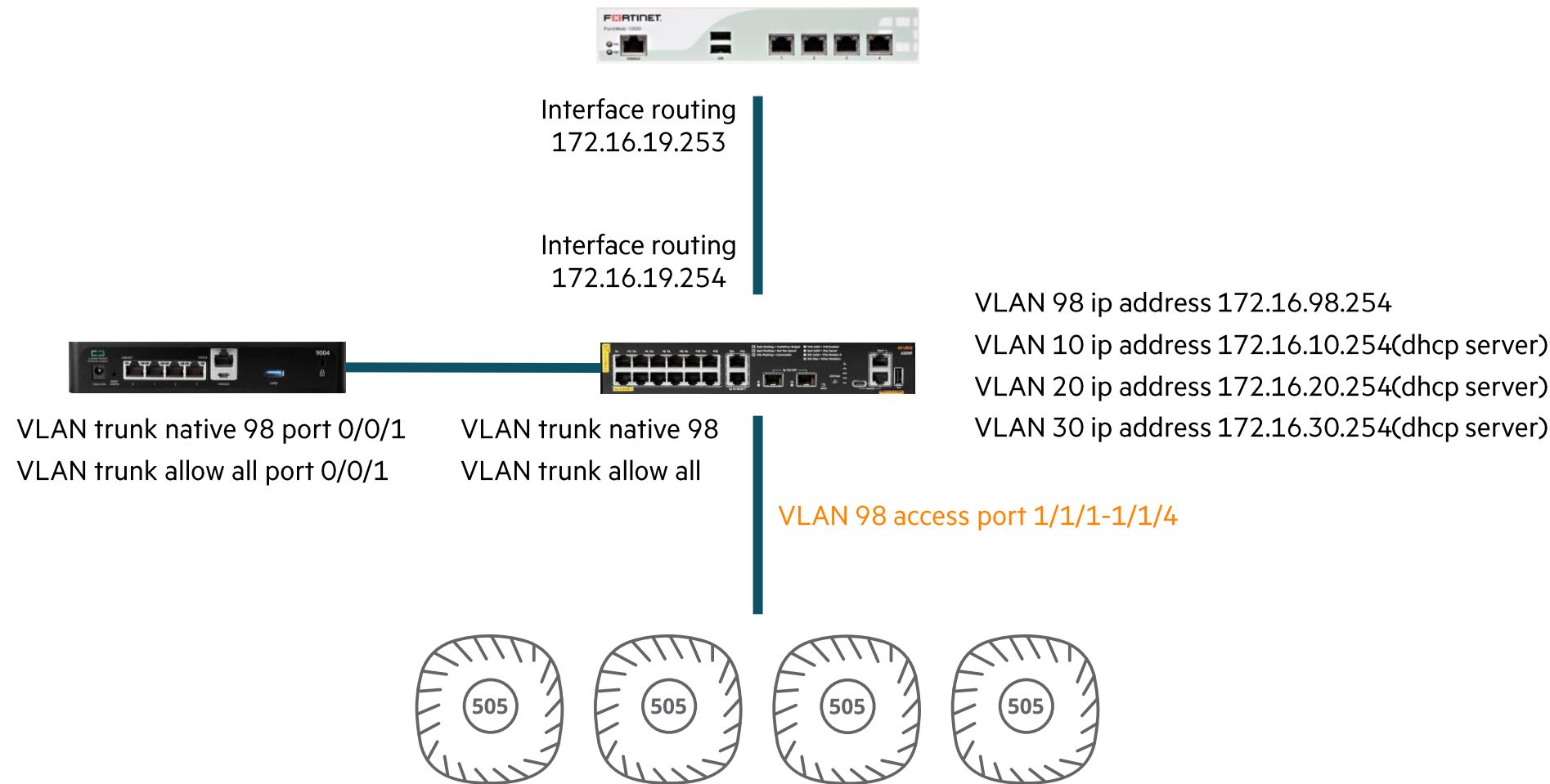


# A. Diagram

---



# Diagram





# B.Console Cable

---



# C.Initial Mobility Controller(MC)

---

- **Initial Controller**
- **Generate trail license AP & PEF**
- **Add License AP & PEF**
- **Set Time**
- **Configure CPSec**



# Enter initial command : Full-setup

```
Auto-provisioning is in progress. It requires DHCP and Activate servers
Choose one of the following options to override or debug auto-provisioning...
'enable-debug' : Enable auto-provisioning debug logs
'disable-debug' : Disable auto-provisioning debug logs
'mini-setup' : Start mini setup dialog. Provides minimal customization and requires DHCP server
'full-setup' : Start full setup dialog. Provides full customization
'static-activate' : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): full-setup

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): yes
```

Auto-provisioning is in progress. It requires DHCP and Activate servers  
Choose one of the following options to override or debug auto-provisioning....

- 'enable-debug' : Enable auto-provisioning debug logs
- 'disable-debug' : Disable auto-provisioning debug logs
- 'mini-setup' : Start mini setup dialog. Provides minimal customization and requires DHCP server
- ' **full - setup** ' : Start full setup dialog. Provides full customization
- 'static-activate' : Provides customization for static or PPPoE ip assignment. Uses activate for conductor information

Enter Option (partial string is acceptable): **full-setup**

Are you sure that you want to stop auto-provisioning and start full setup dialog? (yes/no): **yes**



# Initial MC-0X

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help  
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end  
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line  
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_Group99
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]:
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 192.168.98.253
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 192.168.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los_Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00
Enter Date (MM/DD/YYYY) [6/14/2025]:
Enter Password for admin login (up to 32 chars): *****
Re-type Password for admin login: *****
```

Commands: <Enter> Submit input or use [default value], <ctrl-I> Help  
<ctrl-B> Back, <ctrl-F> Forward, <ctrl-A> Line begin, <ctrl-E> Line end  
<ctrl-D> Delete, <BackSpace> Delete back, <ctrl-K> Delete to end of line  
<ctrl-P> Previous question <ctrl-X> Restart beginning <ctrl-R> Reload box

```
Enter System name [Aruba7210_01_B1_F8]: ArubaMC_GroupXX
Enter Switch Role (conductor|standalone|md) [md]: standalone
Enter Controller VLAN ID [1]: 98
Enter Controller VLAN port [GE 0/0/0]: [Enter]
Enter Controller VLAN port mode (access|trunk) [access]: trunk
Enter Native VLAN ID [1]: 98
Do you wish to configure IPV4 address on vlan (yes|no) [yes]: yes
Enter VLAN interface IP address [172.16.0.254]: 172.16.98.1-4 //เรียงตาม group
Enter VLAN interface subnet mask [255.255.255.0]: 255.255.255.0
Enter IP Default gateway [none]: 172.16.98.254
Enter DNS IP address [none]: 8.8.8.8
Do you wish to configure IPV6 address on vlan (yes|no) [yes]: no
Enter Country code (ISO-3166), <ctrl-I> for supported list: TH
You have chosen Country code TH for Thailand (yes|no)?: yes
Enter the controller's IANA Time zone [America/Los Angeles]: Asia/Bangkok
Enter Time in UTC [07:01:44]: 14:04:00 //Current Time HH:MM:SS
Enter Date (MM/DD/YYYY) [6/14/2025]: [Enter] // Check Date Current
Enter Password for admin login (up to 32 chars): P@ssw0rd
Re-type Password for admin login: P@ssw0rd
```

# Hierarchy

## Mobility Conductor

Leave as default  
configuration

**ArubaOS\_MM01\_Handon**

VLAN/IP Address

**ArubaOS\_MM02\_Handon**

VLAN/IP Address

## Manage Network

Leave as default  
configuration

**ArubaOS\_GroupHQ**

Company-wide  
settings, eg Corp  
SSID

**ArubaOS\_GroupBr**

Company-wide  
settings, eg Corp  
SSID

**ArubaOS\_MC01\_HQ**

VLAN/IP Address

**ArubaOS\_MC02\_HQ**

VLAN/IP Address

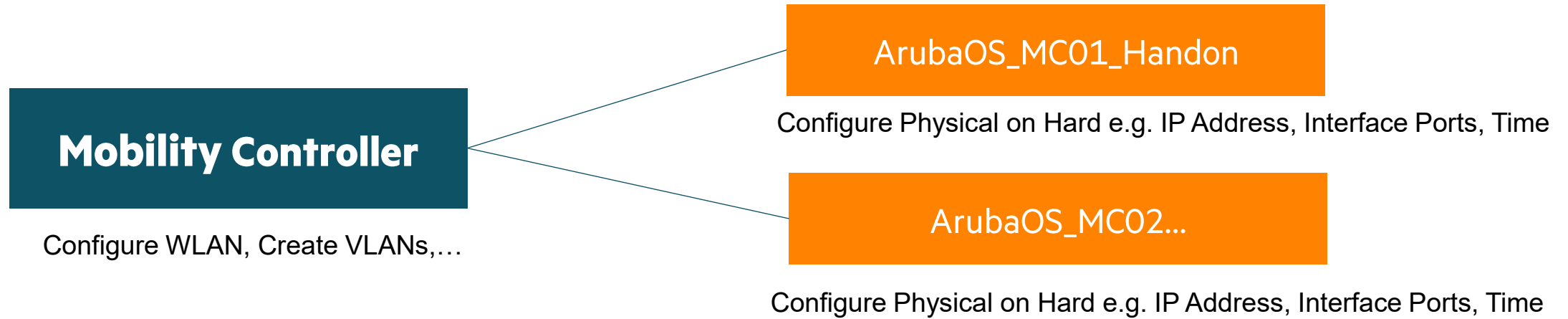
**ArubaOS\_MC01\_Br**

VLAN/IP Address

**ArubaOS\_MC02\_Br**

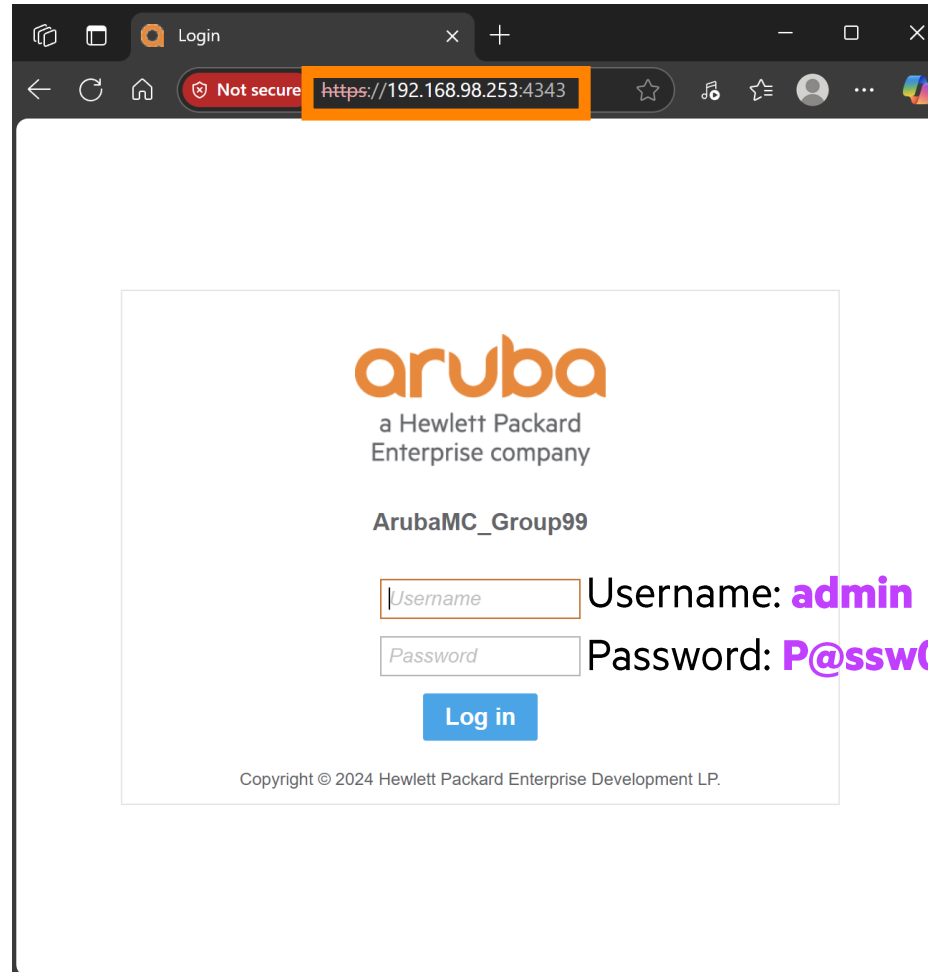
VLAN/IP Address

# Hierarchy



# Add License

Enter Web Browser URL >> **https://172.16.98.(1-4):4343**



The screenshot shows a web browser window with the title "Login". The address bar displays "https://192.168.98.253:4343" with a "Not secure" warning. The main content area features the Aruba logo, the text "a Hewlett Packard Enterprise company", and the identifier "ArubaMC\_Group99". Below this are input fields for "Username" and "Password", a "Log in" button, and a copyright notice: "Copyright © 2024 Hewlett Packard Enterprise Development L.P.".

Username: **admin**

Password: **P@ssw0rd**

# Add License

**aruba** MOBILITY CONTROLLER ArubaMC\_Group99

ACCESS POINTS 0 0 CLIENTS 0 0 0 ALERTS 0

admin

**1** Mobility Controller

Dashboard

**2** Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

**3** License

Redundancy

Connect to external license server: ☐

**Usage**

| Usage Summary                | Usage by This Controller            |                                    |                                             | Usage by Other Controller    |                                     |                                     |
|------------------------------|-------------------------------------|------------------------------------|---------------------------------------------|------------------------------|-------------------------------------|-------------------------------------|
|                              | AP<br>Access Points                 | PEF<br>Policy Enforcement Firewall | RF Protect<br>Wireless Intrusion Protection | ACR<br>Advanced Cryptography | WebCC<br>Web Content Classification | VIA<br>Virtual Intranet Access      |
| Feature Enabled              | <input checked="" type="checkbox"/> | <input type="checkbox"/>           | <input type="checkbox"/>                    | <input type="checkbox"/>     | <input type="checkbox"/>            | <input checked="" type="checkbox"/> |
| Scope                        | Per-AP                              | Per-AP                             | Per-AP                                      | Per-Session                  | Per-AP                              | Per-Session                         |
| Licenses Installed           | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Expired Licenses             | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Active Licenses              | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Licenses Used                | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |
| Licenses Remaining Available | 0                                   | 0                                  | 0                                           | 0                            | 0                                   | 0                                   |

**4** > Inventory

# Set Time

**1**

Mobility Controller

Dashboard

**Configuration**

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

ArubaMC\_Group99

**2**

Admin AirWave CPSec Certificates SNMP Logging Profiles

**Basic Info**

Password for user admin:

Retype password:

> Domain Name System



# CPSec with auto cert provisioning

The screenshot displays the Aruba Mobility Controller web interface. The interface is divided into a left sidebar and a main content area. The sidebar contains a menu with the following items: Dashboard, Configuration (highlighted in orange), WLANs, Roles & Policies, Access Points, AP Groups, Authentication, Services, Interfaces, System (highlighted in orange), Tasks, License, and Redundancy. The main content area has a top navigation bar with tabs: General, Admin, AirWave, CPsec (highlighted in orange), Certificates, SNMP, Logging, and Profiles. Below the tabs, the 'Control Plane Security' section is expanded, showing three toggle switches: 'Enable CPsec' (checked), 'Enable auto cert provisioning' (checked), and 'Only accept APs from specified ranges' (unchecked). A link for 'Cluster for Allowlist Propagation' is also visible. In the top right corner, there is a 'Pending Changes' button with a refresh icon. Numbered callouts (1-6) are placed over the interface: 1 points to the 'Mobility Controller' header, 2 points to the 'Configuration' menu item, 3 points to the 'System' menu item, 4 points to the 'CPsec' tab, 5 points to the 'Enable auto cert provisioning' toggle, and 6 points to the 'Pending Changes' button.

1 Mobility Controller

2 Configuration

3 System

4 CPsec

5 Enable auto cert provisioning

6 Pending Changes



# D. Configure VLANs

---

- Create VLANs
- Configure VLAN Interface
- Assign VLANs on port



# Create VLANs

The screenshot displays the Aruba Mobility Controller web interface. At the top, the header includes the Aruba logo, the system name 'ArubaMC\_Group99', and status indicators for Access Points (0), Clients (0), and Alerts (0). The user is logged in as 'admin'. The left sidebar shows a menu with 'Configuration' and 'Interfaces' highlighted. The main content area shows the 'VLANs' tab selected, with a table containing one entry (ID 1) and a '+ 5' button to add more.

**MOBILITY CONTROLLER**  
ArubaMC\_Group99

**ACCESS POINTS** 0 0  
**CLIENTS** 0 0 0  
**ALERTS** 0

**Mobility Controller**

Dashboard  
**Configuration**  
WLANs  
Roles & Policies  
Access Points  
AP Groups  
Authentication  
Services  
**Interfaces**  
System  
Tasks  
License  
Redundancy

Ports **VLANs** IP Routes GRE Tunnels Pool Management OSPF Multicast

**VLANs**

| NAME | ID(S) |
|------|-------|
| --   | 1     |

+ 5

# Create VLANs

The image shows the Aruba Mobility Controller web interface. A 'New Vlan' dialog box is open, with three numbered orange circles indicating the steps: 1. Enter the VLAN name 'Client'. 2. Enter the VLAN ID/Range '10,20,30'. 3. Click the 'Submit' button. The background interface includes a top navigation bar with 'aruba' logo, 'MOBILITY CONTROLLER ArubaMC\_Group99', and status indicators for ACCESS POINTS, CLIENTS, and ALERTS. A left sidebar lists navigation options: Dashboard, Configuration (with sub-items: WLANs, Roles & Policies, Access Points, AP Groups, Authentication, Services), Interfaces (with sub-items: System, Tasks, License, Redundancy), Diagnostics, and Maintenance. The main content area shows a table with one entry for VLAN 1 and a '+ Add' button.

**aruba** MOBILITY CONTROLLER ArubaMC\_Group99

ACCESS POINTS 0 0 CLIENTS 0 0 ALERTS 0

admin

Mobility Controller >

Dashboard

**Configuration**

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services

**Interfaces**

- System
- Tasks
- License
- Redundancy

Diagnostics

Maintenance

**New Vlan**

VLAN name: Client **1**

VLAN ID/Range: 10,20,30 **2**

Cancel Submit **3**

|    | 1 |
|----|---|
| -- |   |

+ Add

# Configure VLAN Interface

Mobility Controller > **ArubaMC\_Group99**

Dashboard

**Configuration**

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces**
- System
- Tasks
- Redundancy
- IoT

Diagnostics

Maintenance

Ports **VLANs** Routes IPv6 Neighbors GRE Tunnels Pool Management

**VLANs**

| NAME   | ID(S)    |  |
|--------|----------|--|
| Client | 10,20,30 |  |
| --     | 1,98     |  |

**VLANs > Client**

| ID | IPV4 AD... | IPV6 AD... | ENABLE ... | PORT M... | ADMIN ... | OPERATI... | PD CLIE... | DHCP SE... |
|----|------------|------------|------------|-----------|-----------|------------|------------|------------|
| 10 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |
| 20 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |
| 30 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |

# Configure VLAN Interface

Ports **VLANs** IP Routes IPv6 Neighbors GRE Tunnels Pool Management

VLANs > Client

| ID | IPv4 AD... | IPv6 AD... | ENABLE ... | PORT M... | ADMIN ... | OPERATI... | PD CLIE... | DHCP SE... |                                                                                     |
|----|------------|------------|------------|-----------|-----------|------------|------------|------------|-------------------------------------------------------------------------------------|
| 10 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |  |
| 20 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |                                                                                     |
| 30 | --         | --         | --         | 0/0/0     | --        | --         | Disabled   | None       |                                                                                     |

+

Port Members IPv4 IPv6 More

IP Address Assignment

IP assignment: Static

IP address: 192.168.10.253

Netmask: 255.255.255.0

IP DHCP settings: None

MTU:

Suppress ARP: ☐

> IGMP

Cancel Submit



# Assign VLANs on port

Mobility Controller > ArubaMC\_Group99

Dashboard  
Configuration  
WLANs  
Roles & Policies  
Access Points  
AP Groups  
Authentication  
Services  
Interfaces  
System  
Tasks  
Redundancy  
IoT  
Diagnostics  
Maintenance

Ports VLANs IP Routes IPv6 Neighbors GRE Tunnels Pool Management

Port Channel

| NAME | MEMBERS | PROTOCOL | TRUSTED | POLICY | MODE | NATIVE VLAN | TRUNK VLANS |
|------|---------|----------|---------|--------|------|-------------|-------------|
| +    |         |          |         |        |      |             |             |

Ports

| PORT     | ADMI...  | TRUST... | POLICY   | MODE   | NATIV... | ACCES... | TRUN... | SPAN... | MONI... | DESC... |
|----------|----------|----------|----------|--------|----------|----------|---------|---------|---------|---------|
| GE-0/... | Enabl... | ✓        | Not-d... | trunk  | 98       | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |
| GE-0/... | Enabled  | ✓        | Not-d... | access | 1        | 1        | 1-4094  | ✓       | --      | --      |

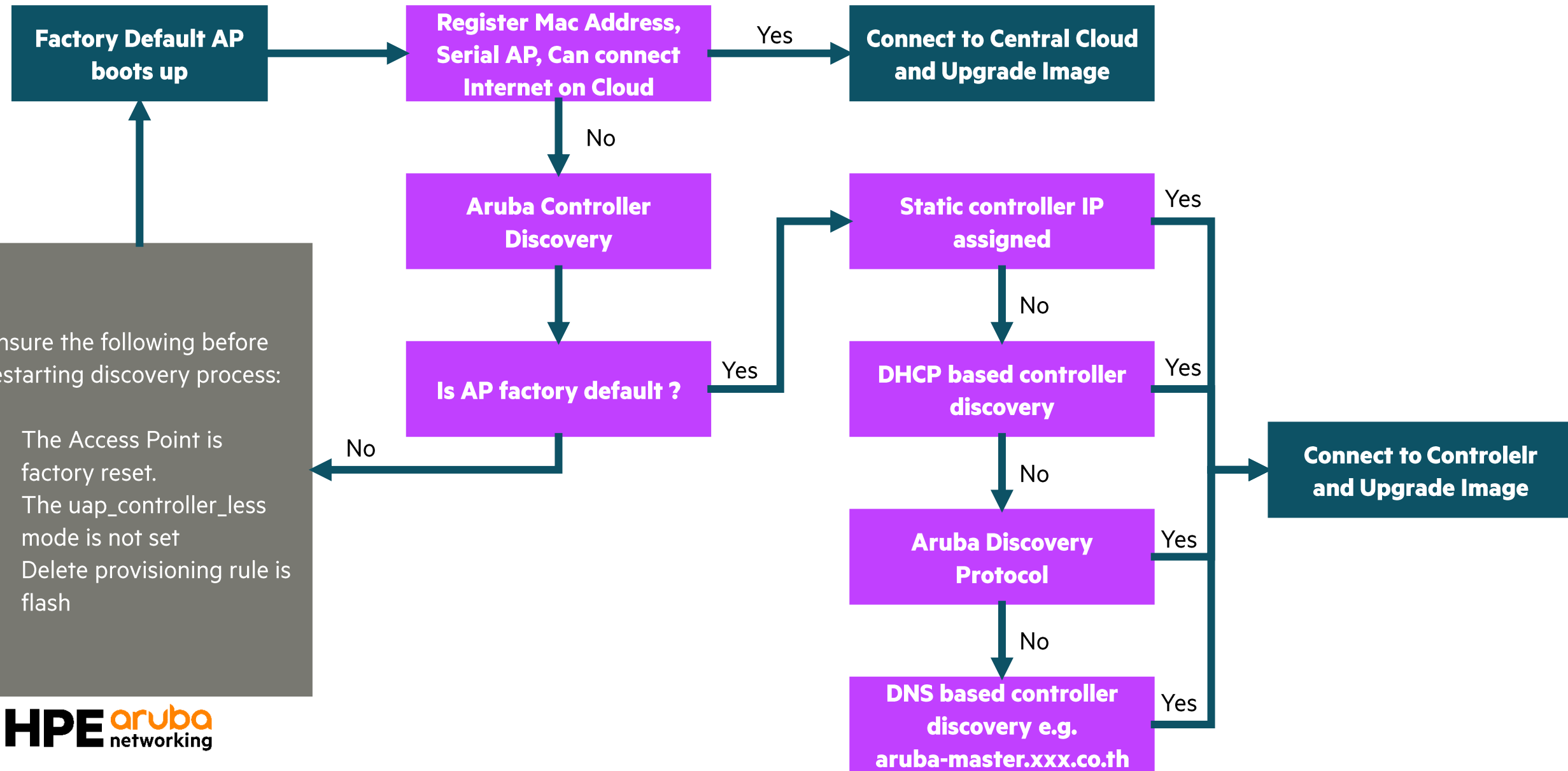


# **E.Initial Access Point(AP)**

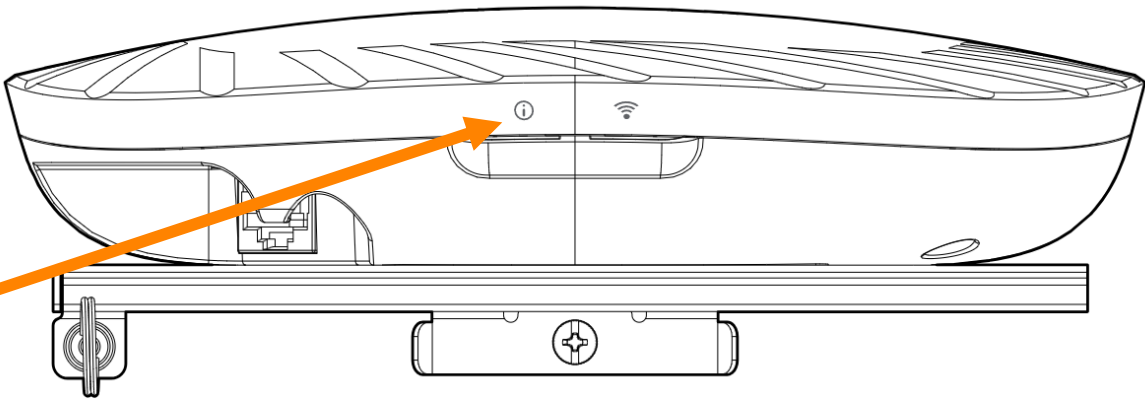
- **Understand Flow Unified**
- **LED Status**
- **Reset Setup**
- **AP convert to Controller**



# Understand Flow Unified AP



# LED Status

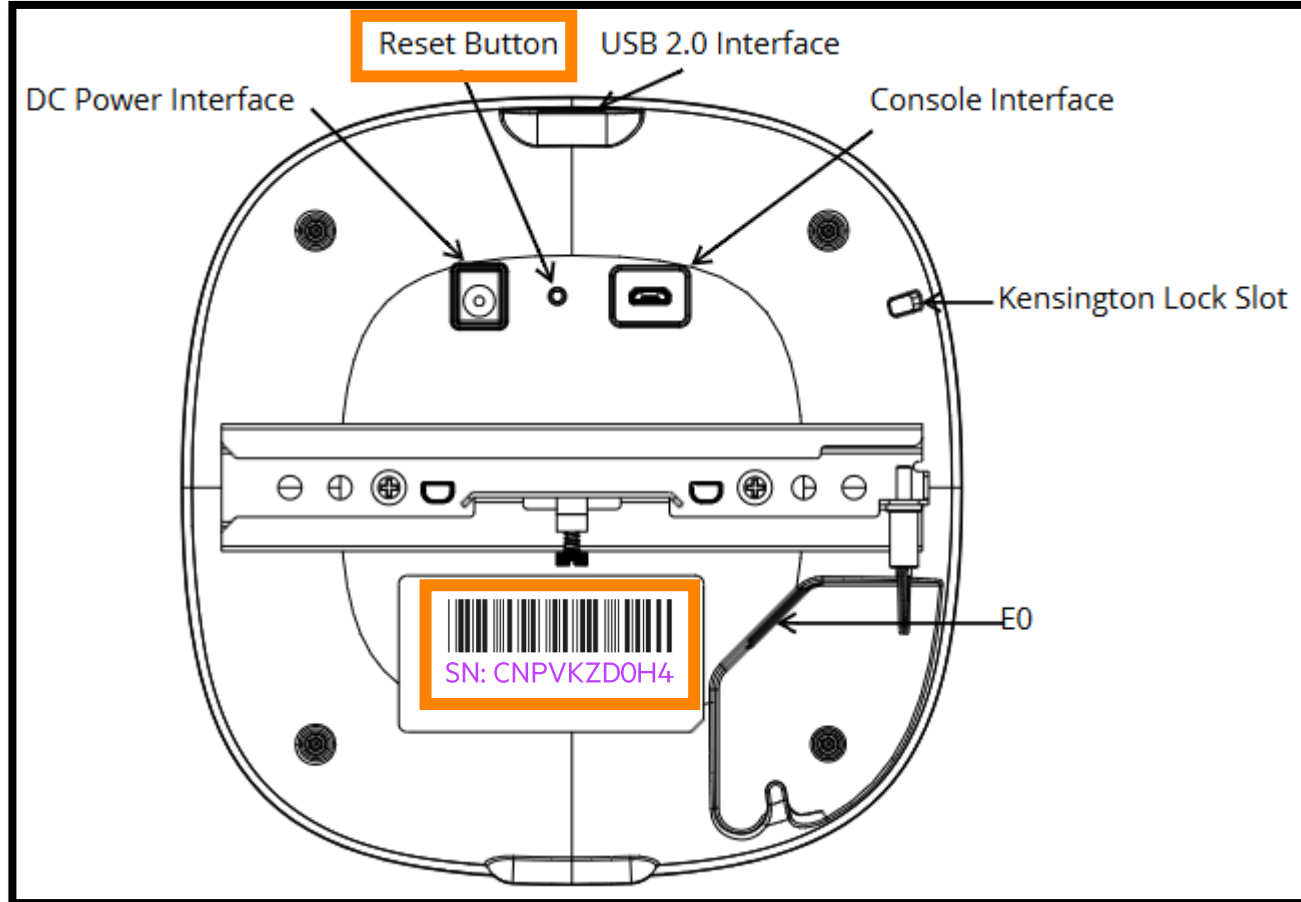


System Status LED ⓘ

| Color/State                    | Meaning                                                                                                         |
|--------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Off                            | Device Powered off                                                                                              |
| Green - solid                  | Device ready, fully functional, no network restrictions                                                         |
| Green - blinking <sup>1</sup>  | Device booting, not ready                                                                                       |
| Green - flash on <sup>2</sup>  | Device ready, fully functional, uplink negotiated in sub-optimal speed (<1Gbps)                                 |
| Green - flash off <sup>3</sup> | Device in deep-sleep mode                                                                                       |
| Amber - solid                  | Device ready, restricted power mode (IPM restrictions applied), no network restrictions                         |
| Amber - flashing off           | Device ready, restricted power mode (IPM restrictions applied), uplink negotiated in sub-optimal speed (<1Gbps) |
| Red                            | System error conditions – Immediate action required                                                             |

- 1. Blinking: one second on, one second off, 2 seconds cycle.
- 2. Flashing off: mostly on, fraction of a second off, 2 seconds cycle.
- 3. Flashing on: mostly off, fraction of a second on, 2 seconds cycle.

# Reset Setup



## Opt-1 Reset the AP during normal operation

press and hold down the reset button using a small, narrow object such as a paper clip for more than 10 seconds during normal operation.

## Opt-2 Reset the AP while powering up

1. Press and hold down the reset button using a small, narrow object such as a paper clip while the access point is not powered on (either via DC power or PoE).
2. Connect the power supply (DC or PoE) to the access point while the reset button is being held down.
3. Release the reset button on the access point after 15 seconds.

## Next Step

**Access Point broadcast SSID : SetMeUp-XX:XX:XX**

 SetMeUp-C2:69:D4

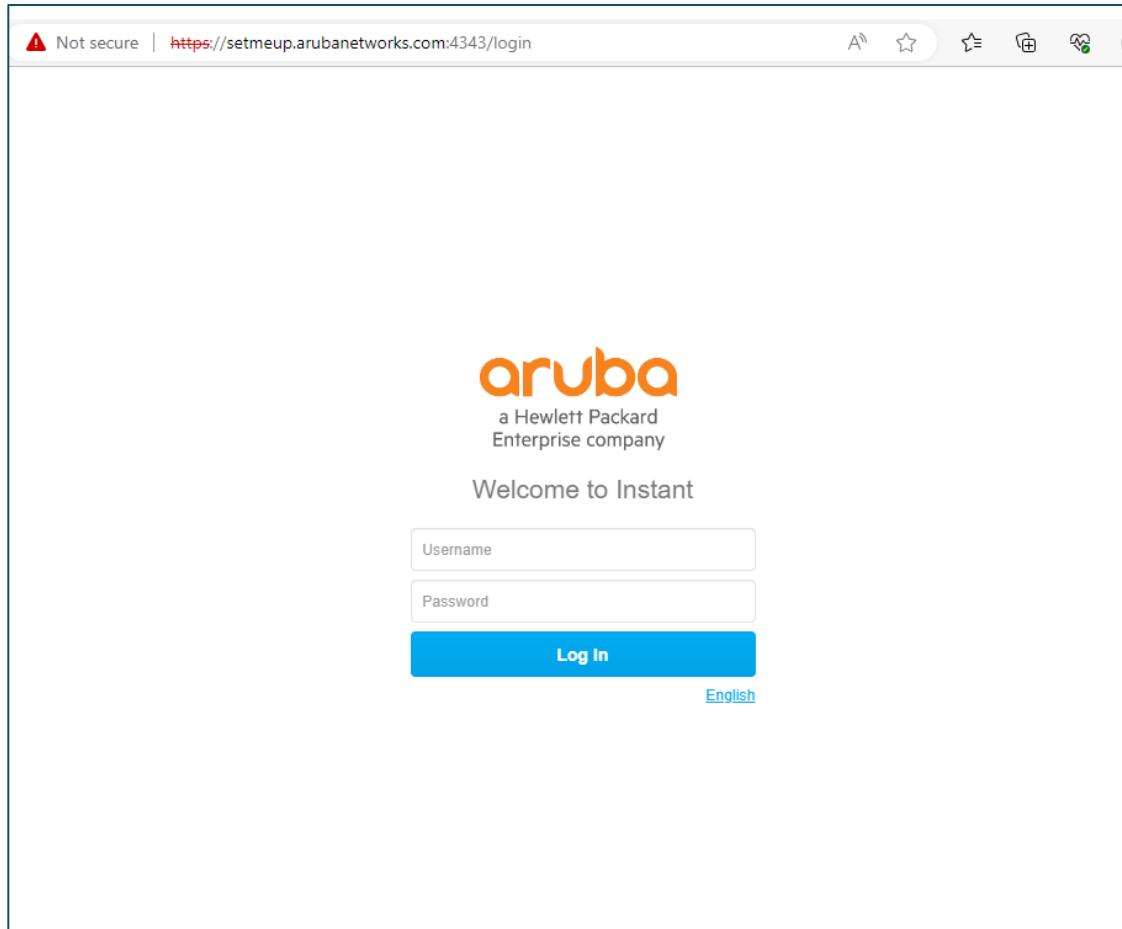
## Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)

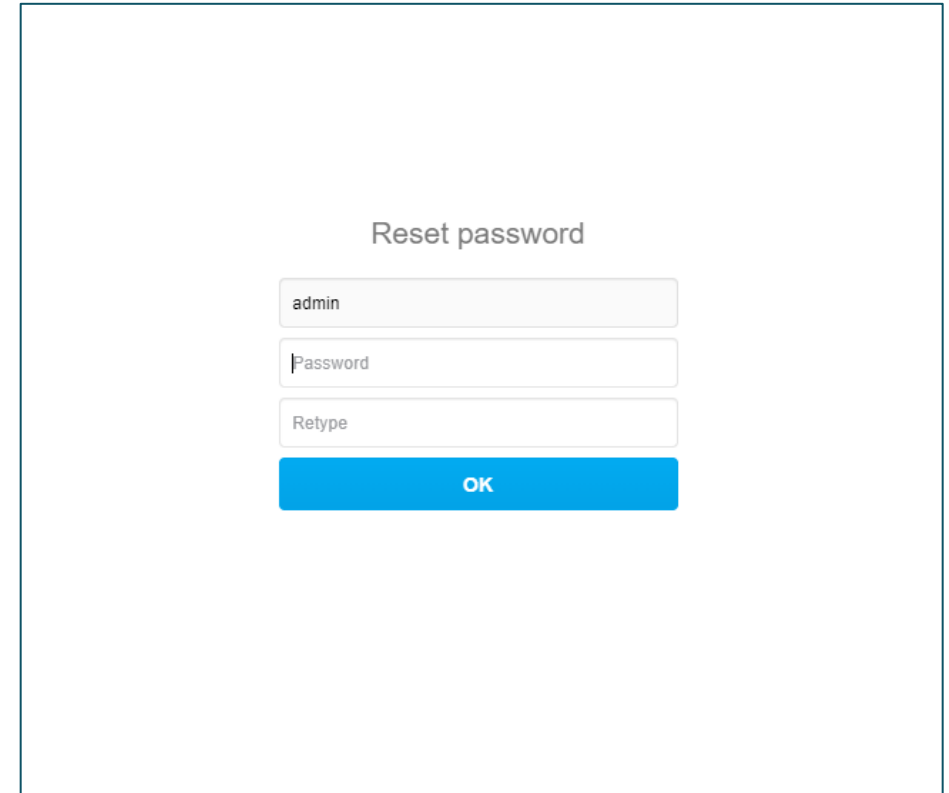
# Login GUI and Changes password

## Default User/Pass

Username : **admin** Password : **CNPVKZD0H4** //S/N (Uppercase)



A screenshot of a web browser showing the Aruba login page. The browser's address bar displays "https://setmeup.arubanetworks.com:4343/login". The page features the Aruba logo (a Hewlett Packard Enterprise company) and the text "Welcome to Instant". Below this, there are input fields for "Username" and "Password", followed by a blue "Log In" button. A link for "English" is visible at the bottom right of the login area.



A screenshot of the "Reset password" page. It contains three input fields: the first is pre-filled with "admin", the second is labeled "Password", and the third is labeled "Retype". Below these fields is a blue button labeled "OK".

Enter new **Password: P@ssw0rd** and **Retype password: P@ssw0rd**



# Select Country Code

Select : **TH-Thailand**

Overview

Info

Networks

Access Points

Clients

Name

SetMeUp-C2:69:D4

Country code

Management

Uplink type

Welcome to Instant

Please specify the Country Code

TH - Thailand

1

RO - Romania

RS - Serbia

RU - Russia

RW - Rwanda

SA - Saudi Arabia

SE - Sweden

SG - Singapore

SH - Saint Helena

SI - Slovenia

SJ - Svalbard and Jan Mayen

SK - Slovak Republic

SM - San Marino

SN - Senegal

SR - Suriname

SS - South Sudan

SV - El Salvador

SX - Sint Maarten

TC - Turks And Caicos Islands

TF - French Southern Territories

TH - Thailand

please try [Upgrading your AP](#) to the latest version and try

ision, please contact technical support for mor

**Dashboards-> Overview**

aruba

VIRTUAL CONTROLLER

SetMeUp-C2:69:D4

Search

Alerts

Help

Profile

Dashboard

Overview

Networks

Access Points

Clients

Mesh Devices

Configuration

Maintenance

Support

Overview

Networks

Access Points

Clients

Mesh Devices

Active

Inactive

Up

Down

Wireless

Wired

1

0

1

0

1

0

Dashboard

IDS

AirGroup

0 Alerts

Clients

2

1

0

12:28

12:29

12:30

12:31

12:32

Throughput (bps)

1M

10K

100

0

100

10K

1M

12:28

12:29

12:30

12:31

12:32

Out

In

RF Dashboard

Clients

Signal

Speed

Access Points

Utilization

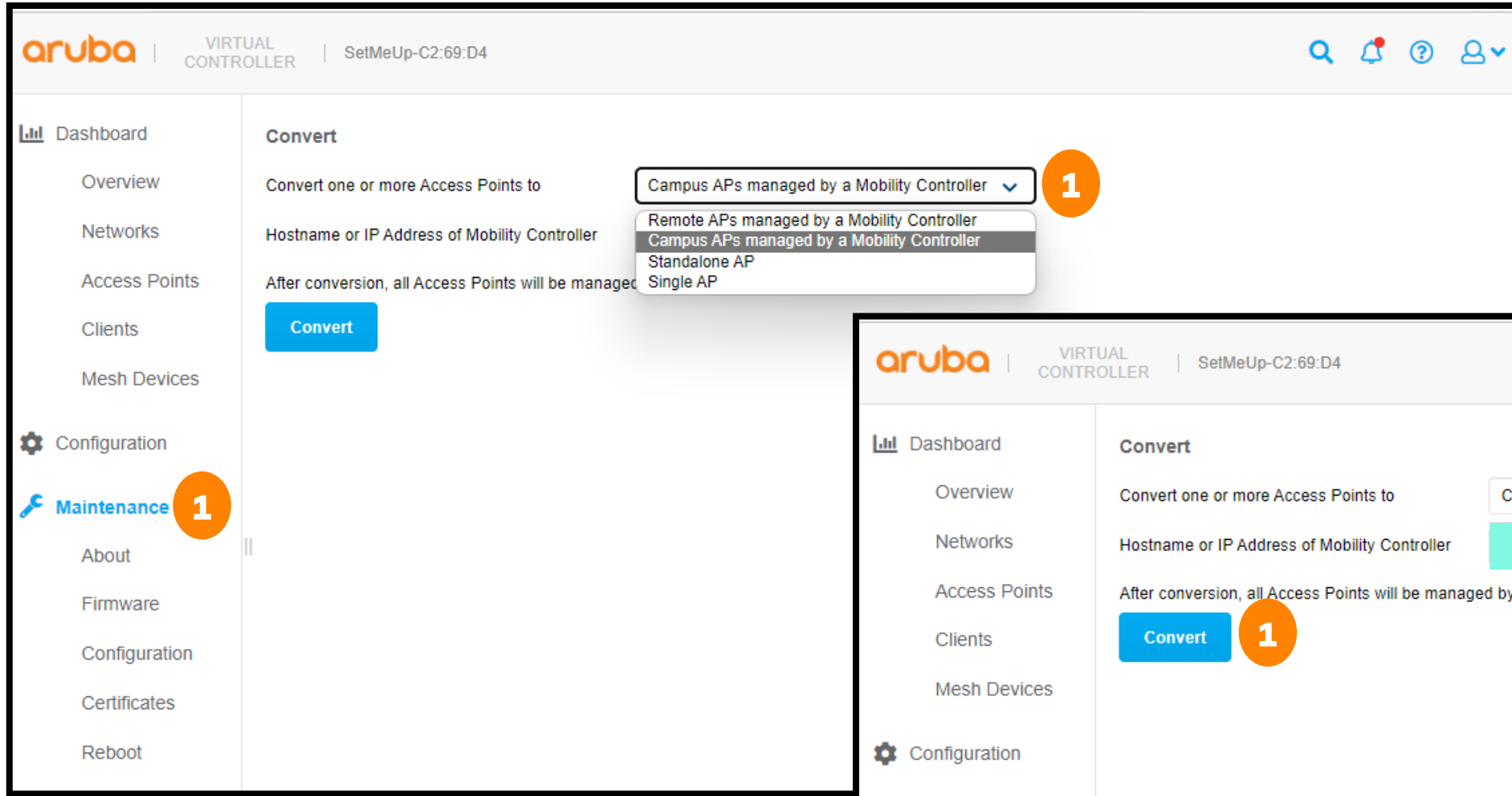
Noise

Errors

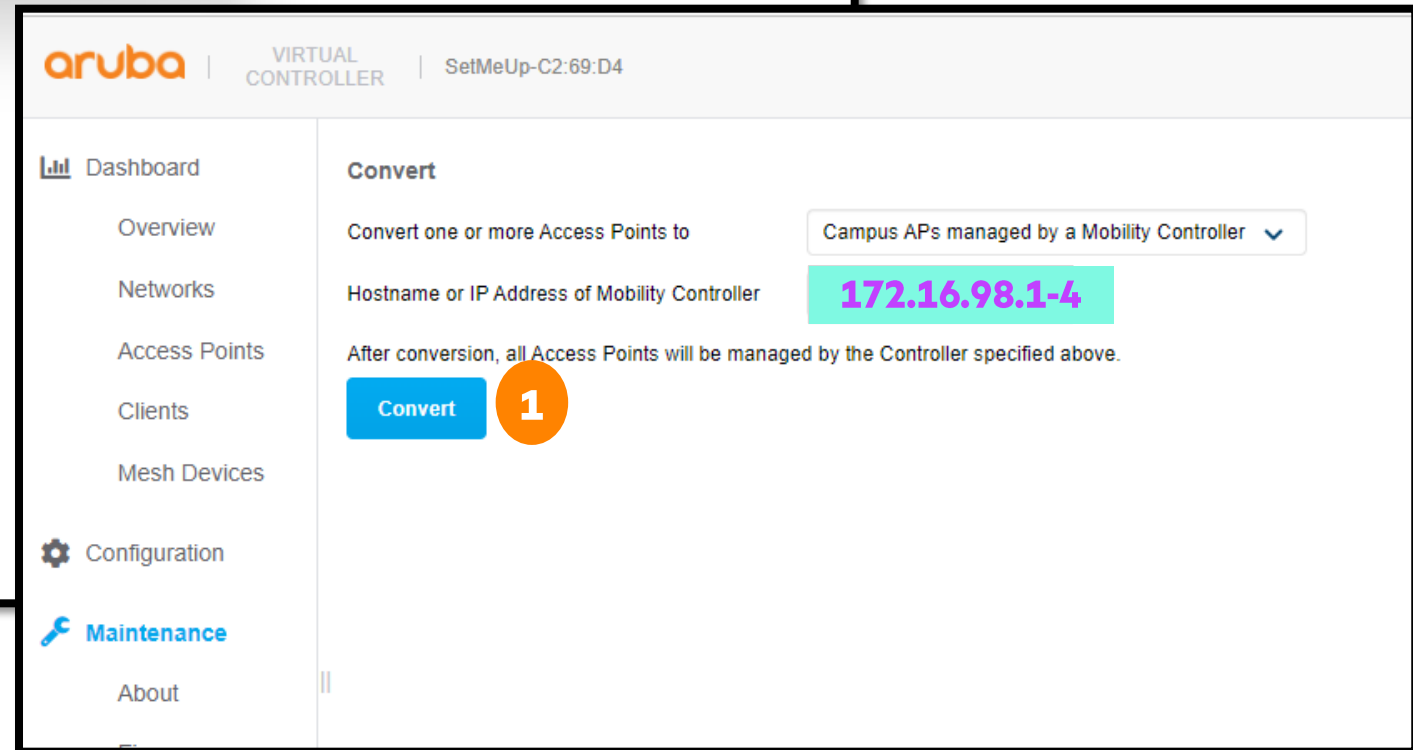
All Clients

9c:8c:d8:c2:6...

# Convert Access Point Campus mode

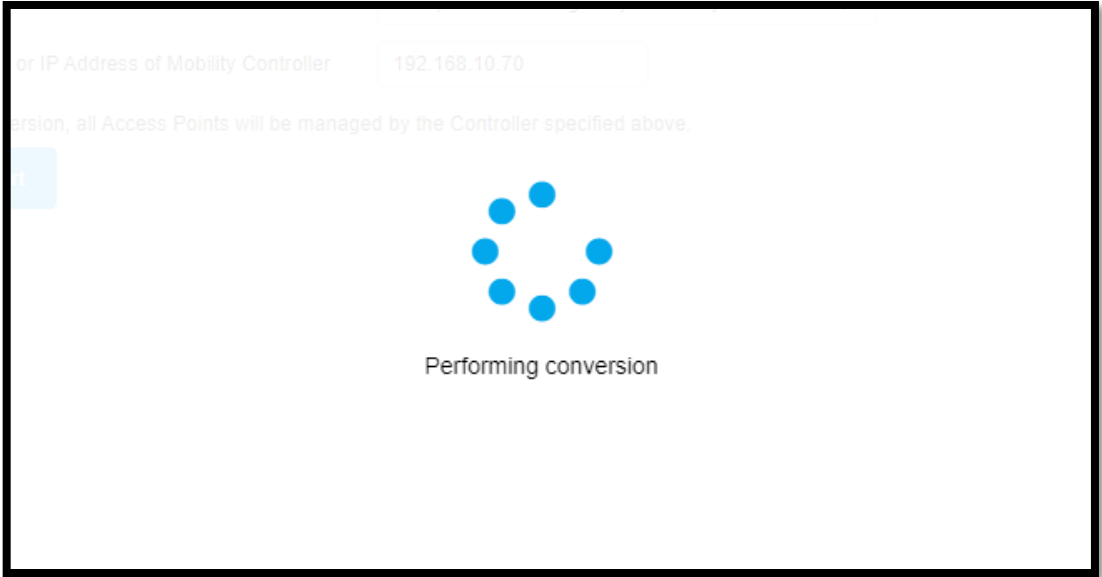
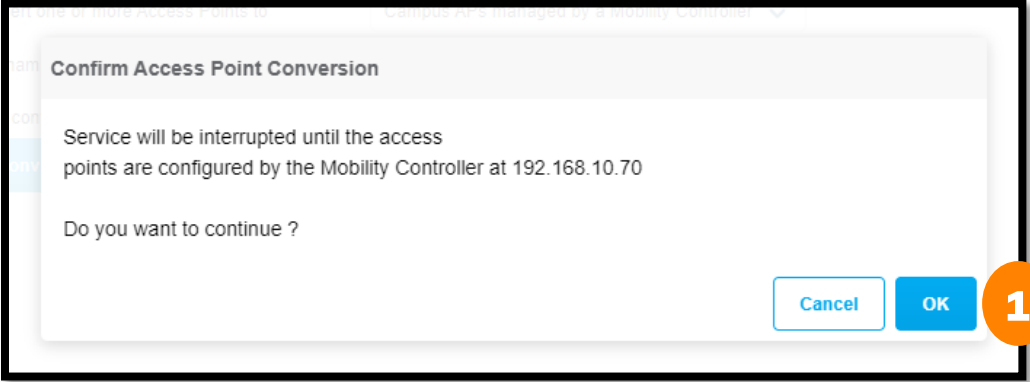


The screenshot shows the Aruba Virtual Controller web interface. The top header includes the Aruba logo, 'VIRTUAL CONTROLLER', and the device ID 'SetMeUp-C2:69:D4'. The left sidebar contains navigation links: Dashboard, Overview, Networks, Access Points, Clients, Mesh Devices, Configuration, and Maintenance (highlighted with a blue circle and the number '1'). The main content area is titled 'Convert' and contains the following text: 'Convert one or more Access Points to', 'Hostname or IP Address of Mobility Controller', and 'After conversion, all Access Points will be managed by the Controller specified above.' A blue 'Convert' button is present. A dropdown menu is open, showing four options: 'Campus APs managed by a Mobility Controller' (selected), 'Remote APs managed by a Mobility Controller', 'Campus APs managed by a Mobility Controller', and 'Standalone AP'. A blue circle with the number '1' is placed next to the selected option.

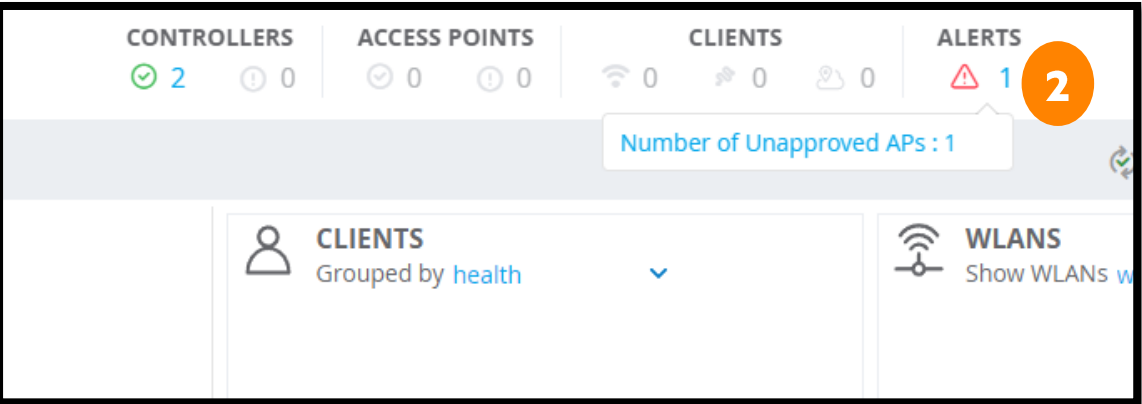


The screenshot shows the Aruba Virtual Controller web interface. The top header includes the Aruba logo, 'VIRTUAL CONTROLLER', and the device ID 'SetMeUp-C2:69:D4'. The left sidebar contains navigation links: Dashboard, Overview, Networks, Access Points, Clients, Mesh Devices, Configuration, and Maintenance (highlighted with a blue circle and the number '1'). The main content area is titled 'Convert' and contains the following text: 'Convert one or more Access Points to', 'Hostname or IP Address of Mobility Controller', and 'After conversion, all Access Points will be managed by the Controller specified above.' A blue 'Convert' button is present. The IP address '172.16.98.1-4' is entered in the 'Hostname or IP Address of Mobility Controller' field. A blue circle with the number '1' is placed next to the 'Convert' button.

# Convert Access Point Campus mode (Cont.)



Approve device on Mobility Controller(MC)  
**In-Case: enable auto cert provisioning not yet.**



# Approve Access Point to Allowlist

Top bar show one ALERTS “Number of Unapproved APs : 1”

CONTROLLERS

2

0

ACCESS POINTS

0

0

CLIENTS

0

0

0

ALERTS

1

Number of Unapproved APs : 1

1

CLIENTS

Grouped by health

WLANS

Show WLANS w

Campus APsRemote APsMesh APsAllowlistProvisioning Rules

Campus AP AllowlistRemote AP Allowlist

|              | MAC ADDRESS                              | NAME | AP GROUP | DESCRIPTION | STATUS   | REVOKE TEXT | APPROVED | UPDATED          |
|--------------|------------------------------------------|------|----------|-------------|----------|-------------|----------|------------------|
| <div>2</div> | <div><div></div></div> 9c:8c:d8:c2:69:d4 | --   | --       | --          | Accepted | --          | No       | Mon Jul 10 12:40 |

3

Delete

Revoke

Approve

50

<

1

>



# F. Create AP Group

---

- Provision AP Change to Group



# Create Group AP

Pending Changes ↻ 1

| AP Groups 3   |     |
|---------------|-----|
| NAME          | APs |
| default       | 1   |
| Group-99      | --  |
| NoAuthApGroup | --  |

+

Pending Changes

☒ Pending Changes for 1 Group

☒  Mobility Controller

System > Profiles > AP > AP group:  
Profile Name = Group-99

Tasks > Provision new Campus APs:  
Name = Group-99

AP Groups:  
Name = Group-99

Additional changes from CLI

Close

Discard changes

Deploy changes 2



# Change AP to New Group

Mobility Controller > 1

Dashboard

Configuration 2

WLANs

Roles & Policies

Access Points 3

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Campus APs Remote APs Mesh APs Allowlist Provisioning Rules

Campus APs 1

| <input checked="" type="checkbox"/> | AP NAME       | AP GROUP | IPV4 ADD...  | IPV6 ADD... | SWITCH IP    | MAC ADD...    | SERIAL #   | T |
|-------------------------------------|---------------|----------|--------------|-------------|--------------|---------------|------------|---|
| <input checked="" type="checkbox"/> | 7c:57:3c:c... | default  | 192.168.9... | --          | 192.168.9... | 7c:57:3c:c... | CNPBK51... | 3 |

5 Provision

Flags:  
U = Unprovisioned, N = Duplicate name, G = No such group, L = Unice  
Mismatch, X = Maintenance Mode, P = PPPoE AP, B = Built-in AP, s = LA  
RAP, c = CERT-based RAP, 1 = 802.1x authenticated AP use EAP-PEAP, 1  
u = Custom-Cert RAP, S = Standby-mode AP, J = USB cert at AP, f = No S  
Mesh Recovery, z = Datazone AP, e = Custom EST cert, p = In deep-sleep  
ShutDown, F = AP failed 802.1x authentication

7c:57:3c:c3:e0:e4

MAC address: 7c:57:3c:c3:e0:e4

LLDP neighbor chassis ID / port ID: 6200 / 1/1/3

Name: 7c:57:3c:c3:e0:e4

AP group: Group-99 6 Select New Group

Controller discovery: ☒ Use AP discovery protocol (ADP) ☐ Static

Controller discovery preference: ☒ IPv4 ☐ IPv6

IP: ☒ DHCP ☐ Static

Deployment: ☒ Campus ☐ Remote ☐ Mesh ☐ Remote mesh portal

Wi-Fi uplink: ☐ Wi-Fi uplink

Cancel Submit 6

Access Points will be Rebooted

CAUTION: Applying this configuration change will interrupt service while the affected Access Points are rebooted.

Do you want to continue ?

Cancel Continue & Reboot 7

## LAB 8 Initial Setup Mobility Controller

Controller VM ของชุด ตั้ง controller ของชุดตัวเองให้ขึ้นมาพร้อมใช้ — ชื่อ, IP, country code, เวลา และ hierarchy

1. ต่อ console เข้า Controller VM ของชุดตัวเอง แล้วเรียกตัวช่วยตั้งค่าครั้งแรก  
full-setup
2. ตอบ wizard ตามนี้ — System name = MC-POD-n · Role = standalone · Country code = TH
3. ตั้ง IP ของ controller ตามชุดตัวเอง — POD n ใช้ 192.168.10.(220+n) netmask 255.255.255.0 gateway ตาม management network {{mgmtNet}} · DNS 8.8.8.8
4. ตั้งรหัส admin แล้วให้ระบบ reboot รอจนขึ้นมาใหม่
5. เข้าเว็บ https://192.168.10.22n:4343 ด้วย admin ที่เพิ่งตั้ง — จากนั้นทำผ่าน Web UI เป็นหลัก
6. ตั้งเวลาให้ตรงก่อนทำอย่างอื่น — Configuration » System » General » Clock ถ้าเวลาเพี้ยน certificate ของ AP จะไม่ผ่านแล้วจะไปติดตอน onboard  
clock timezone ICT +07:00  
ntp server pool.ntp.org
7. ดูโครง hierarchy ที่ Configuration » Network — จำไว้ว่า config ที่วางไว้ระดับบนจะไหลลงระดับล่าง

### ตรวจสอบผล

```
show switches
show clock
show ip interface brief
show license
```

### ระวัง

ตั้ง country code ผิดเป็นเรื่องใหญ่ — ช่องสัญญาณที่ใช้ได้จะไม่ตรงกับที่ กสทช. อนุญาต และแก้ทีหลังต้องรื้อ AP ใหม่ทั้งหมด ต้องเป็น TH เท่านั้น

## LAB 9 VLAN และ Interface บน Controller

Controller VM ของชุด สร้าง AP VLAN ของชุดตัวเอง ให้มี IP, DHCP และ NAT ครบ — เตรียมให้ AP มาลง

1. Configuration » Interfaces » VLANs » + สร้าง VLAN 260n (n = หมายเลขชุด) ตั้งชื่อ AP-POD-n
2. ใส่ IP ให้ VLAN นั้น — 10.26.n.1/24 คือ gateway ของ AP ในชุดตัวเอง
3. สร้าง DHCP pool ของ VLAN นี้ แจก .{{poolRange}} เวน .1–.10 และ .241–.254 lease {{leaseHours}} ชั่วโมง
4. เปิด NAT ขาเข้าให้ VLAN นี้ ออกเน็ตผ่าน uplink ของ controller
5. เอา config ที่พิมพ์แล้วจากหัวข้อ Config ด้านบนมาวางแทนก็ได้ — เลือกแท็บชุดตัวเอง กดคัดลอก แล้ววางลง CLI
6. Configuration » Interfaces » Ports — ให้พอร์ตที่ AP จะเข้ามาผ่าน VLAN 260n ได้

### ตรวจสอบผล

```
show vlan
show ip interface brief
show ip dhcp database
show running-config | include nat
```

### ระวัง

ทั้งห้องใช้เลข VLAN คนละเบอร์กันด้วยเหตุผลเดียว — วาง VLAN ของชุดอื่นลงเครื่องตัวเองแล้วจะไล่หาไม่เจอว่าทำไม AP ไม่ขึ้น เช็ค n ของตัวเองก่อนวางทุกครั้ง

## LAB 10 AP Provisioning — เอา AP ของชุดเข้ามา

Controller VM ของชุด    แปลง Campus AP ของชุดให้มาจบที่ controller ตัวเอง แล้วจัดเข้า AP group ของชุด

1. ดู LED ของ AP ก่อน — สีและจังหวะกะพริบบอกได้ว่าค้างอยู่ขั้นไหน ยังไม่ได้ IP หรือหา controller ไม่เจอ
2. ถ้า AP เคยถูกตั้งค่ามาก่อน ให้ reset ด้วยปุ่ม reset ค้างไว้ตอนจ่ายไฟ จนไฟกะพริบแล้วปล่อย
3. เข้าเว็บ AP ครั้งแรก เปลี่ยนรหัสผ่าน แล้วเลือก Country code = TH ให้ตรงกับ controller
4. แปลง AP เป็น Campus mode ให้ชี้มาที่ controller ของชุดตัวเอง — จาก CLI ของ AP  
`convert-aos-ap campus-ap 192.168.10.22n`
5. AP จะ reboot แล้วมาขอเข้า controller — ที่ controller ไปที่ Configuration » Access Points » Allowlist แล้วกด Approve ตัวของชุดตัวเอง
6. สร้าง AP group ของชุด — Configuration » AP Groups » + ตั้งชื่อ AP-GROUP-POD-n
7. ย้าย AP ที่เพิ่งเข้ามาไปอยู่ group นั้น แล้วรอ AP reboot อีกรอบ

### ตรวจสอบ

```
show ap database
show ap active
show ap-group
show ip dhcp binding
```

### ระวัง

AP ไม่ขึ้นสักที ให้ไล่จากล่างขึ้นบน — show ip dhcp binding  
เห็น AP ได้ IP ใหม่ → ถ้าไม่ได้ปัญหาลงอยู่ที่ VLAN/DHCP ใน LAB 9 ไม่ใช่ที่ตัว AP



# A. Configure WLAN PSK+MAC

- Add local user

# Configure WLAN PSK

aruba

MOBILITY  
CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller

1

Dashboard

Configuration

2

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |  |
|-------------|----------|----------------|-------------|--|
| + 3         |          |                |             |  |



# Configure WLAN PSK

WLANs 1

| NAME (SSID) | AP GROUP          | KEY MANAGEMENT | INFORMATION |  |
|-------------|-------------------|----------------|-------------|--|
| PSK-G99     | default, Group-99 | WPA2-Personal  | --          |  |

+

PSK-G99

GeneralVLANsSecurityAccessProfiles

Name (ssid):

PSK-G99

1

Create WiFi Name

Primary usage:

☒ Employee

☐ Guest

Select AP Groups

Broadcast on:

☐ default

☒ Group-99

2

Select New GroupAP

Forwarding mode:

Tunnel

Cancel

Submit

3

# Configure WLAN PSK

### New WLAN

General

VLANs

Security

Access

VLAN:

20

▼

1

Select vlan 20

[Show VLAN details](#)

Cancel

Back

Next

2

# Configure WLAN PSK

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

1

0

CLIENTS

1

0

0

ALERTS

0

admin

Mobility Controller

Search

Dashboard

Overview

Infrastructure

Traffic Analysis

Security

Services

Configuration

Diagnostics

Maintenance

Client

WLAN

265 kB

Radios

Wireless Clients 1

| NAME         | IP ADDRESS   | HEALTH | BAND  | ROLE          | SNR   | USAGE  | WLAN    | CONNECTED...      |
|--------------|--------------|--------|-------|---------------|-------|--------|---------|-------------------|
| 192.168.20.1 | 192.168.20.1 | Good   | 5 GHz | authenticated | 54 dB | 104 kB | PSK-G99 | 7c:57:3c:c3:e0:e4 |

DETAILS

Name  
192.168.20.1

IP address  
192.168.20.1

MAC address  
a4:d9:31:b8:57:75

Health score  
92%

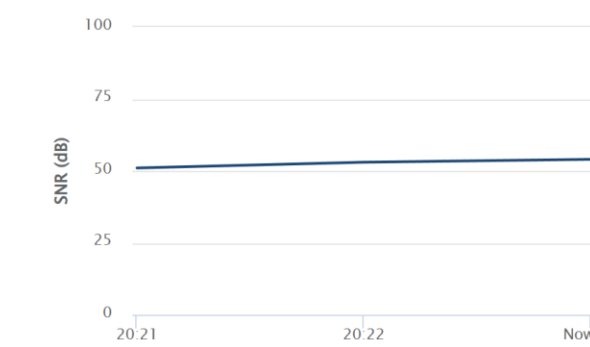
Speed  
842 Mbps

Max speed  
866 Mbps

Frames in the last minute  
121

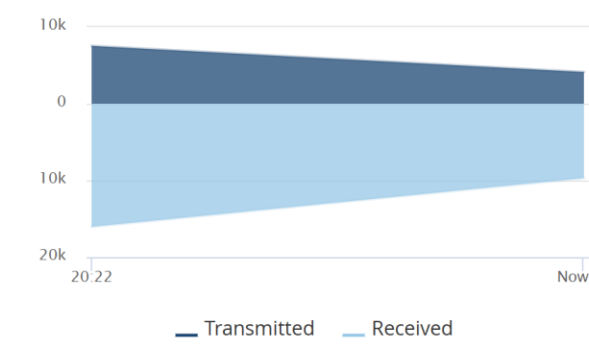
SIGNAL

Show information about signal quality



USAGE

Show information about throughput



2

Copy mac address

# Add mac address to local user

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS  
1 0

CLIENTS  
1 0 0

ALERTS  
0

admin

Mobility Controller

Dashboard

Configuration1

WLANs

Roles & Policies

Access Points

AP Groups

Authentication2

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Auth Servers3

Profiles

L2 Authentication

L3 Authentication

User Rules

Advanced

Server Groups 2

| NAME     | SERVICES | FAIL THROUGH | LOAD BALANCE | SERVICES RULES |   |
|----------|----------|--------------|--------------|----------------|---|
| default  | 1        | --           | --           | 1              |   |
| internal | 1        | --           | --           | 1              | 4 |

+

Server Group > internal

ServersOptionsServer Rules

Drag rows to re-order

| NAME     | TYPE | IP ADDRESS | TRIM FQDN | MATCH RULES |   |
|----------|------|------------|-----------|-------------|---|
| Internal | --   | --         | --        | 0           | 5 |

+

Server Group > internal > Internal

UsersOptionsImportExportServer Group Trim FQDNServer Group Match RulesGuest user page

| USER NAME | ROLE | EMAIL | EXPIRATION | STATIC IP FOR RAPS | ENABLED |  |
|-----------|------|-------|------------|--------------------|---------|--|
|-----------|------|-------|------------|--------------------|---------|--|

+

6

# Add mac address to local user

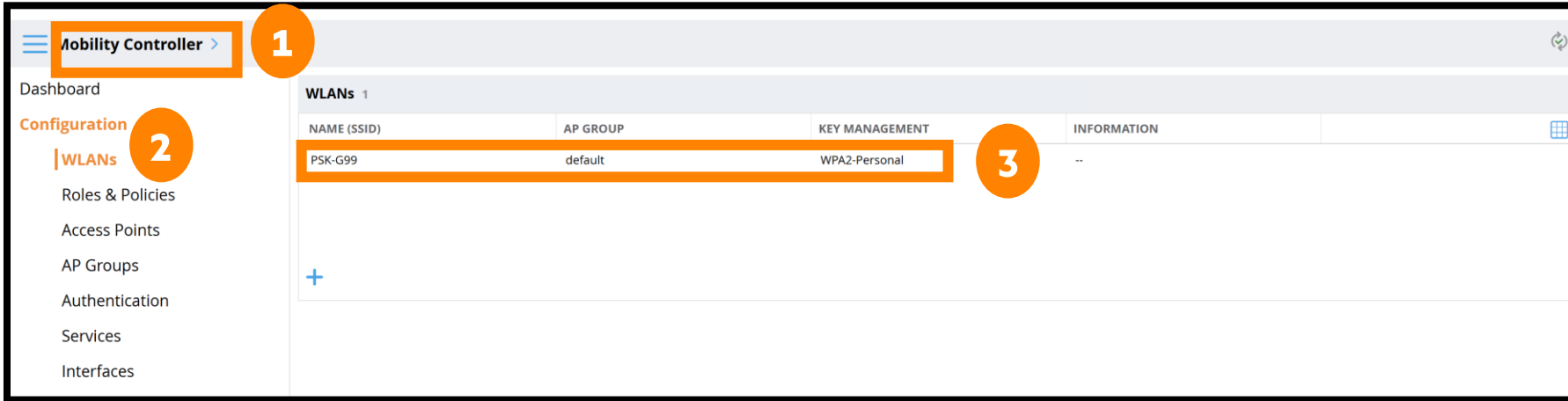
1 Past mac address to username, password, retype password

Internal Server > Add New User

|                           |                                            |                                         |
|---------------------------|--------------------------------------------|-----------------------------------------|
| User name:                | <input type="text" value="a4d931b85775"/>  | <input type="button" value="Generate"/> |
| Password:                 | <input type="password" value="....."/>     | <input type="button" value="Generate"/> |
| Retype password:          | <input type="password" value="....."/>     |                                         |
| Role:                     | <input type="text" value="authenticated"/> | 2                                       |
| E-mail:                   | <input type="text"/>                       |                                         |
| Static inner IP for RAPs: | <input type="text"/>                       |                                         |
| Enabled:                  | <input checked="" type="checkbox"/>        |                                         |
| Expiration:               | <input type="text" value="None"/>          |                                         |

Cancel Submit 3

# Configure WLAN Add MAC Authentication



1. Mobility Controller

2. Configuration

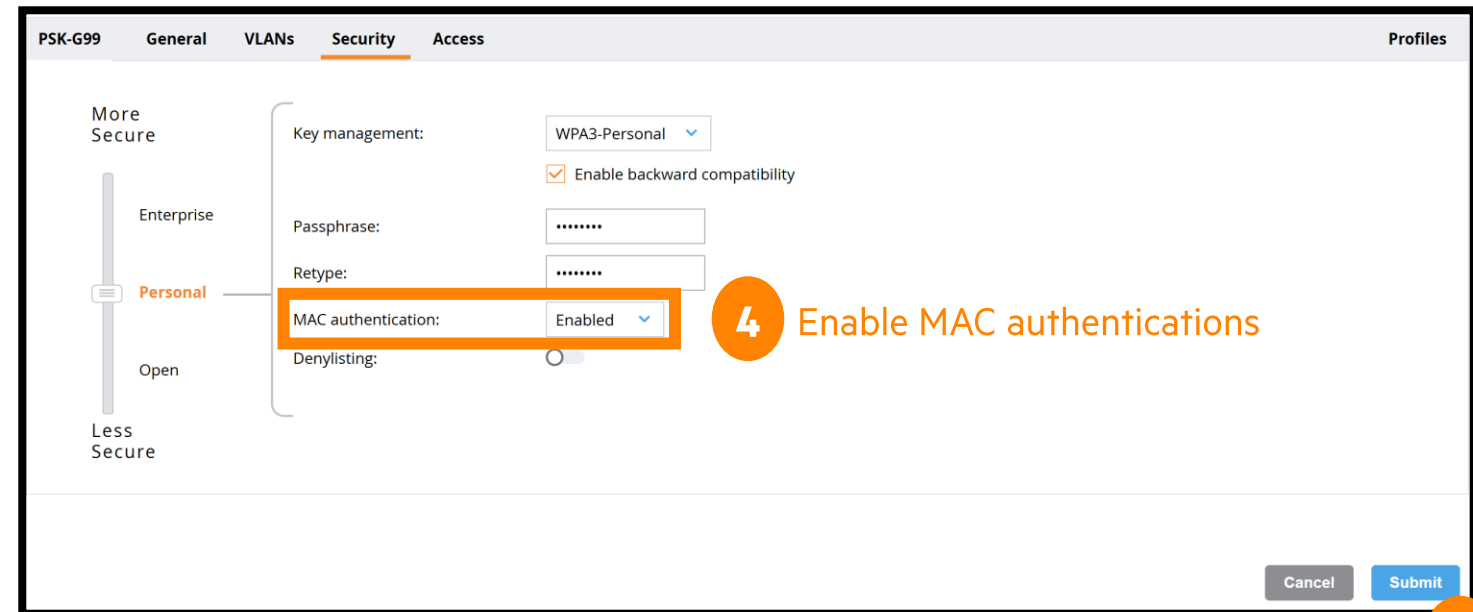
3. WLANs

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

Dashboard

Configuration

- WLANs
- Roles & Policies
- Access Points
- AP Groups
- Authentication
- Services
- Interfaces



PSK-G99

General VLANs Security Access Profiles

More Secure

Enterprise

Personal

Open

Less Secure

Key management: WPA3-Personal

Enable backward compatibility

Passphrase: .....

Retype: .....

MAC authentication: Enabled

Denylisting: ☐

4. Enable MAC authentications

Cancel Submit



# Configure WLAN Add MAC Authentication

PSK-G99   General   VLANs   Security   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Cancel** **Submit**

**Pending Changes**

**WLANs 1**

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

**General**   **VLANs**   **Security**   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Pending Changes**

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

**Close** **Discard changes** **Deploy changes**

# Configure WLAN Add MAC Authentication

PSK-G99   General   VLANs   Security   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Cancel** **Submit**

**Pending Changes**

**WLANs 1**

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| PSK-G99     | default  | WPA2-Personal  | --          |

**General**   **VLANs**   **Security**   **Access**   Profiles

Default role: **denyall**

MAC authentication role: **guest**

Show [roles](#)

**Pending Changes**

☒ Pending Changes for 1 Group

☒ [+](#) Mobility Controller

**Close** **Discard changes** **Deploy changes**



# B. Configure WLAN 802.1X

---

# Configure WLAN 802.1X

The screenshot displays the Aruba Mobility Controller web interface. At the top, the Aruba logo is on the left, and the title 'MOBILITY CONTROLLER ArubaMC\_Group99' is in the center. To the right of the title are three status sections: 'ACCESS POINTS' with a green checkmark and '1', 'CLIENTS' with a Wi-Fi icon and '0', and 'ALERTS' with a warning triangle and '0'. Further right is a help icon and a user dropdown menu showing 'admin'. Below the title bar is a navigation bar with a hamburger menu icon and the text 'Mobility Controller'. The left sidebar contains a 'Dashboard' link and a 'Configuration' link, which is highlighted with an orange box and a '2'. Under 'Configuration' is a sub-link 'WLANs'. The main content area shows a table titled 'WLANs 0'. The table has four columns: 'NAME (SSID)', 'AP GROUP', 'KEY MANAGEMENT', and 'INFORMATION'. Below the table is a blue plus sign icon with a '3' next to it, indicating a button to add a new WLAN.

aruba

MOBILITY CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS  
✓ 1 0

CLIENTS  
0 0 0

ALERTS  
0

admin

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |
|-------------|----------|----------------|-------------|
| + 3         |          |                |             |



# Configure WLAN 802.1X

**New WLAN**

General VLANs Security Access

Name (SSID): 802.1X-G99 **1**

Primary usage: ☒ Employee ☐ Guest

Select AP Groups **2**

Broadcast on: ☐ default ☒ Group-99

Forwarding mode: Tunnel

Cancel Back Next **3**

**New WLAN**

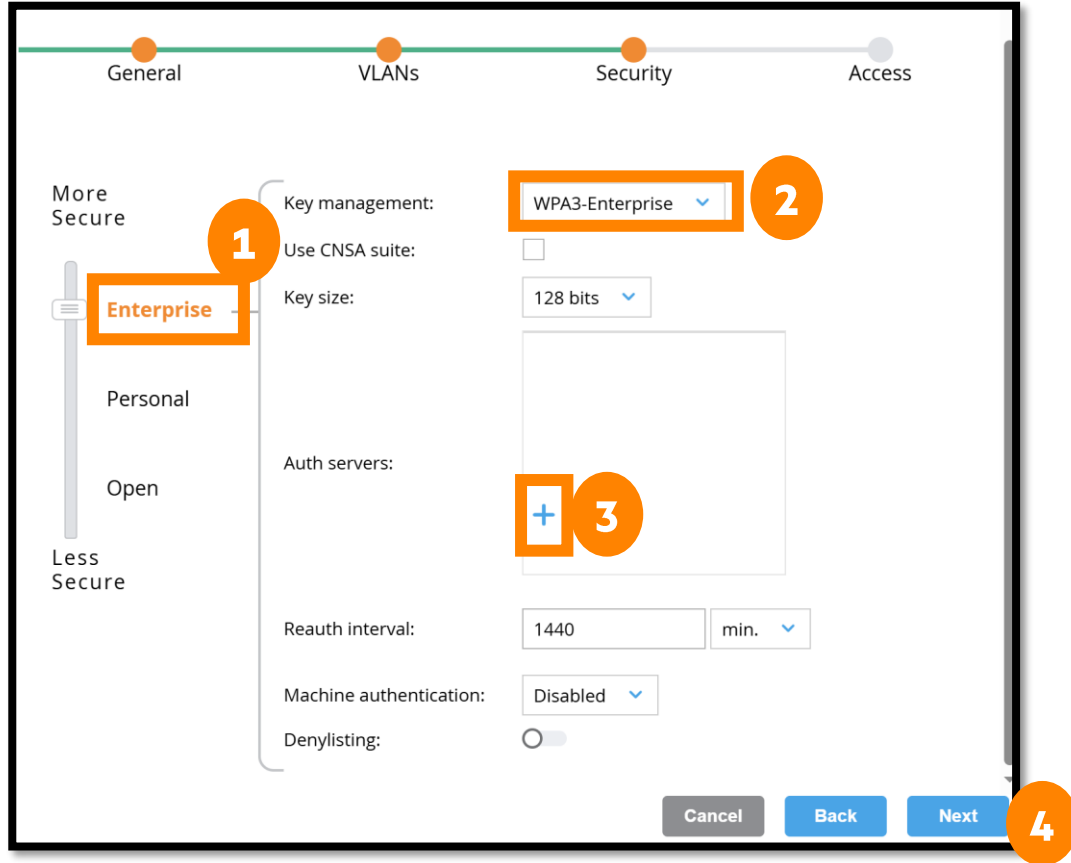
General VLANs Security Access

VLAN: 10 **4** Select vlan 10

[Show VLAN details](#)

Cancel Back Next **5**

# Configure WLAN 802.1X



General VLANs Security Access

More Secure

Enterprise

Personal

Open

Less Secure

Key management: WPA3-Enterprise

Use CNSA suite: ☐

Key size: 128 bits

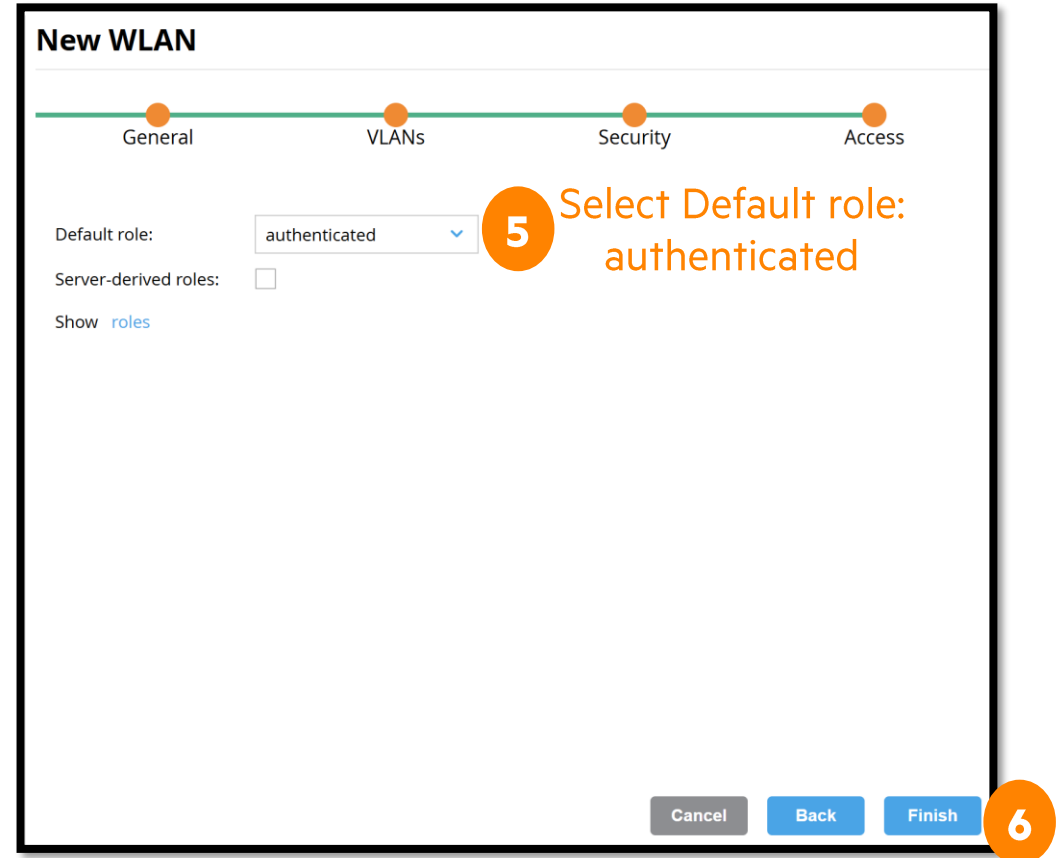
Auth servers: +

Reauth interval: 1440 min.

Machine authentication: Disabled

Denylisting: ☐

Cancel Back Next



New WLAN

General VLANs Security Access

Default role: authenticated

Server-derived roles: ☐

Show roles

Cancel Back Finish

Select Default role: authenticated



# Configure add local user for 802.1X

**Mobility Controller**

Dashboard **Configuration** (1) **Auth Servers** (3) AAA Profiles L2 Authentication L3 Authentication

Server Groups 3

| NAME              | SERVICES | FAIL THROUGH | LOAD BALANCE | SERVICES RULES |
|-------------------|----------|--------------|--------------|----------------|
| default           | 1        | --           | --           | 1              |
| Internal          | 1        | --           | --           | 1              |
| 802.1X-G99_dot... | 1        | --           | --           | 0              |

All Servers 1

| NAME     | TYPE | IP ADDRESS / HOST | SERVICES GROUP           |
|----------|------|-------------------|--------------------------|
| Internal | --   | --                | default internal 802.... |

**Auth Servers** AAA Profiles L2 Authentication L3 Authentication

**Server > Internal** **Users** Options Import Export [Guest user page](#)

| USER NAME     | ROLE          | EMAIL | EXPIRATION | STATIC IP FO... | ENABLED |
|---------------|---------------|-------|------------|-----------------|---------|
| a4d931b857... | authenticated | --    | --         | 0.0.0.0         | Yes     |

+ (5)

50 < 1 >

**Internal Server > Add New User** (6)

User name: group99 **Generate**

Password: ..... **Generate**

Retype password: .....

Role: authenticated (7)

E-mail:

Static inner IP for RAPs:

Enabled: ☒

Expiration: None

Create User  
User: group0X  
Pass: P@sswOrd

Cancel Submit (8)



# C. Configure WLAN Guest

- Add local user



# Configure WLAN Guest

aruba

MOBILITY  
CONTROLLER  
ArubaMC\_Group99

ACCESS POINTS

✓ 1

⌚ 0

CLIENTS

📶 0

👤 0

📍 0

ALERTS

⚠ 0

?

admin ▾

Mobility Controller 1

Dashboard

Configuration 2

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

WLANs 0

| NAME (SSID) | AP GROUP | KEY MANAGEMENT | INFORMATION |  |
|-------------|----------|----------------|-------------|--|
| + 3         |          |                |             |  |

# Configure WLAN Guest

## New WLAN

General VLANs Security Access

Name (SSID):  1

Primary usage: ☐ Employee ☒ Guest 2

Select AP Groups

Broadcast on: ☐ default ☒ Group-99 3

Forwarding mode:

Cancel Back Next 4

## New WLAN

General VLANs Security Access

VLAN:  5 Select vlan 30

[Show VLAN details](#)

Cancel Back Next 6

# Configure WLAN Guest

## New WLAN

General

VLANs

Security

Access

### Select Internal Captive Portal on Controller

ClearPass or other external Captive Portal

**Internal Captive Portal with authentication**

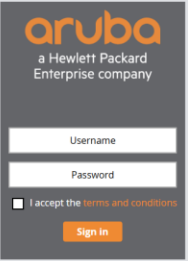
Internal Captive Portal with email registration

Internal Captive Portal, no auth or registration

No Captive Portal

**Captive Portal Options:**

Template [Custom HTML](#)



Click thumbnail above to edit [Preview](#)

Redirect URL:

Use purple wi-fi: ☐

Cancel

Back

Next

## New WLAN

General

VLANs

Security

Access

Default role: Guest-99-guest-logon

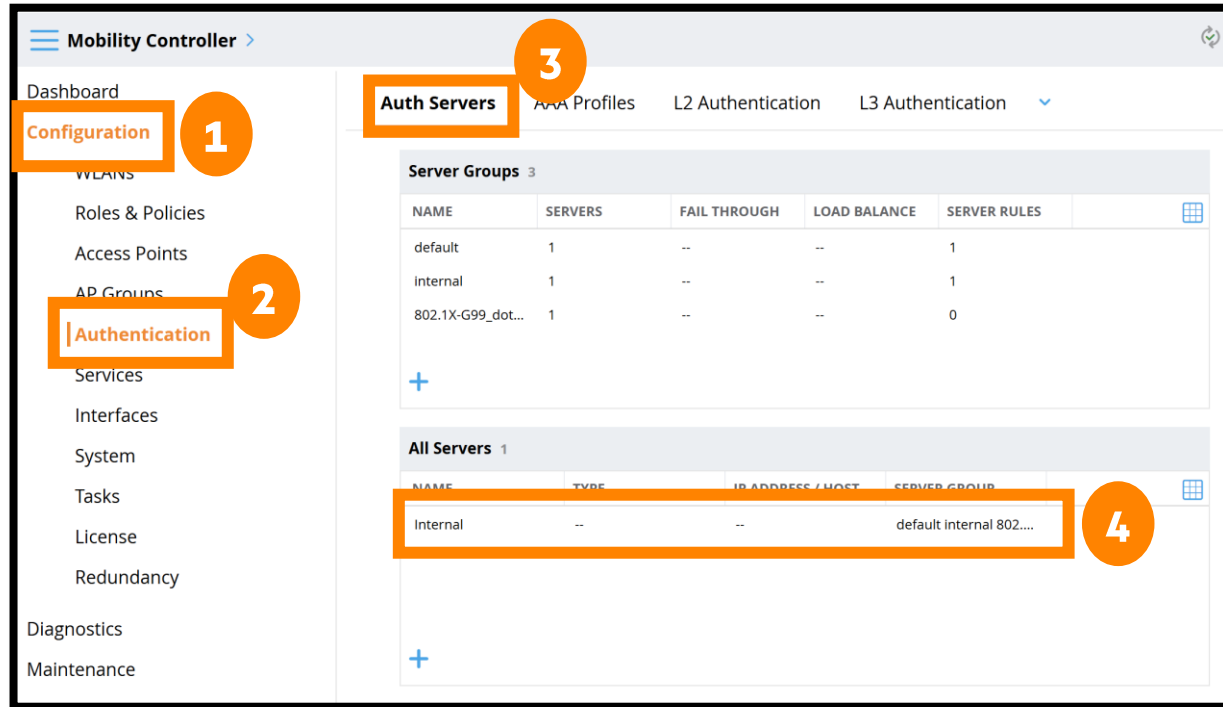
Cancel

Back

Finish

**HPE** aruba  
networking

# Configure add local user for 802.1X



Mobility Controller >

Dashboard

**Configuration** 1

Roles & Policies

Access Points

AP Groups

**Authentication** 2

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

**Auth Servers** 3

AAA Profiles

L2 Authentication

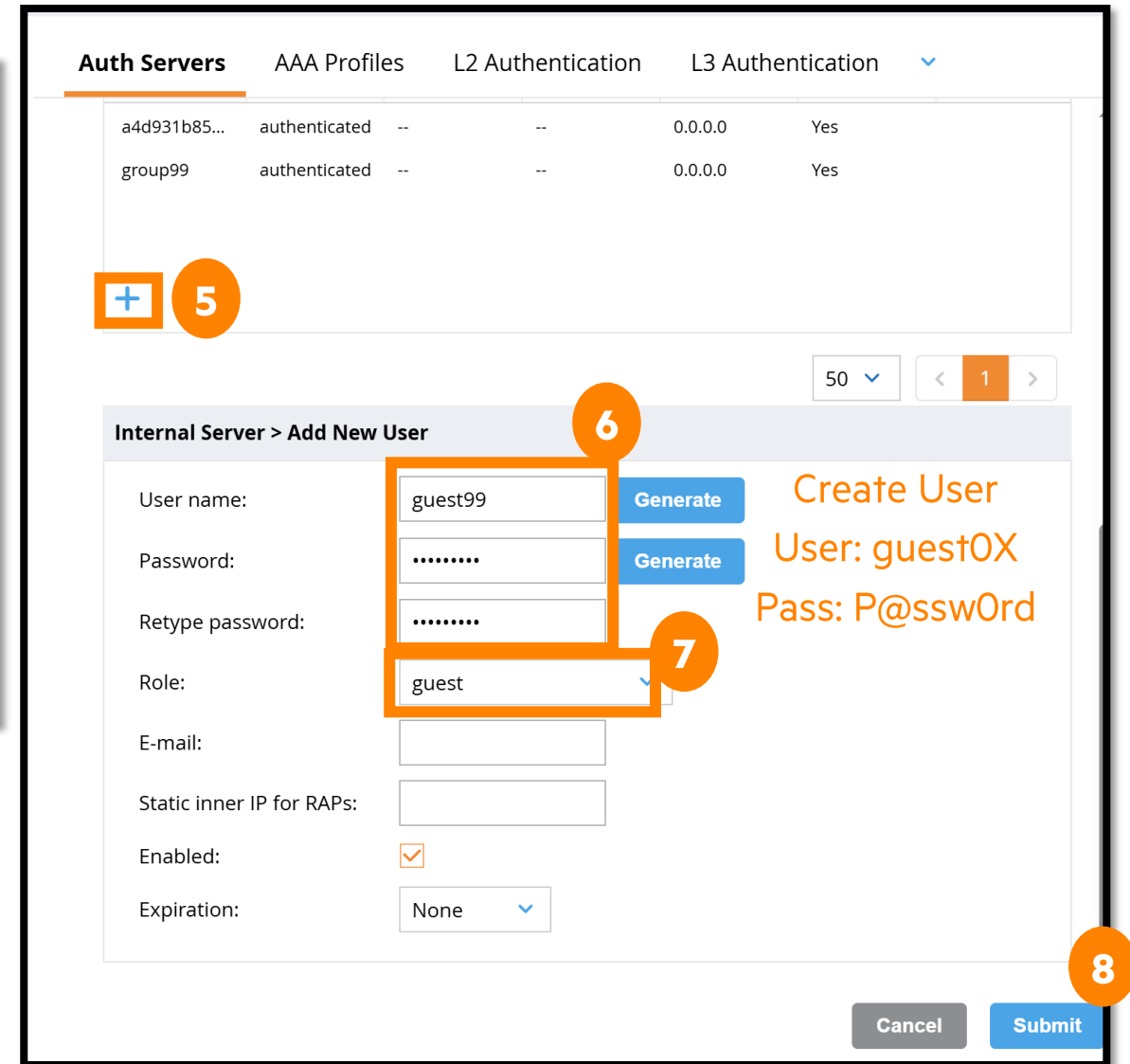
L3 Authentication

Server Groups 3

| NAME              | SERVERS | FAIL THROUGH | LOAD BALANCE | SERVER RULES |
|-------------------|---------|--------------|--------------|--------------|
| default           | 1       | --           | --           | 1            |
| Internal          | 1       | --           | --           | 1            |
| 802.1X-G99_dot... | 1       | --           | --           | 0            |

All Servers 1

| NAME     | TYPE | IP ADDRESS / HOST | SERVER GROUP             |
|----------|------|-------------------|--------------------------|
| Internal | --   | --                | default internal 802.... |



Auth Servers

AAA Profiles

L2 Authentication

L3 Authentication

|              |               |    |    |         |     |
|--------------|---------------|----|----|---------|-----|
| a4d931b85... | authenticated | -- | -- | 0.0.0.0 | Yes |
| group99      | authenticated | -- | -- | 0.0.0.0 | Yes |

50

< 1 >

**Internal Server > Add New User** 6

User name: guest99

Password: .....

Retype password: .....

Role: guest

E-mail:

Static inner IP for RAPs:

Enabled: ☒

Expiration: None

Generate

Generate

7

Create User  
User: guest0X  
Pass: P@ssw0rd

8

Cancel

Submit



## LAB 11 WLAN แบบ PSK และเพิ่ม MAC Authentication

Controller VM ของชุด ปล่อง SSID แรกของชุดด้วยรหัสผ่านร่วม แล้วเพิ่มการเช็ค MAC ซ่อนเข้าไปอีกชั้น

1. Configuration » WLANs » + สร้าง SSID ชื่อ POD-n-PSK ผูกกับ AP group ของชุด
2. เลือก Security = Personal · Key management = WPA2/WPA3 Personal · ตั้ง passphrase ที่ตกลงกันในชุด
3. เลือก VLAN ปลายทางเป็น 260n ของชุดตัวเอง แล้วกด Submit และ Deploy changes
4. เอามือถือหรือโน้ตบุ๊กเชื่อม SSID นี้ ต้องได้ IP ในวง 10.26.n.x และออกเน็ตได้
5. เพิ่มผู้ใช้แบบ MAC เข้าฐานข้อมูลในตัว controller — ใช้ MAC ของเครื่องที่เพิ่งต่อ โดยใส่ MAC เป็นทั้งชื่อผู้ใช้และรหัสผ่าน (Configuration » Authentication » Local users)
6. กลับไปที่ SSID เดิม เปิด MAC Authentication แล้วชี้ auth server เป็น Internal
7. ทดสอบสองทาง — เครื่องที่ MAC อยู่ในฐานะต้องต่อได้ เครื่องที่ไม่อยู่ต้องโดนปฏิเสธ ทั้งที่รู้ passphrase

### ตรวจสอบผล

```
show wlan ssid-profile
show user-table
show local-userdb
show ap active
```

### ระวัง

MAC address ต้องใส่แบบไม่มีขีดไม่มีจุดคั่น และเป็นตัวพิมพ์เล็กทั้งหมด — ใส่ผิดรูปแบบจะไม่ error แต่จะไม่แมตช์อะไรเลย

## LAB 12 WLAN แบบ 802.1X

Controller VM ของชุด ปล่อง SSID ที่ต้องล็อกอินรายคน โดยใช้ฐานผู้ใช้ในตัว controller เป็น RADIUS

1. Configuration » WLANs » + สร้าง SSID ชื่อ POD-n-DOT1X ผูกกับ AP group ของชุด
2. เลือก Security = Enterprise · Key management = WPA2/WPA3 Enterprise
3. ตั้ง Authentication server เป็น Internal — หลักสูตรนี้ไม่มี ClearPass จึงใช้ฐานในตัว
4. เลือก VLAN ปลายทาง 260n แล้ว Submit และ Deploy changes
5. สร้างผู้ใช้สำหรับทดสอบ — Configuration » Authentication » Local users เพิ่ม user1 รหัส Aruba123!
6. ต่อจากเครื่องลูกข่าย ใส่ user1 กับรหัส แล้วเช็คว่าได้ IP ในวง 10.26.n.x
7. เทียบให้เห็นความต่าง — show user-table จะบอกว่าแต่ละเครื่องเข้ามาด้วยวิธีไหน SSID PSK จะไม่มีชื่อผู้ใช้ แต่ SSID นี้มี

### ตรวจสอบ

```
show user-table
show aaa authentication-server internal
show wlan ssid-profile
```

### ระวัง

เครื่องลูกข่ายมักไม่ยอมต่อเพราะไม่เชื่อ certificate ของ controller — ในแลบให้เลือกยอมรับต่อไป แต่ของจริงต้องเอา CA ไปติดตั้งที่เครื่องลูกข่ายก่อน

## LAB 13 WLAN แบบ Guest ผ่าน Captive Portal

Controller VM ของชุด ปล่อง SSID เปิดที่บังคับต้องหน้าล็อกอินก่อนใช้เน็ต — แบบที่ใช้กับผู้มาติดต่อ

1. Configuration » WLANs » + สร้าง SSID ชื่อ POD-n-GUEST ผูกกับ AP group ของชุด
2. เลือก Security = Captive portal (Guest) — ตัว SSID เองเปิดโล่ง ไม่มี passphrase
3. เลือก Captive portal profile แบบ Internal และ Authentication server = Internal
4. เลือก VLAN ปลายทาง 260n แล้ว Submit และ Deploy changes
5. สร้างผู้ใช้ guest — Configuration » Authentication » Local users เพิ่ม guest1 รหัส Guest123!
6. ต่อจากมือถือ ต้องได้ IP ก่อน แล้วเปิดเบราว์เซอร์ต้องต้องหน้าล็อกอิน ใส่ guest1 จึงจะออกเน็ตได้
7. ดูใน show user-table ว่าก่อนล็อกอินเครื่องอยู่ใน role logon และหลังล็อกอินเปลี่ยนเป็น guest

### ตรวจสอบผล

```
show user-table
show aaa authentication captive-portal
show rights
```

### ระวัง

หน้า portal ไม่เด้งเกือบทุกครั้งเพราะเครื่องลูกข่ายยัง resolve DNS ไม่ได้ — ต้องปล่อยให้ role logon ยิง DNS ออกไปได้ ไม่งั้นเบราว์เซอร์ไม่มีอะไรให้ redirect



# **D. Configure Role & Policy**

- **Network aliases**



# Network Aliases

Configure on node “Mobility Controller”-> Configuration-> Roles & Policies -> Aliases -> +

The screenshot displays the Aruba Mobility Conductor web interface. At the top, the header includes the Aruba logo, the text "MOBILITY CONDUCTOR ArubaOS\_MM01\_Ha...", and several status indicators: "CONTROLLERS" (2 green, 0 grey), "ACCESS POINTS" (0 green, 0 grey), "CLIENTS" (1 Wi-Fi icon, 0 user icon, 0 group icon), and "ALERTS" (0 warning icon). A user dropdown menu shows "admin".

The left sidebar contains a navigation menu with the following items: "Dashboard", "Configuration" (highlighted), "WLANs", "Roles & Policies" (highlighted), "Access Points", "AP Groups", "Authentication", "Services", "Interfaces", "Controllers", "System", and "Tasks".

The main content area is titled "Mobility Controller" and features a sub-menu with "Roles", "Policies", "Applications", and "Aliases" (highlighted). Under the "Aliases" tab, there is a section for "Network Aliases" which contains a table with the following data:

| NAME        | ITEMS | DESCRIPTION | IP VERSION | INVERT |
|-------------|-------|-------------|------------|--------|
| mswitch     | 1     | --          | IPv4       | --     |
| controller  | 1     | --          | IPv4       | --     |
| vrrp_ip     | 1     | --          | IPv4       | --     |
| any         | 1     | --          | IPv4       | --     |
| user        | 1     | --          | IPv4       | --     |
| localip     | 1     | --          | IPv4       | --     |
| auth-google | 2     | --          | IPv4       | --     |

Below the table is a blue "+" icon to add a new alias. A small grid icon is visible in the top right corner of the table area.

# Network Aliases (cont.)

Enter Name "Local\_Network" +

**Destination Local\_Network**

IP version:  1

Name:

Description:

Invert: ☐

**Items**

| TYPE | IP ADDRESS | NETMASK/RANGE/... | DOMAIN NAME |
|------|------------|-------------------|-------------|
|------|------------|-------------------|-------------|

+ 2

Rule type : **Network**  
IP Address : **192.168.0.0**  
Network mask : **255.255.0.0**  
**OK -> Submit**

**Add New Destination**

Rule type:  3

IP address:

Network mask:

4



# Policies

## Create Policies

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Roles

Policies

Applications

Aliases

| NAME           | RULES COU... | TYPE    | POLICY US...     | DESCRIPTION |
|----------------|--------------|---------|------------------|-------------|
| validuseret... | 1            | eth     | --               | --          |
| global-sacl    | 0            | session | guest, statef... | --          |
| sdn-acl        | 0            | session | --               | --          |
| sys-control    | 13           | session | sys-ap-role, ... | --          |
| sys-ap-acl     | 12           | session | sys-ap-role      | --          |
| switch-logo... | 1            | session | switch-logon     | --          |
| sys-switch-acl | 12           | session | sys-switch-r...  | --          |

+

New Policy

Policy type:

Session

Policy name:

Local\_Network-Polic

Description:

Cancel

Submit

# Policies (cont.)

Roles **Policies** Applications Aliases

**Policies**

| NAME                          | RULES C... | TYPE           | POLICY ...   | DESCRIP... |          |
|-------------------------------|------------|----------------|--------------|------------|----------|
| apprf-sys-ap-role-sacl        | 0          | session        | sys-ap-role  | --         |          |
| apprf-sys-switch-role-sacl    | 0          | session        | sys-switc... | --         |          |
| apprf-guest-99-guest-log...   | 0          | session        | Guest-99...  | --         |          |
| <b>Local_Network-Policies</b> | 0          | <b>session</b> | --           | --         | <b>1</b> |
| uplink-lb-cfg-racl            | 0          | routing        | --           | --         |          |
| uplink-lb-sys-racl            | 0          | routing        | --           | --         |          |
| master-boc-traffic            | 0          | routing        | --           | --         |          |

**Policy > Local\_Network-Policies Rules** ⓘ Drag rows to re-order

| IP VERSI... | SOURCE | DESTINA... | SERVICE/... | ACTION | DESCRIP... |  |
|-------------|--------|------------|-------------|--------|------------|--|
|-------------|--------|------------|-------------|--------|------------|--|

**2**

## New Rule for Local\_Network-Policies

Rule type: ☒ Access control **3** ☐ Application

Cancel

OK **4**

## Local\_Network-Policies > New forwarding Rule

IP version: IPv4 **5**  
Source: Any  
Destination: Alias  
Destination alias: local\_network  
Service/app: Service  
Services alias: svc-icmp  
Action: Deny

TOS:  
Time range: - None - Reset  
802.1p priority:  
Options:  
☐ Log ☐ Mirror ☐ Denylist ☐ Disable scanning  
Queue:  
Description:

Cancel

Submit **6**

Create Policies Block ping

IP Version : **IPV4**

Source : **Any**

Destination : **Alias**

Destination alias : **Local\_Network**

Service/app : **Service**

Service alias : **svc-icmp**

Action : **Deny**

**Submit**

# Role

## Create Role

Mobility Controller

Dashboard

Configuration

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Roles

Policies

Applications

Aliases

Roles 15

| NAME            | RULES    |
|-----------------|----------|
| logon           | 32 Rules |
| guest           | 11 Rules |
| ap-role         | 35 Rules |
| stateful-dot1x  | 0 Rules  |
| guest-logon     | 27 Rules |
| sys-ap-role     | 25 Rules |
| sys-switch-role | 25 Rules |

+



New Role

Name:

engineer

Cancel

Submit

# Role

## Assign policies in Role engineer

Mobility Controller >

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Roles

ies

Applications

Aliases

Roles 16

| NAME                 | RULES    |
|----------------------|----------|
| denyall              | 1 Rules  |
| default-via-role     | 3 Rules  |
| default-vpn-role     | 4 Rules  |
| authenticated        | 4 Rules  |
| voice                | 45 Rules |
| Guest-99-guest-logon | 26 Rules |
| engineer             | 0 Rules  |

+

engineer

Show Advanced View

Global Rules

| IP VERSI... | SOURCE | DESTINA... | SERVICE/... | ACTION | DESCRIP... |
|-------------|--------|------------|-------------|--------|------------|
|-------------|--------|------------|-------------|--------|------------|

+

engineer

Policies

Bandwidth

Captive Portal

Show Basic View

| NAME           | RULES COU... | TYPE    | POLICY US...    | DESCRIPTI... |
|----------------|--------------|---------|-----------------|--------------|
| global-sacl    | 0            | session | guest, state... | --           |
| apprf-engin... | 0            | session | engineer        | --           |
| engineer       | 0            | session | engineer        | --           |

+

# Role

Assign policies > local\_network(first), allowall(second)

**New Policy**

☒ Add an existing policy **1** ☐ Create a new policy

Policy type: Session

Policy name: Local\_Network-Policies **2**

Description:

Position:

**3** Cancel Submit

first

| engineer       | Policies     | Bandwidth | Captive Portal  | Show Basic View |
|----------------|--------------|-----------|-----------------|-----------------|
| NAME           | RULES COU... | TYPE      | POLICY US...    | DESCRIPTI...    |
| global-sacl    | 0            | session   | guest, state... | --              |
| apprf-engin... | 0            | session   | engineer        | --              |
| engineer       | 0            | session   | engineer        | --              |
| Local_Netw...  | 1            | session   | --              | --              |

**4**

**5**

second

**New Policy**

☒ Add an existing policy **6** ☐ Create a new policy

Policy type: Session

Policy name: allowall **7**

Description:

Position:

**8** Cancel Submit

save

**Pending Changes** **10**

**Roles** Policies Applications Aliases

**Roles 16**

| NAME                 | RULES    |
|----------------------|----------|
| denyall              | 1 Rules  |
| default-via-role     | 3 Rules  |
| default-vpn-role     | 4 Rules  |
| authenticated        | 4 Rules  |
| voice                | 45 Rules |
| Guest-99-guest-logon | 26 Rules |
| engineer             | 3 Rules  |

**9**

| engineer       | Policies     | Bandwidth | Captive Portal | Show Basic View |
|----------------|--------------|-----------|----------------|-----------------|
| NAME           | RULES COU... | TYPE      | POLICY US...   | DESCRIPTI...    |
| apprf-engin... | 0            | session   | engineer       | --              |
| engineer       | 0            | session   | engineer       | --              |
| Local_Netw...  | 1            | session   | engineer       | --              |
| allowall       | 2            | session   | default-iap... | --              |

# Role

## Assign role engineer to user local

Mobility Controller

Dashboard

Configuration

WLANs

Roles & Policies

Access Points

AP Groups

Authentication

Services

Interfaces

System

Tasks

License

Redundancy

Diagnostics

Maintenance

Auth Servers

AAA Profiles

L2 Authentication

Server Groups

| NAME          | SERVERS | FAIL THRO... | LOAD BAL... | SERVER RU... |
|---------------|---------|--------------|-------------|--------------|
| default       | 1       | --           | --          | 1            |
| internal      | 1       | --           | --          | 1            |
| 802.1X-G99... | 1       | --           | --          | 0            |

All Servers 1

| NAME     | TYPE | IP ADDRESS / ... | SERVER GROUP      |
|----------|------|------------------|-------------------|
| Internal | --   | --               | default intern... |

Server > Internal

Users

Options

Import

Guest user page

| USER NA... | ROLE        | EMAIL | EXPIRATI... | STATIC IP... | ENABLED |
|------------|-------------|-------|-------------|--------------|---------|
| a4d931h    | authenti... | --    | --          | 0.0.0.0      | Yes     |
| group99    | authenti... | --    | --          | 0.0.0.0      | Yes     |
| guest99    | guest       | --    | --          | 0.0.0.0      | Yes     |



Edit User group99

User name:group99

Password:\*\*\*\*\*Generate

Role:engineer

E-mail:

Static inner IP for RAPS:0.0.0.0

Enabled:☒

Expiration:None

CancelSubmit

Change to Role engineer



# Role

## Test connect WiFi > 802.1X-G99

The screenshot shows the Aruba Network Controller interface. At the top, there are tabs for ACCESS POINTS (1), CLIENTS (1), and ALERTS (0). The CLIENTS tab is selected, and a search bar is visible. Below the search bar, there are statistics for wireless clients: 1 user, 3 wireless clients, 82.1 kB bandwidth, and 2 wireless networks. A table titled 'Wireless Clients 1' lists the following client:

| NAME    | IP ADDR...   | HEALTH   | BAND  | ROLE     |
|---------|--------------|----------|-------|----------|
| group99 | 192.168.10.2 | Unkno... | 5 GHz | engineer |

Annotations: A red circle with '1' is next to the CLIENTS tab. A red circle with '2' is next to the 'engineer' role. A red circle with '3' is next to the 'group99' name.

## Test ping gateway user

```
Command Prompt
Microsoft Windows [Version 10.0.19045.5965]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sakkr>ping 192.168.10.254

Pinging 192.168.10.254 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.254:
 Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),

C:\Users\sakkr>ping 192.168.10.253

Pinging 192.168.10.253 with 32 bytes of data:
Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.10.253:
 Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

Annotation: A red circle with '4' is next to the first ping command.

## LAB 14 Role, Policy และ Dashboard

Controller VM ของชุด จำกัดว่าใครไปไหนได้บ้างด้วย Role กับ Policy แล้วดูผลจริงบน Dashboard

1. สร้าง network alias ของปลายทางที่จะคุม — Configuration » Advanced » Aliases ตั้งชื่อ INTERNAL-NET ชี้ไปที่ 192.168.10.0/24
2. สร้าง policy ใหม่ — Configuration » Roles & Policies » Policies » + ใส่กฎ deny ไปที่ INTERNAL-NET ไว้บนสุด แล้วต่อด้วยกฎ permit any
3. ลำดับของกฎสำคัญกว่าตัวกฎ — กฎแรกที่แมตช์คือกฎที่ใช้ ถ้าเอา permit any ไว้บน กฎ deny ข้างล่างจะไม่มีวันทำงาน
4. สร้าง Role ชื่อ guest-restricted แล้วผูก policy นี้เข้าไป
5. เอา role ไปผูกกับ SSID POD-n-GUEST เป็น initial role หลังล็อกอิน แล้ว Deploy changes
6. ทดสอบจากเครื่อง guest — ออกเน็ตได้ แต่ ping 192.168.10.x ต้องไม่ผ่าน
7. เปิด Dashboard » Overview ดูจำนวน client และ AP · Dashboard » Clients ดูรายเครื่องว่าอยู่ role ไหน · Dashboard » Applications ดูว่า traffic เป็นแอปอะไรบ้าง

### ตรวจสอบผล

```
show rights
show netdestination
show acl brief
show user-table verbose
```

### ระวัง

แก้ policy แล้วเครื่องที่ต่ออยู่ไม่เปลี่ยนพฤติกรรม เป็นเรื่องปกติ — role ผูกกับ session ที่ยังค้างอยู่ ต้องตัดเครื่องออกแล้วต่อใหม่ถึงจะได้ role ใหม่

**CUSTOMER FIRST**  
**CUSTOMER LAST**

The image features the text "CUSTOMER FIRST" on the top line and "CUSTOMER LAST" on the bottom line. The letters are large, bold, and white. Each letter is filled with a different photograph of people at what appears to be a conference or event. The photos show various individuals, some smiling, some looking at devices, and others in group settings. The overall theme is customer-centricity.